

63. Electrical Characteristics

63.1 Absolute Maximum Ratings

Table 63.1 Absolute Maximum Rating

Conditions: VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0 V

Item		Symbol	Value	Unit
Power supply voltage		VCC, VCC_USB	−0.3 to +4.0	V
V _{BATT} power supply voltage		V _{BATT}	−0.3 to +4.0	V
Input voltage (except for ports for 5 V tolerant*1)		V _{in}	−0.3 to VCC + 0.3 (up to 4.0)	V
Input voltage (ports for 5 V tolerant*1)		V _{in}	−0.3 to VCC + 4.0 (up to 5.8)	V
Reference power supply voltage		VREFH0	−0.3 to AVCC0 + 0.3 (up to 4.0)	V
Analog power supply voltage		AVCC0, AVCC1*2	−0.3 to +4.0	V
Analog input voltage		V _{AN}	−0.3 to AVCC + 0.3 (up to 4.0)	V
Junction temperature	D version	T _j	−40 to +105	°C
	G version	T _j	−40 to +125	°C
Storage temperature		T _{stg}	−55 to +125	°C

Caution: Permanent damage to the LSI may result if absolute maximum ratings are exceeded.

Note 1. P07, P11 to P17, P20, P21, P30 to P33, P67, and PC0 to PC3 are 5 V tolerant.

Note 2. Connect the AVCC0, AVCC1, and VCC_USB pins to VCC, and the AVSS0, AVSS1, and VSS_USB pins to VSS.

When the A/D converter unit 0 is not to be used, connect the VREFH0 pin to VCC and the VREFL0 pin to VSS, respectively.

Do not leave these pins open. Insert capacitors of high frequency characteristics between the AVCC0 and AVSS0 pins, or AVCC1 and AVSS1 pins. Place capacitors of about 0.1 μF as close as possible to every power supply pin and use the shortest and heaviest possible traces.

63.2 Recommended Operating Conditions

Table 63.2 Recommended Operating Conditions (1)

Item	Symbol	Min.	Typ.	Max.	Unit
Power supply voltage*1	VCC	2.7	—	3.6	V
	VSS	—	0	—	
V _{BATT} power supply voltage	V _{BATT}	1.62*2	—	3.6	V
USB power supply voltage	VCC_USB	—	VCC	—	V
	VSS_USB	—	0	—	
Analog power supply voltage*1, *3	AVCC0	—	VCC	—	V
	AVSS0	—	0	—	
	AVCC1	—	VCC	—	
	AVSS1	—	0	—	
	VREFH0	2.7	—	AVCC0	
	VREFL0	—	0	—	
Input voltage (except for 5 V tolerant ports, except for P03, P05 and P40 to P47)*4	V _{in}	−0.3	—	VCC + 0.3	V
Input voltage (P03, P05 and P40 to P47)	V _{in}	−0.3	—	AVCC0 + 0.3	V
Input voltage (5V tolerant ports: P11 to P17, P20, P21, P30 to P33, P67, and PC0 to PC3)*5	V _{in}	−0.3	—	VCC + 3.6 (up to 5.5)	V
Input voltage (5V tolerant port: P07)	V _{in}	−0.3	—	AVCC0 + 3.6 (up to 5.5)	V
Operating temperature (D version)	T _{opr}	−40	—	85	°C
Operating temperature (G version)	T _{opr}	−40	—	105	°C

Note 1. Comply with the following potential condition: VCC = AVCC0 = AVCC1 = VCC_USB

Note 2. The low CL crystal unit cannot be used when the V_{BATT} voltage is less than 2.0 V.

Note 3. For details, refer to section 56.6.11, Voltage Range of Analog Power Supply Pins.

Note 4. P07, P11 to P17, P20, P21, P30 to P33, P67, and PC0 to PC3 are 5 V tolerant.

Note 5. For P30 to P32, input as follows when the V_{BATT} power supply is selected.

V_{in} Min. = −0.3, Max. = V_{BATT} + 0.3 (V_{BATT} = 1.62 to 3.6 V)

Table 63.3 Recommended Operating Conditions (2)

Item	Symbol	Value
Decoupling capacitance for stabilizing the internal voltage	C _{VCL}	0.22 μF ± 30%*1

Note 1. Use a multilayer ceramic capacitor with a nominal capacitance of 0.22 μF, for which the sum of the capacitance tolerance and change in the capacitance under the usage conditions will be no greater than ±30%.

63.3 DC Characteristics

Table 63.4 DC Characteristics (1)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Schmitt trigger input voltage	IRQ input pin*1, MTU input pin*1, POE input pin*1, TPU input pin*1, TMR input pin*1, CMTW input pin*1, SCI input pin*1, CAN input pin*1, CAC input pin*1, ADTRG# input pin*1, QSPI input pin*1, SSIE input pin*1, GPTW input pin*1, POEG input pin*1, RES#, NMI, TCK	V_{IH}	$0.8 \times V_{CC}$	—	—	V	
		V_{IL}	—	—	$0.2 \times V_{CC}$		
		ΔV_T	$0.06 \times V_{CC}$	—	—		
	RIIC input pin (except for SMBus)	V_{IH}	$0.7 \times V_{CC}$	—	—		
		V_{IL}	—	—	$0.3 \times V_{CC}$		
		ΔV_T	$0.05 \times V_{CC}$	—	—		
	Ports for 5 V tolerant*2	V_{IH}	$0.8 \times V_{CC}$	—	—		
		V_{IL}	—	—	$0.2 \times V_{CC}$		
	Other input pins excluding ports for 5 V tolerant*3	V_{IH}	$0.8 \times V_{CC}$	—	—		
		V_{IL}	—	—	$0.2 \times V_{CC}$		
High level input voltage (except for Schmitt trigger input pin)	MD pin, EMLE	V_{IH}	$0.9 \times V_{CC}$	—	—	V	
	EXTAL, RSPI input pin, EXDMAC input pin, WAIT#, SDHI input pin, MMC input pin, PDC input pin, PMGI input pin		$0.8 \times V_{CC}$	—	—		
	ETHERC input pin		2.3	—	—		
	D0 to D31		$0.7 \times V_{CC}$	—	—		
	RIIC (SMBus)		2.1	—	—		
Low level input voltage (except for Schmitt trigger input pin)	MD pin, EMLE	V_{IL}	—	—	$0.1 \times V_{CC}$	V	
	EXTAL, RSPI input pin, ETHERC input pin, EXDMAC input pin, WAIT#, SDHI input pin, MMC input pin, PDC input pin, PMGI input pin		—	—	$0.2 \times V_{CC}$		
	D0 to D31		—	—	$0.3 \times V_{CC}$		
	RIIC (SMBus)		—	—	0.8		

Note 1. This does not include the pins, which are multiplexed as ports for 5 V tolerant.

Note 2. P07, P11 to P17, P20, P21, P30 to P33, P67, and PC0 to PC3 are 5 V tolerant.

Note 3. For P30 to P32, input as follows when the V_{BATT} power supply is selected.

V_{IH} Min. = $V_{BATT} \times 0.8$, V_{IL} Max. = $V_{BATT} \times 0.2$ ($V_{BATT} = 1.62$ to 3.6 V)

Table 63.5 DC Characteristics (2)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Output high voltage	All output pins	V _{OH}	VCC – 0.5	—	—	V	I _{OH} = –1 mA
Output low voltage	All output pins (except for RIIC pins and ETHERC output pin)	V _{OL}	—	—	0.5	V	I _{OL} = 1.0 mA
	RIIC output pin		—	—	0.4		I _{OL} = 3.0 mA
			—	—	0.6		I _{OL} = 6.0 mA
	RIIC output pin (only P12 and P13 in channel 0)	V _{OL}	—	—	0.4	V	I _{OL} = 15.0 mA (ICFER.FMPE = 1)
			—	0.4	—		I _{OL} = 20.0 mA (ICFER.FMPE = 1)
	ETHERC output pin	V _{OL}	—	—	0.4	V	I _{OL} = 1.0 mA
Input leakage current	RES#, MD pin, EMLE*1, BSCANP*1, NMI	I _{in}	—	—	1.0	μA	V _{in} = 0 V V _{in} = VCC
Three-state leakage current (off state)	Other than ports for 5 V tolerant	I _{TSL}	—	—	1.0	μA	V _{in} = 0 V V _{in} = VCC
	Ports for 5 V tolerant		—	—	5.0		V _{in} = 0 V V _{in} = 5.5 V
Input pull-up resistor current	Other than P35	I _p	–300	—	–10	μA	VCC = 2.7 to 3.6 V V _{in} = 0 V
Input pull-down resistor current	EMLE, BSCANP	I _p	10	—	300	μA	V _{in} = VCC
Input capacitance	All input pins (except for P03, P05, P12, P13, P16, P17, P20, P21, EMLE, BSCANP, USB0_DP, and USB0_DM)	C _{in}	—	—	8	pF	V _{bias} = 0 V V _{amp} = 20 mV f = 1 MHz T _a = 25°C
	P03, P05, P12, P13, P16, P17, P20, P21, EMLE, BSCANP, USB0_DP, and USB0_DM		—	—	16		
Output voltage of the VCL pin		V _{CL}	—	1.18	—	V	

Note 1. The input leakage current value at the EMLE and BSCANP pins are only when $V_{in} = 0$ V.

Table 63.6 DC Characteristics (3)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	D version		G version		Unit	Test Conditions		
				Typ.	Max.	Typ.	Max.				
Supply current *1	High-speed operating mode	Full operation*2		I _{CC} *3	—	261	—	319	mA	ICLK = 240 MHz, PCLKA = 120 MHz, PCLKB = 60 MHz, PCLKC = 60 MHz, PCLKD = 60 MHz, FCLK = 60 MHz, BCLK = 120 MHz, BCLK pin = 60 MHz	
		Normal operation	Peripheral module clocks are supplied *4		61	—	61	—			
			Peripheral module clocks are stopped *4, *5		30	—	30	—			
		Core Mark	Peripheral module clocks are stopped *4, *5		37	—	37	—			
		Sleep mode: Peripheral module clocks are supplied*4			42	144	42	196			
		All module clock stop mode (reference value)			14	115	14	167			
		Increased by BGO operation*8	Reading from the code flash memory while the data flash memory is being programmed		6	—	6	—			
			Reading from the code flash memory while the code flash memory is being programmed		7	—	7	—			
		Increased by Trusted Secure IP operation			—	15	—	15			
		Low-speed operating mode 1: Peripheral module clocks are stopped*4			4.2	—	4.2	—			All clocks 1 MHz
	Low-speed operating mode 2: Peripheral module clocks are stopped*4		4.2	—	4.2	—	All clocks 32.768 kHz				
	Software standby mode		3.95	107	3.95	155					
	Deep software standby mode	Power is supplied to the standby RAM and USB resume detecting unit (USB0 only)		I _{RUSH}	15.5	70	15.5	98	μA	V _{BATT} = 2.0 V, VCC = 0 V V _{BATT} = 3.3 V, VCC = 0 V V _{BATT} = 1.62 V, VCC = 0 V V _{BATT} = 2.0 V, VCC = 0 V V _{BATT} = 3.3 V, VCC = 0 V	
		Power is not supplied to the standby RAM and USB resume detecting unit (USB0 only)	Low power consumption function of the power-on reset circuit is disabled*6		11.5	42	11.5	58			
			Low power consumption function of the power-on reset circuit is enabled*7		4.9	32	4.9	47			
		Increase current by operating RTC	When a low C _L crystal is in use		1	—	1	—			
			When a standard C _L crystal is in use		2	—	2	—			
		When the RTC is operating while VCC is not supplied (Only the RTC and sub-clock oscillator operate with the battery backup function)	When a low C _L crystal is in use		0.9	—	0.9	—			
					1.6	—	1.6	—			
			When a standard C _L crystal is in use		1.6	—	1.6	—			
					1.7	—	1.7	—			
					3.3	—	3.3	—			
	Inrush current on returning from deep software standby mode		Inrush current*9	I _{RUSH}	—	211	—	211	mA		

Note 1. Supply current values are measured when all output pins are unloaded and all input pull-up resistors are disabled.

Note 2. Peripheral module clocks are supplied.

Note 3. I_{CC} depends on the f (ICLK) as follows.

(when ICLK : PCLKA : PCLKB/PCLKC/PCLKD : BCLK : BCLK pin = 4 : 2 : 1 : 2 : 1 and EXTAL = 12 MHz)

- D version

I_{CC} max. = $0.62 \times f + 113$ (full operation in high-speed operating mode)

I_{CC} typ. = $0.22 \times f + 7$ (normal operation in high-speed operating mode)

I_{CC} typ. = $0.50 \times f + 3.7$ (ICLK 1 MHz max) (low-speed operating mode 1)

I_{CC} max. = $0.13 \times f + 113$ (sleep mode)

- G version

I_{CC} max. = $0.65 \times f + 164$ (full operation in high-speed operating mode)

I_{CC} typ. = $0.22 \times f + 7$ (normal operation in high-speed operating mode)

I_{CC} typ. = $0.50 \times f + 3.7$ (ICLK 1 MHz max) (low-speed operating mode 1)

I_{CC} max. = $0.13 \times f + 164$ (sleep mode)

Note 4. Whether the peripheral module clocks are supplied or stopped is controlled only by the bit settings in the module stop control registers A to D.

Note 5. When the peripheral module clock is stopped, the settings of the clock frequency are as follows:

ICLK = 240 MHz and PCLKA = PCLKB = PCLKC = PCLKD = FCLK = BCLK = BCLK pin = 3.75 MHz (divided by 64).

Note 6. When the low power consumption function is disabled, the DEEPCUT[1:0] bits are set to 01b.

Note 7. When the low power consumption function is enabled, the DEEPCUT[1:0] bits are set to 11b.

Note 8. These are the increases during programming of the code flash memory after the code flash memory (limitations apply to the combinations of address ranges of the program area and the readable area) or the data flash memory has been programmed or erased.

Note 9. Reference value

Table 63.7 DC Characteristics (4)

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	D version			G version			Unit	Test Conditions
				Min.	Typ.	Max.	Min.	Typ.	Max.		
Analog power supply current*1, *3	During 12-bit A/D conversion (unit 0)		AI _{CC}	—	0.8	1	—	0.8	1	mA	IAVCC0_AD
	During 12-bit A/D conversion (unit 0) with channel-dedicated sample-and-hold circuits (3 channels)			—	1.7	2.5	—	1.7	2.5		IAVCC0_AD + SH
	During 12-bit A/D conversion (unit 1)			—	0.6	1	—	0.6	1		IAVCC1_AD
	During 12-bit A/D conversion (unit 1) + temperature sensor			—	0.7	1.1	—	0.7	1.1		IAVCC1_AD + TEMP
	During D/A conversion (2 channels)	Unbuffered output		—	0.25	0.4	—	0.25	0.4		IAVCC1_DA
		Buffered output		—	0.75	1.1	—	0.75	1.1		
	Waiting for A/D, D/A, and temperature sensor conversion (all units)			—	0.9	1.4	—	0.9	1.4	IAVCC0 + IAVCC1	
A/D, D/A, and temperature sensor are in standby mode (all units)		—	1.4	6.7	—	1.4	9.0	μA	IAVCC0 + IAVCC1		
Reference power supply current	During 12-bit A/D conversion (unit 0)		AI _{REFH}	—	38	60	—	38	60	μA	IVREFH0
	Waiting for 12-bit A/D conversion (unit 0)			—	0.07	0.5	—	0.07	0.6		IVREFH0
	12-bit A/D converter in module stop mode (unit 0)			—	0.07	0.4	—	0.07	0.5		IVREFH0
USB operating current	Low speed	USB0	I _{CCUSBLS}	—	3.7	6.5	—	3.7	6.5	mA	VCC_USB
	Full speed	USB0	I _{CCUSBFS}	—	4.2	10	—	4.2	10	mA	VCC_USB
RAM retention voltage			V _{RAM}	2.7	—	—	2.7	—	—	V	
VCC rising gradient			SrVCC	8.4	—	20000	8.4	—	20000	μs/V	
VCC falling gradient*2			SfVCC	8.4	—	—	8.4	—	—	μs/V	

Note 1. The reference power supply current is included in the power supply current value for 12-bit A/D converter (unit 1) and D/A converter.

Note 2. This applies when V_{BATT} is used.

Note 3. Supply current values are measured when all output pins are unloaded.

Table 63.8 Permissible Output Currents

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit
Permissible output low current (average value per pin)	All output pins*1	Normal drive	I_{OL}	—	—	2.0	mA
	All output pins*2	High drive		—	—	3.8	
	All output pins*3	High-speed interface high-drive		—	—	7.5	
Permissible output low current (max. value per pin)	All output pins*1	Normal drive	I_{OL}	—	—	4.0	mA
	All output pins*2	High drive		—	—	7.6	
	All output pins*3	High-speed interface high-drive		—	—	15	
Permissible output low current (total)	Total of all output pins		ΣI_{OL}	—	—	80	mA
Permissible output high current (average value per pin)	All output pins*1	Normal drive	I_{OH}	—	—	−2.0	mA
	All output pins*2	High drive		—	—	−3.8	
	All output pins*3	High-speed interface high-drive		—	—	−7.5	
Permissible output high current (max. value per pin)	All output pins*1	Normal drive	I_{OH}	—	—	−4.0	mA
	All output pins*2	High drive		—	—	−7.6	
	All output pins*3	High-speed interface high-drive		—	—	−15	
Permissible output high current (total)	Total of all output pins		ΣI_{OH}	—	—	−80	mA

Caution: To protect the MCU's reliability, the output current values should not exceed the values in Table 63.8.

Note 1. This is the value when normal driving ability is set with a pin for which normal driving ability is selectable.

Note 2. This is the value when high driving ability is set with a pin for which normal driving ability is selectable or the value of the pin to which high driving ability is fixed.

Note 3. This is the value when high-speed interface high-driving ability is set with a pin for which high-speed interface high-driving ability is selectable.

Table 63.9 Thermal Resistance Value (Reference)

Item	Package	Symbol	Max.	Unit	Test Conditions
Thermal resistance	176-pin LFQFP (PLQP0176KB-C)	θ_{ja}	31.5	°C/W	JESD51-2 and JESD51-7 compliant
	144-pin LFQFP (PLQP0144KA-B)		32.6		
	100-pin LFQFP (PLQP0100KB-B)		34.0		
	224-pin LFBGA (PLBG0224GA-A)		23.1		JESD51-2 and JESD51-9 compliant
	176-pin LFBGA (PLBG0176GA-A)		30.5		
	145-pin TFLGA (PTLG0145KA-A)		22.9		
	176-pin LFQFP (PLQP0176KB-C)	Ψ_{jt}	0.4	°C/W	JESD51-2 and JESD51-7 compliant
	144-pin LFQFP (PLQP0144KA-B)		0.5		
	100-pin LFQFP (PLQP0100KB-B)		0.6		
	224-pin LFBGA (PLBG0224GA-A)		0.2		JESD51-2 and JESD51-9 compliant
	176-pin LFBGA (PLBG0176GA-A)		0.3		
	145-pin TFLGA (PTLG0145KA-A)		0.2		

Note: The values are reference values when the 4-layer board is used. Thermal resistance depends on the number of layers or size of the board. For details, refer to the JEDEC standards.

63.4 AC Characteristics

Table 63.10 Operating Frequency (High-Speed Operating Mode)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit
Operating frequency	System clock (ICLK)		f	—	—	240	MHz
	Peripheral module clock (PCLKA)			—	—	120	
	Peripheral module clock (PCLKB)			—	—	60	
	Peripheral module clock (PCLKC)			—	—	60	
	Peripheral module clock (PCLKD)			—	—	60	
	Flash-IF clock (FCLK)			—*1	—	60	
	External bus clock (BCLK)	Package of 144 pins or more		—	—	120	
		100-pin package		—	—	60	
	BCLK pin output	Package of 144 pins or more		—	—	80	
		100-pin package		—	—	30	
	SDRAM clock (SDCLK)	Package of 144 pins or more		—	—	80	
	SDCLK pin output	Package of 144 pins or more		—	—	80	

Note 1. The FCLK must run at a frequency of at least 4 MHz when changing the flash memory contents.

Table 63.11 Operating Frequency (Low-Speed Operating Mode 1)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit
Operating frequency	System clock (ICLK)	f	—	—	1	MHz
	Peripheral module clock (PCLKA)		—	—	1	
	Peripheral module clock (PCLKB)		—	—	1	
	Peripheral module clock (PCLKC)*1		—	—	1	
	Peripheral module clock (PCLKD)*1		—	—	1	
	Flash-IF clock (FCLK)		—	—	1	
	External bus clock (BCLK)		—	—	1	
	BCLK pin output		—	—	1	
	SDRAM clock (SDCLK)		—	—	1	
	SDCLK pin output		—	—	1	

Note 1. When the 12-bit A/D converter is used, the frequency must be set to at least 1 MHz.

Table 63.12 Operating Frequency (Low-Speed Operating Mode 2)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.	Unit
Operating frequency	System clock (ICLK)		f	32	—	264	kHz
	Peripheral module clock (PCLKA)			—	—	264	
	Peripheral module clock (PCLKB)			—	—	264	
	Peripheral module clock (PCLKC)*1			—	—	264	
	Peripheral module clock (PCLKD)*1			—	—	264	
	Flash-IF clock (FCLK)			32	—	264	
	External bus clock (BCLK)			—	—	264	
	BCLK pin output			—	—	264	
	SDRAM clock (SDCLK)	Package of 144 pins or more		—	—	264	
	SDCLK pin output	Package of 144 pins or more		—	—	264	

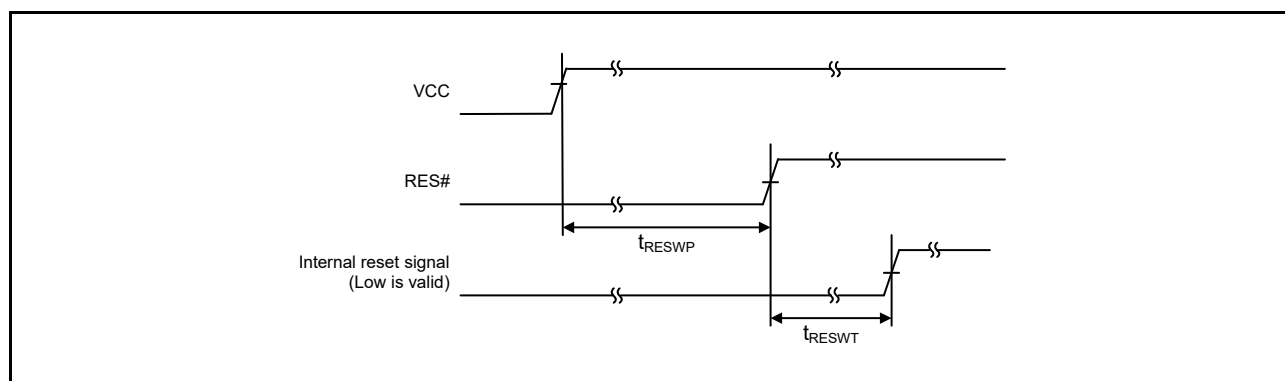
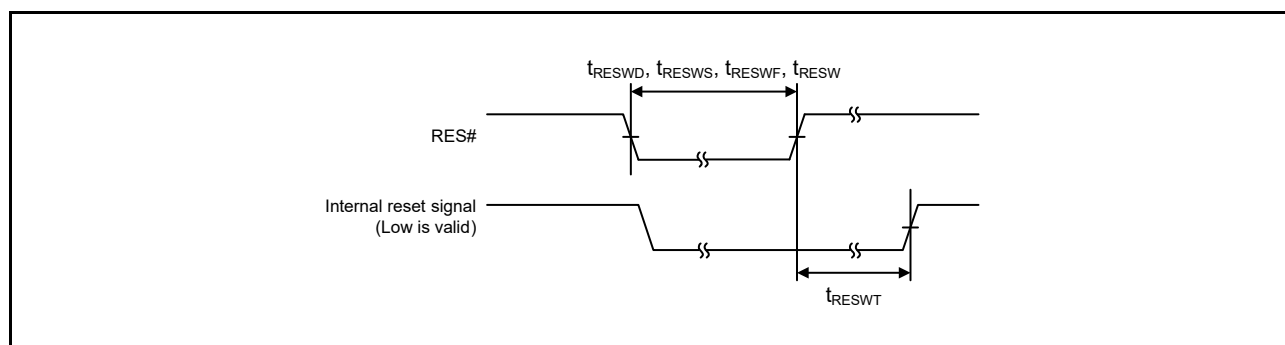
Note 1. The 12-bit A/D converter cannot be used.

63.4.1 Reset Timing

Table 63.13 Reset Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
RES# pulse width	Power-on	t_{RESWP}	1	—	—	ms	Figure 63.1
	Deep software standby mode	t_{RESWD}	0.6	—	—	ms	Figure 63.2
	Software standby mode, low-speed operating mode 2	t_{RESWS}	0.3	—	—	ms	
	Programming or erasure of the code flash memory, or programming, erasure or blank checking of the data flash memory	t_{RESWF}	200	—	—	μ s	
	Other than above	t_{RESW}	200	—	—	μ s	
Waiting time after release from the RES# pin reset		t_{RESWT}	54	—	55	t_{Lcyc}	Figure 63.1
Internal reset time (independent watchdog timer reset, watchdog timer reset, software reset)		t_{RESW2}	100	—	108	t_{Lcyc}	

**Figure 63.1 Reset Input Timing at Power-On****Figure 63.2 Reset Input Timing**

63.4.2 Clock Timing

Table 63.14 BCLK Pin Output, SDCLK Pin Output Clock Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
BCLK pin output cycle time	Package of 144 pins or more	t_{Bcyc}	12.5	—	—	ns	Figure 63.3
	100-pin package		33.2	—	—		
BCLK pin output high pulse width		t_{CH}	3.25	—	—	ns	
BCLK pin output low pulse width		t_{CL}	3.25	—	—	ns	
BCLK pin output rising time		t_{Cr}	—	—	3	ns	
BCLK pin output falling time		t_{Cf}	—	—	3	ns	
SDCLK pin output cycle time	Package of 144 pins or more	t_{Bcyc}	12.5	—	—	ns	
SDCLK pin output high pulse width		t_{CH}	3.25	—	—	ns	
SDCLK pin output low pulse width		t_{CL}	3.25	—	—	ns	
SDCLK pin output rising time		t_{Cr}	—	—	3	ns	
SDCLK pin output falling time		t_{Cf}	—	—	3	ns	

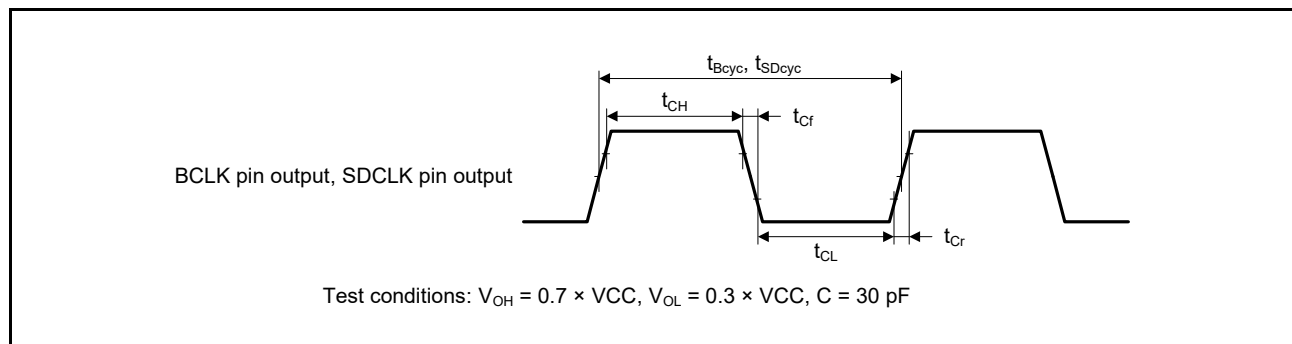
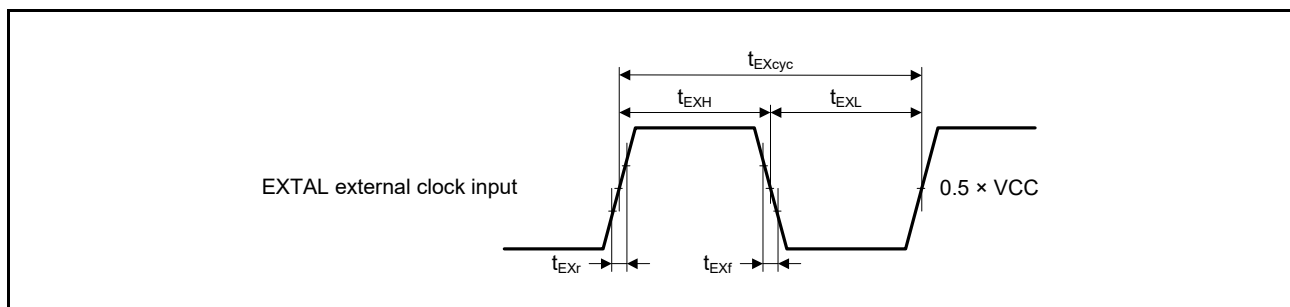
**Figure 63.3 BCLK Pin and SDCLK Pin Output Timing**

Table 63.15 EXTAL Clock Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	$f_{EXMAIN} \leq 24$ MHz			$f_{EXMAIN} > 24$ MHz			Unit	Test Conditions
		Min.	Typ.	Max.	Min.	Typ.	Max.		
EXTAL external clock input cycle time	t_{EXcyc}	41.66	—	—	33.33	—	—	ns	Figure 63.4
EXTAL external clock input frequency	f_{EXMAIN}	—	—	24	—	—	30	MHz	
EXTAL external clock input high pulse width	t_{EXH}	15.83	—	—	13.33	—	—	ns	
EXTAL external clock input low pulse width	t_{EXL}	15.83	—	—	13.33	—	—	ns	
EXTAL external clock rising time	t_{EXr}	—	—	5	—	—	5	ns	
EXTAL external clock falling time	t_{EXf}	—	—	5	—	—	5	ns	

**Figure 63.4 EXTAL External Clock Input Timing****Table 63.16 Main Clock Timing**

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Main clock oscillation frequency	f_{MAIN}	8	—	24	MHz	Figure 63.5
Main clock oscillator stabilization time (crystal)	$t_{MAINOSC}$	—	—	—*1	ms	
Main clock oscillator stabilization wait time (crystal)	$t_{MAINOSCWT}$	—	—	—*2	ms	

Note 1. When using a main clock, ask the manufacturer of the oscillator to evaluate its oscillation. Refer to the results of evaluation provided by the manufacturer for the oscillation stabilization time.

Note 2. The number of cycles selected by the value of the MOSCWTCR.MSTS[7:0] bits determines the main clock oscillation stabilization wait time in accord with the formula below.

$$t_{MAINOSCWT} = [(MSTS[7:0] \text{ bits} \times 32) + 10] / f_{LOCO}$$

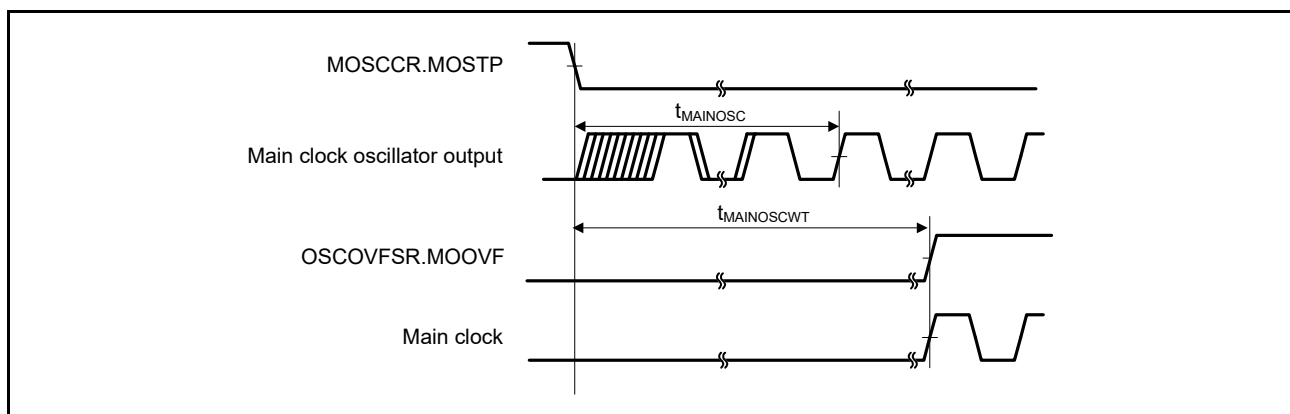
**Figure 63.5 Main Clock Oscillation Start Timing**

Table 63.17 LOCO and IWDT-Dedicated Low-Speed Clock Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
LOCO clock cycle time	t_{Lcyc}	4.63	4.16	3.78	μ s	
LOCO clock oscillation frequency	f_{LOCO}	216	240	264	kHz	
LOCO clock oscillation stabilization wait time	t_{LOCOWT}	—	—	44	μ s	Figure 63.6
IWDT-dedicated low-speed clock cycle time	t_{iLcyc}	9.26	8.33	7.57	μ s	
IWDT-dedicated low-speed clock oscillation frequency	f_{iLOCO}	108	120	132	kHz	
IWDT-dedicated low-speed clock oscillation stabilization wait time	$t_{iLOCOWT}$	—	142	190	μ s	Figure 63.7

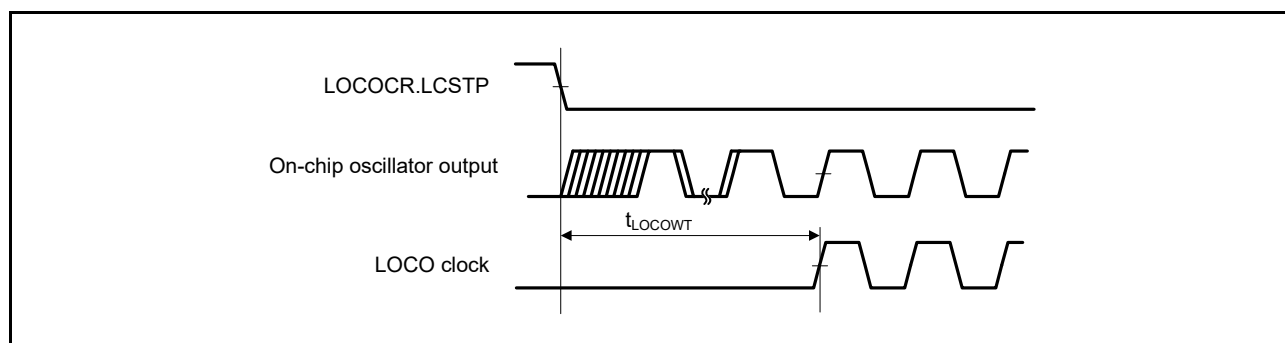
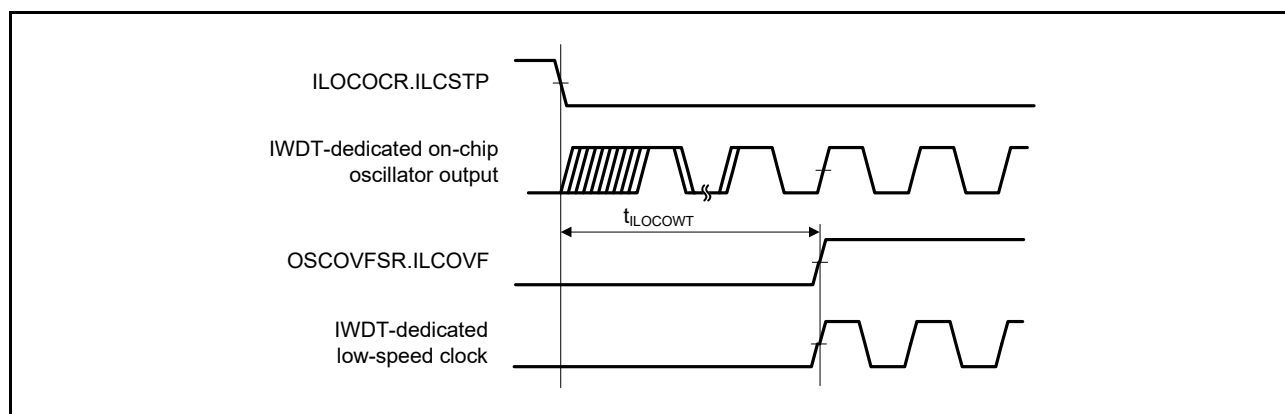
**Figure 63.6 LOCO Clock Oscillation Start Timing****Figure 63.7 IWDT-dedicated Low-Speed Clock Oscillation Start Timing**

Table 63.18 HOCO Clock Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
HOCO clock oscillation frequency	f_{HOCO}	15.61	16	16.39	MHz	$T_a \geq -20^\circ\text{C}$
		17.56	18	18.44		
		19.52	20	20.48		
		15.52	16	16.48		$-40^\circ\text{C} \leq T_a < -20^\circ\text{C}$
		17.46	18	18.54		
		19.4	20	20.6		
HOCO clock oscillation stabilization wait time	t_{HOCOWT}	—	105	149	μs	Figure 63.8
HOCO clock power supply stabilization time	t_{HOCOP}	—	—	150	μs	Figure 63.9

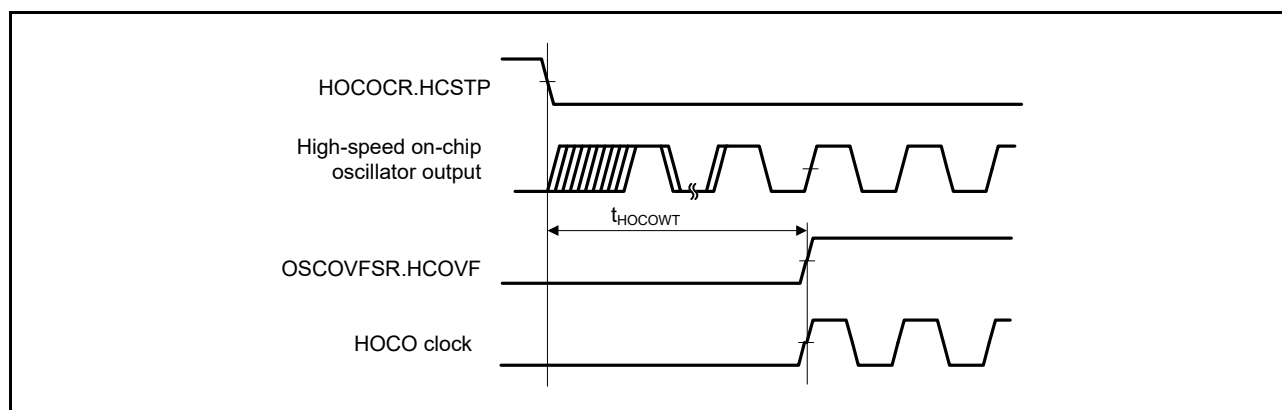
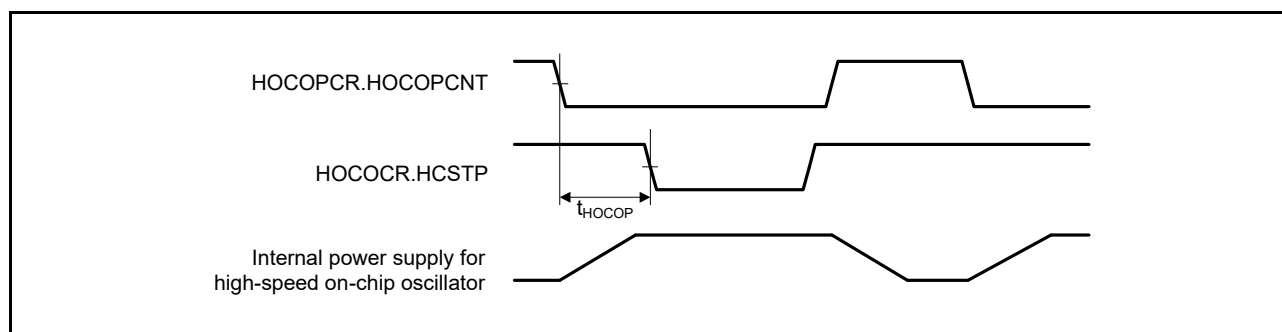
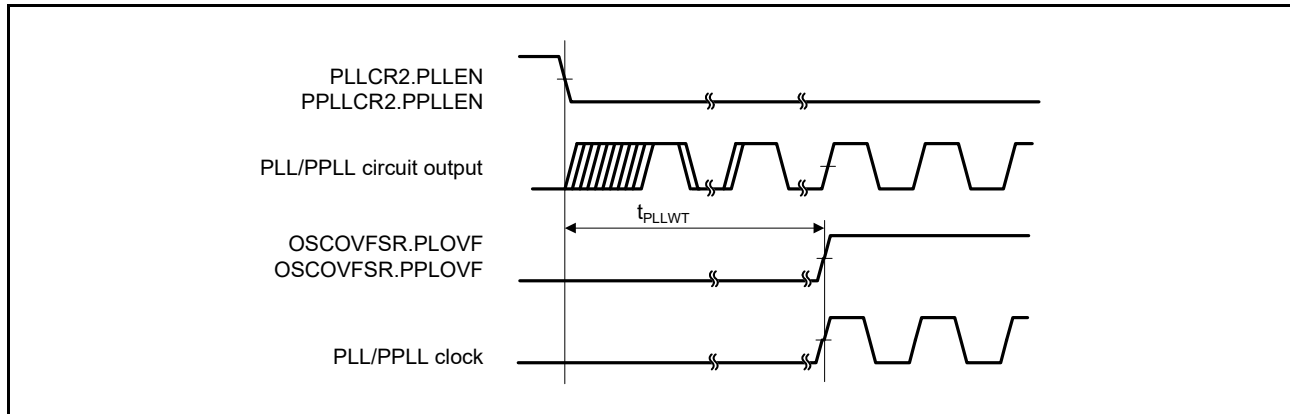
**Figure 63.8 HOCO Clock Oscillation Start Timing (Oscillation is Started by Setting the HOCOCR.HCSTP Bit)****Figure 63.9 High-Speed On-Chip Oscillator Power Supply Control Timing**

Table 63.19 PLL/PPLL Clock Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
PLL/PPLL clock oscillation frequency	f_{PLL}	120	—	240	MHz	
PLL/PPLL clock oscillation stabilization wait time	t_{PLLWT}	—	259	320	μ s	Figure 63.10

**Figure 63.10 PLL/PPLL Clock Oscillation Start Timing****Table 63.20 Sub-Clock Timing**

Conditions: $V_{CC} = AVCC0 = AVCC1 = VCC_USB = 2.7$ to 3.6 V, 2.7 V $\leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $V_{BATT} = 1.62$ to 3.6 V, $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Sub-clock oscillation frequency	f_{SUB}	—	32.768	—	kHz	
Sub-clock oscillation stabilization time	t_{SUBOSC}	—	—	*1	s	Figure 63.11
Sub-clock oscillation stabilization wait time	$t_{SUBOSCWT}$	—	—	*2	s	

Note 1. When using a sub-clock, ask the manufacturer of the oscillator to evaluate its oscillation. Refer to the results of evaluation provided by the manufacturer for the oscillation stabilization time.

Note 2. The number of cycles selected by the value of the $SOSWTCR.SSTS[7:0]$ bits determines the sub-clock oscillation stabilization wait time in accord with the formula below.

$$t_{SUBOSCWT} = [(SSTS[7:0] \text{ bits} \times 16384) + 10] / f_{LOCO}$$

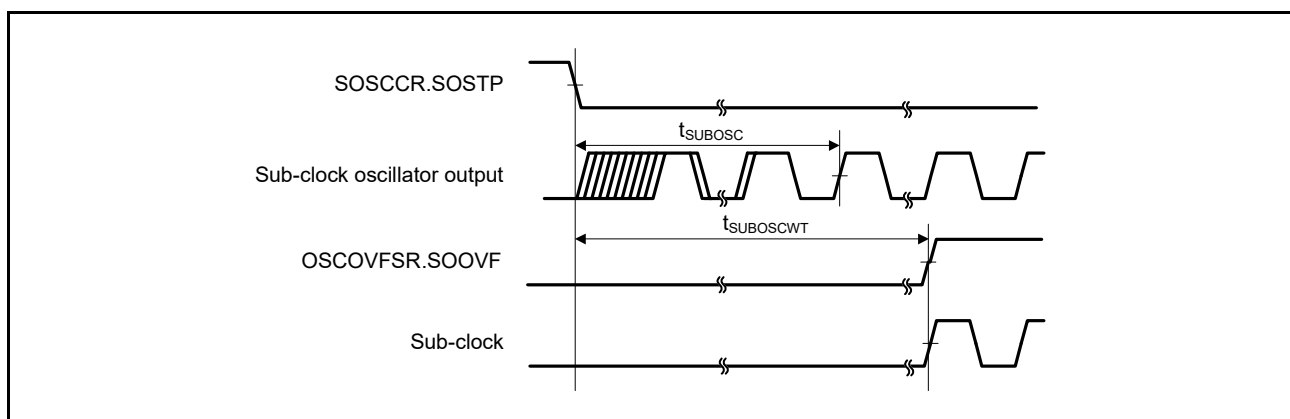
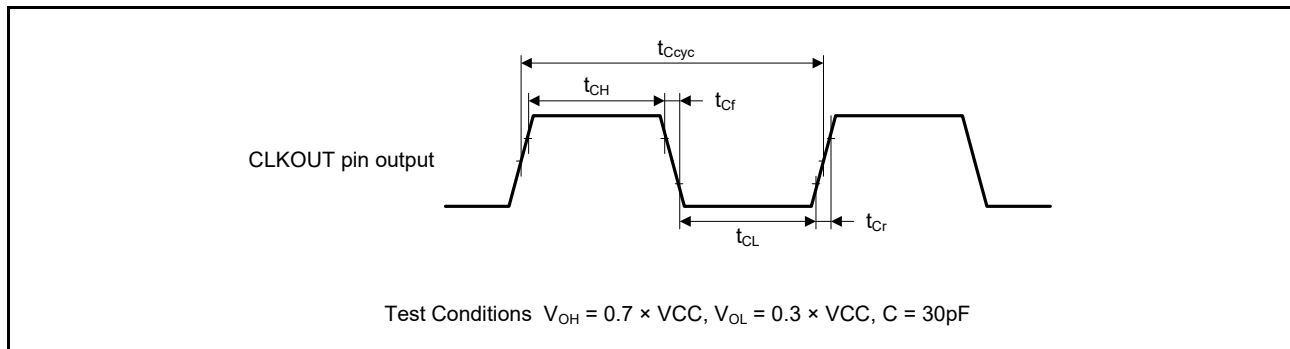
**Figure 63.11 Sub-Clock Oscillation Start Timing**

Table 63.21 CLKOUT Pin Output Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$,
 High-drive output is selected by the drive capacity control register

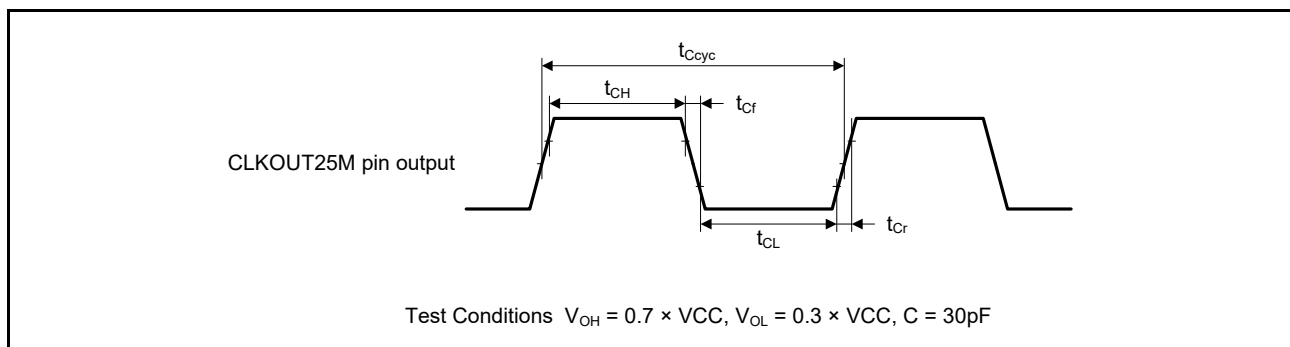
Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
CLKOUT pin output cycle time	t_{Ccyc}	25	—	—	ns	Figure 63.12 $t_{Ccyc} = 25$ ns
CLKOUT pin output high pulse width*1	t_{CH}	5	—	—	ns	
CLKOUT pin output low pulse width*1	t_{CL}	5	—	—	ns	
CLKOUT pin output rising time	t_{Cr}	—	—	5	ns	
CLKOUT pin output falling time	t_{Cf}	—	—	5	ns	

Note 1. If the main clock oscillator is selected by the CLKOUT output source select bit (CKOCR.CKOSEL[2:0]) and the external clock input is selected by the main clock oscillator switching bit (MOFCR.MOSEL), the pulse width depends on the input clock wave form.

**Figure 63.12 CLKOUT Pin Output Timing****Table 63.22 CLKOUT25M Pin Output Timing**

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$,
 High-speed interface high-drive is selected by the drive capacity control register

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
CLKOUT25M pin output cycle time	t_{Ccyc}	—	40	—	ns	Figure 63.13
CLKOUT25M pin output high pulse width	t_{CH}	13	—	—	ns	
CLKOUT25M pin output low pulse width	t_{CL}	13	—	—	ns	
CLKOUT25M pin output rising time	t_{Cr}	—	—	3	ns	
CLKOUT25M pin output falling time	t_{Cf}	—	—	3	ns	

**Figure 63.13 CLKOUT25M Pin Output Timing**

63.4.3 Timing of Recovery from Low Power Consumption Modes

Table 63.23 Timing of Recovery from Low Power Consumption Modes (1)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item			Symbol	Min.	Typ.	Max.		Unit	Test Conditions
						$t_{SBYOSCWT}^{*2}$	t_{SBYSEQ}^{*3}		
Recovery time from software standby mode *1	Crystal resonator connected to main clock oscillator	Main clock oscillator operating	t_{SBYMC}	—	—	$\{(MSTS[7:0] \text{ bit} \times 32) + 76\} / 0.216$	$100 + 7 / f_{ICLK} + 2n / f_{MAIN}$	μs	Figure 63.14
		Main clock oscillator and PLL circuit operating	t_{SBYPC}			$\{(MSTS[7:0] \text{ bit} \times 32) + 138\} / 0.216$	$100 + 7 / f_{ICLK} + 2n / f_{PLL}$		
	External clock input to main clock oscillator	Main clock oscillator operating	t_{SBYEX}			352	$100 + 7 / f_{ICLK} + 2n / f_{EXMAIN}$		
		Main clock oscillator and PLL circuit operating	t_{SBYPE}			639	$100 + 7 / f_{ICLK} + 2n / f_{PLL}$		
	Sub-clock oscillator operating		t_{SBYSC}			$\{(SSTS[7:0] \text{ bit} \times 16384) + 13\} / 0.216 + 10 / f_{FCLK}$	$100 + 4 / f_{ICLK} + 2n / f_{SUE}$		
	High-speed on-chip oscillator operating	High-speed on-chip oscillator operating	t_{SBYHO}			454	$100 + 7 / f_{ICLK} + 2n / f_{HOCO}$		
		High-speed on-chip oscillator operating and PLL circuit operating	t_{SBYPH}			741	$100 + 7 / f_{ICLK} + 2n / f_{PLL}$		
	Low-speed on-chip oscillator operating*4		t_{SBYLO}			338	$100 + 7 / f_{ICLK} + 2n / f_{LOCO}$		

Note 1. The time for recovery from software standby mode is determined by the value obtained by adding the oscillation stabilization waiting time ($t_{SBYOSCWT}$) and the time required for operations by the software standby release sequencer (t_{SBYSEQ}).

Note 2. When several oscillators were running before the transition to software standby, the greatest value of the oscillation stabilization waiting time $t_{SBYOSCWT}$ is selected.

Note 3. For n, the greatest value is selected from among the internal clock division settings.

Note 4. This condition applies when $f_{ICLK}:f_{FCLK} = 1:1, 2:1$, or $4:1$.

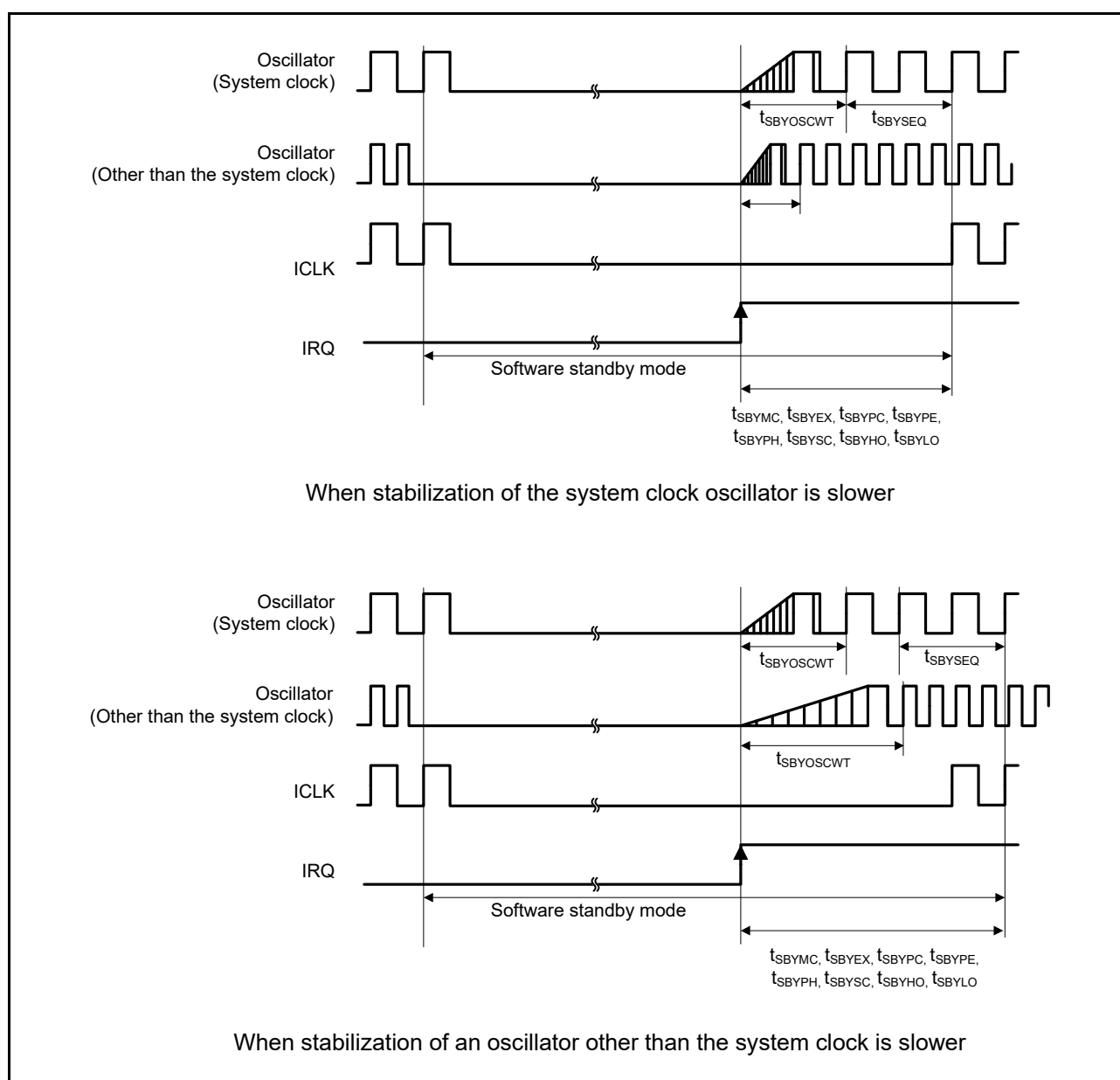
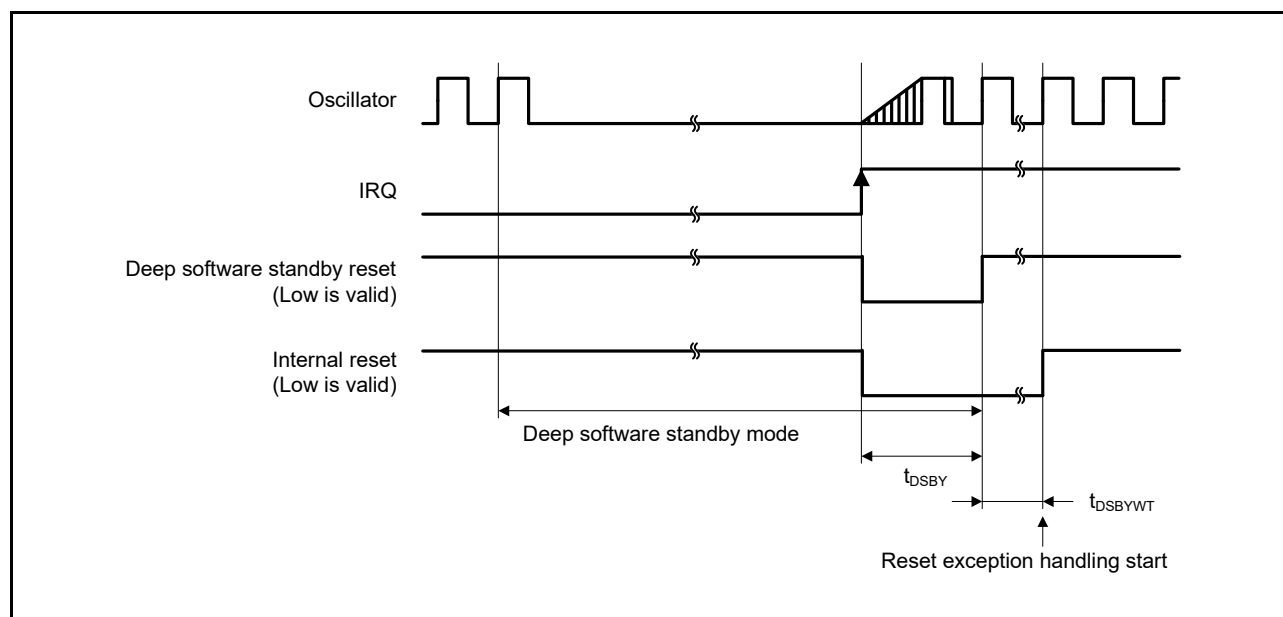


Figure 63.14 Software Standby Mode Recovery Timing

Table 63.24 Timing of Recovery from Low Power Consumption Modes (2)

Conditions: $VCC = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Recovery time from deep software standby mode	t_{DSBY}	—	—	0.9	ms	Figure 63.15
Wait time after recovery from deep software standby mode	t_{DSBYWT}	23	—	24	t_{Lcyc}	

**Figure 63.15 Deep Software Standby Mode Recovery Timing**

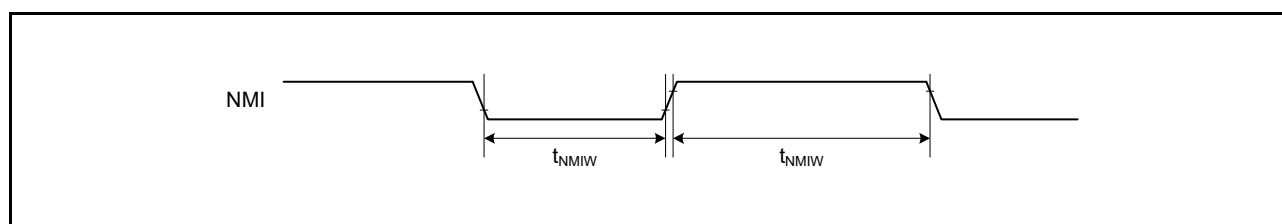
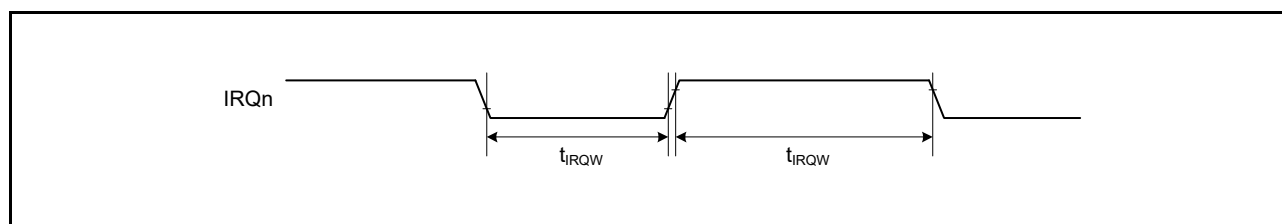
63.4.4 Control Signal Timing

Table 63.25 Control Signal Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$

Item	Symbol	Min.*1	Typ.	Max.	Unit	Test Conditions*1
NMI pulse width	t_{NMIW}	200	—	—	ns	$t_{PBcyc} \times 2 \leq 200$ ns, Figure 63.16
		$t_{PBcyc} \times 2$	—	—		$t_{PBcyc} \times 2 > 200$ ns, Figure 63.16
IRQ pulse width	t_{IRQW}	200	—	—	ns	$t_{PBcyc} \times 2 \leq 200$ ns, Figure 63.17
		$t_{PBcyc} \times 2$	—	—		$t_{PBcyc} \times 2 > 200$ ns, Figure 63.17

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.16 NMI Interrupt Input Timing****Figure 63.17 IRQ Interrupt Input Timing**

63.4.5 Bus Timing

Table 63.26 Bus Timing

Conditions 1: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $ICLK = 8$ to 240 MHz, $PCLKA = 8$ to 120 MHz, $PCLKB = BCLK = SDCLK = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Conditions 2: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 3.0$ to 3.6 V, 3.0 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $ICLK = 60$ to 240 MHz, $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, 60 MHz $< BCLK = SDCLK \leq 80$ MHz,
 $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$,
 $C = 15$ pF for the SDCLK pin, $C = 30$ pF for other pins.

To control the drive capacity when using the SDRAM: set the PFBCR3.SDCLKDRV bit in external bus control register 1 to 1 to select the drive capacity of the SDCLK pin, and set the SDRAM pins other than the SDCLK pin as high-speed-interface driving outputs.

Item	Symbol	Conditions 1		Conditions 2		Unit	Test Conditions
		Min.	Max.	Min.	Max.		
Address delay time	t_{AD}	—	12.5	—	12.5	ns	Figure 63.18 to Figure 63.23
Byte control delay time	t_{BCD}	—	12.5	—	12.5	ns	
CS# delay time	t_{CSD}	—	12.5	—	12.5	ns	
ALE delay time	t_{ALED}	—	12.5	—	12.5	ns	
RD# delay time	t_{RSD}	—	12.5	—	12.5	ns	
Read data setup time	t_{RDS}	12.5	—	12.5	—	ns	
Read data hold time	t_{RDH}	0	—	0	—	ns	
WR# delay time	t_{WRD}	—	12.5	—	12.5	ns	
Write data delay time	t_{WDD}	—	12.5	—	12.5	ns	
Write data hold time	t_{WDH}	0	—	0	—	ns	
WAIT# setup time	t_{WTS}	12.5	—	12.5	—	ns	Figure 63.24
WAIT# hold time	t_{WTH}	0	—	0	—	ns	
Address delay time 2 (SDRAM)	t_{AD2}	1	12.5	1	10.0	ns	Figure 63.25 to Figure 63.31
CS# delay time 2 (SDRAM)	t_{CSD2}	1	12.5	1	10.0	ns	
DQM delay time (SDRAM)	t_{DQMD}	1	12.5	1	10.0	ns	
CKE delay time (SDRAM)	t_{CKED}	1	12.5	1	10.0	ns	
Read data setup time 2 (SDRAM)	t_{RDS2}	10	—	6.0	—	ns	
Read data hold time 2 (SDRAM)	t_{RDH2}	0	—	0	—	ns	
Write data delay time 2 (SDRAM)	t_{WDD2}	—	12.5	—	10.0	ns	
Write data hold time 2 (SDRAM)	t_{WDH2}	1	—	1	—	ns	
WE# delay time (SDRAM)	t_{WED}	1	12.5	1	10.0	ns	
RAS# delay time (SDRAM)	t_{RASD}	1	12.5	1	10.0	ns	
CAS# delay time (SDRAM)	t_{CASD}	1	12.5	1	10.0	ns	

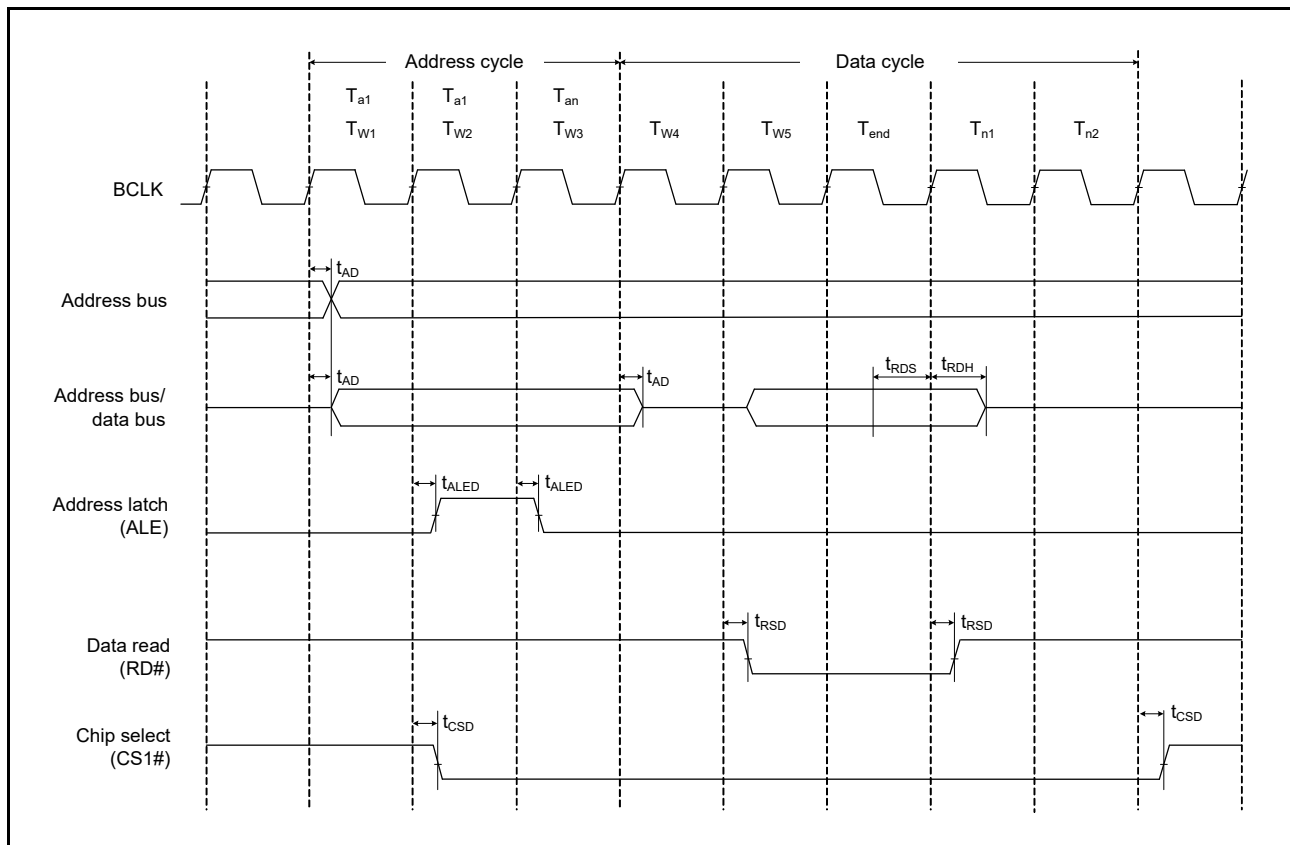


Figure 63.18 Address/Data Multiplexed Bus Read Access Timing

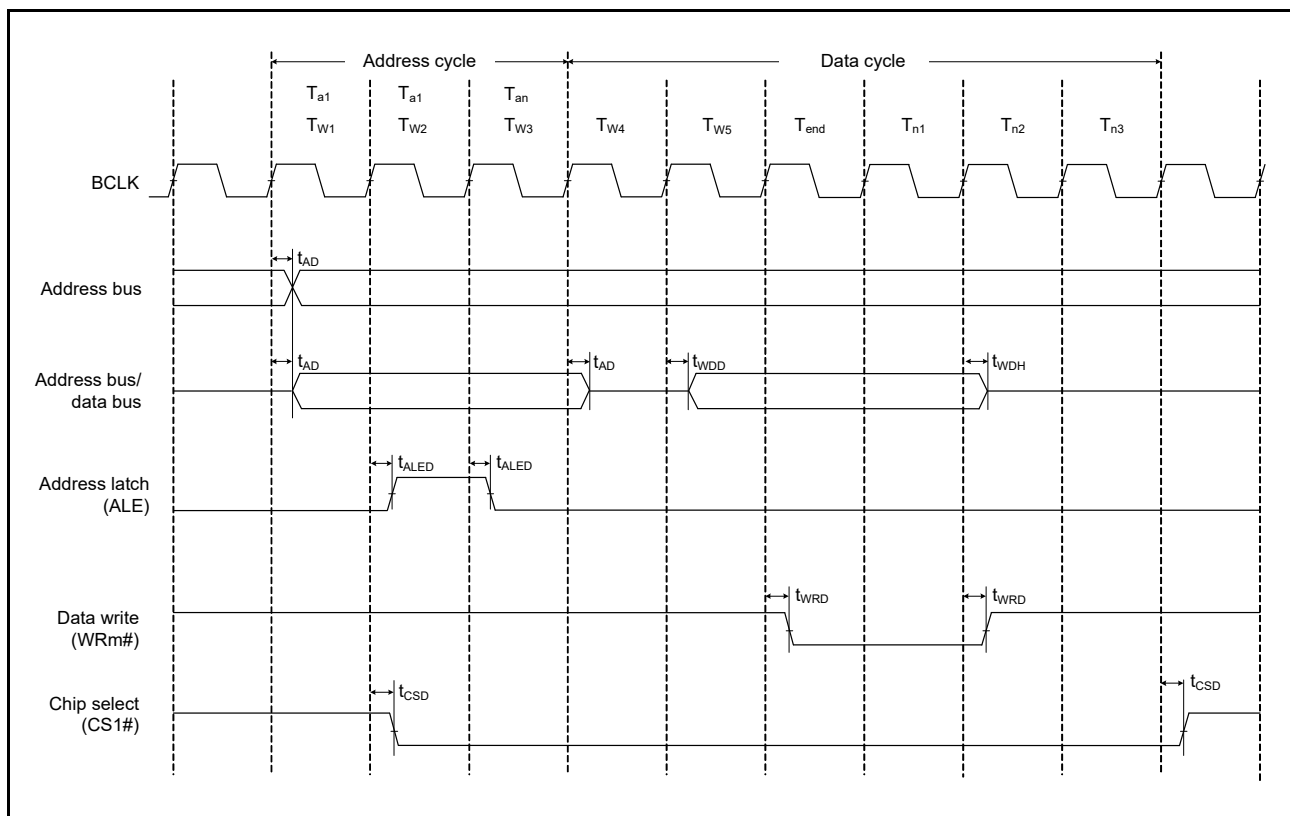


Figure 63.19 Address/Data Multiplexed Bus Write Access Timing

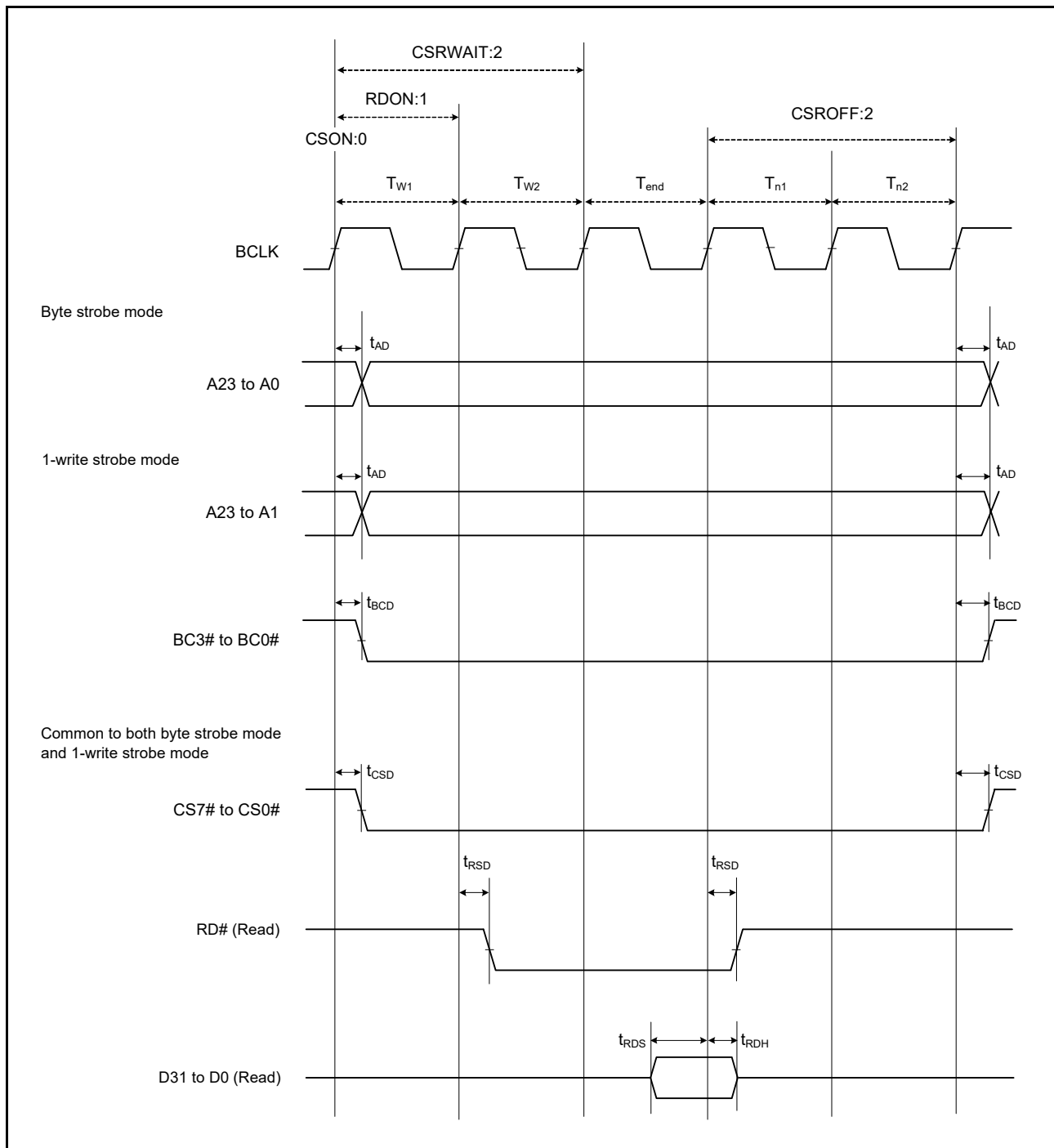


Figure 63.20 External Bus Timing/Normal Read Cycle (Bus Clock Synchronized)

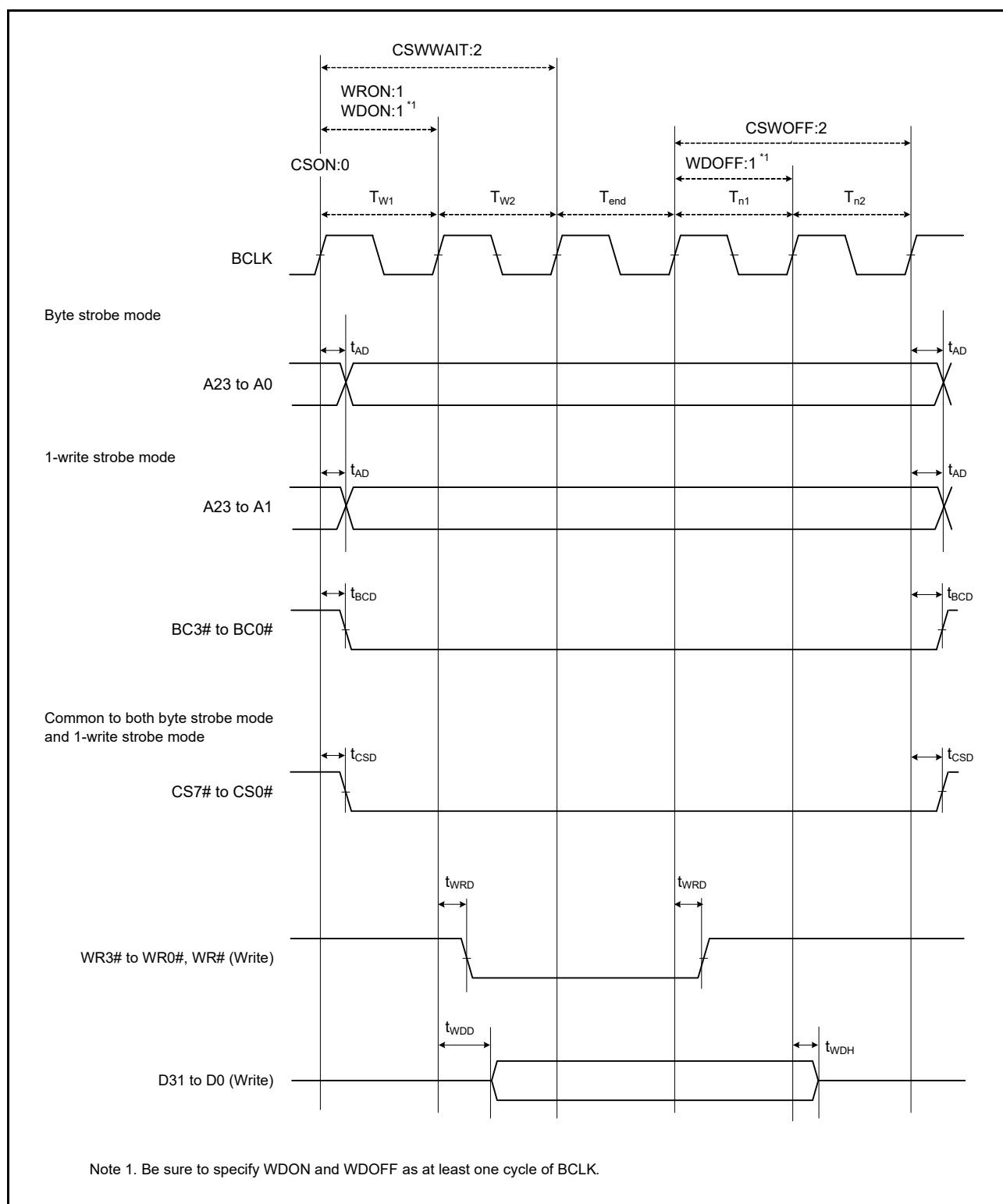


Figure 63.21 External Bus Timing/Normal Write Cycle (Bus Clock Synchronized)

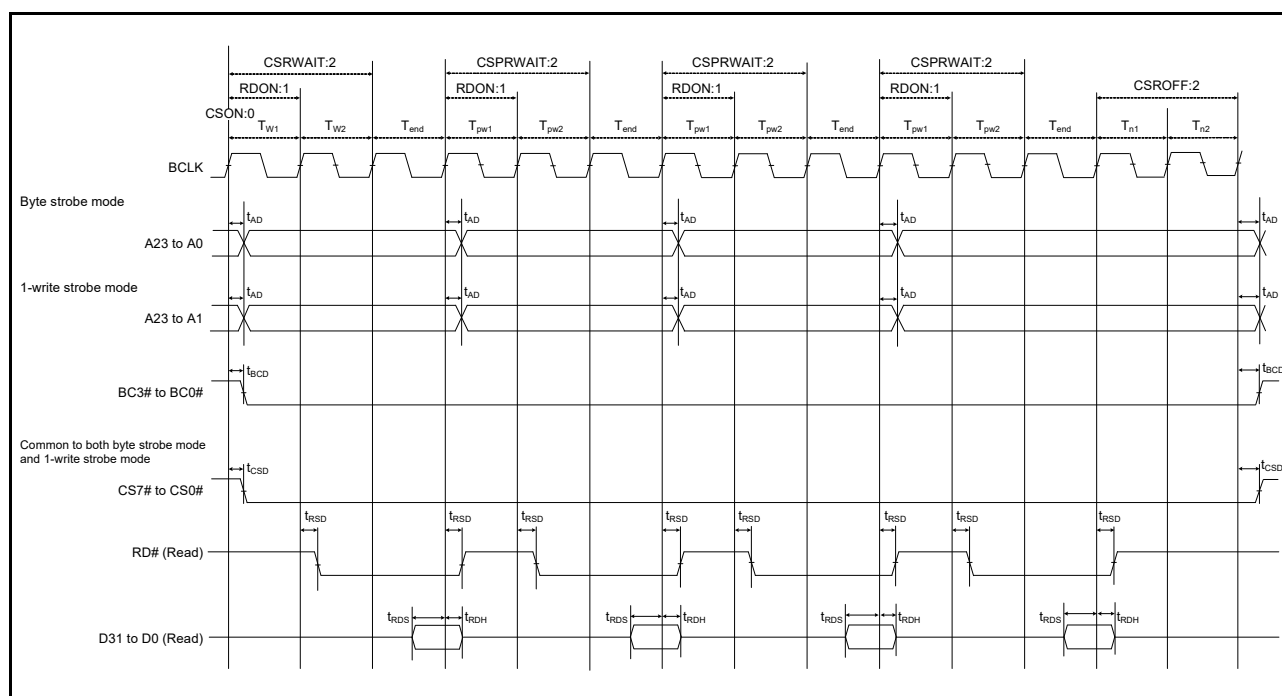
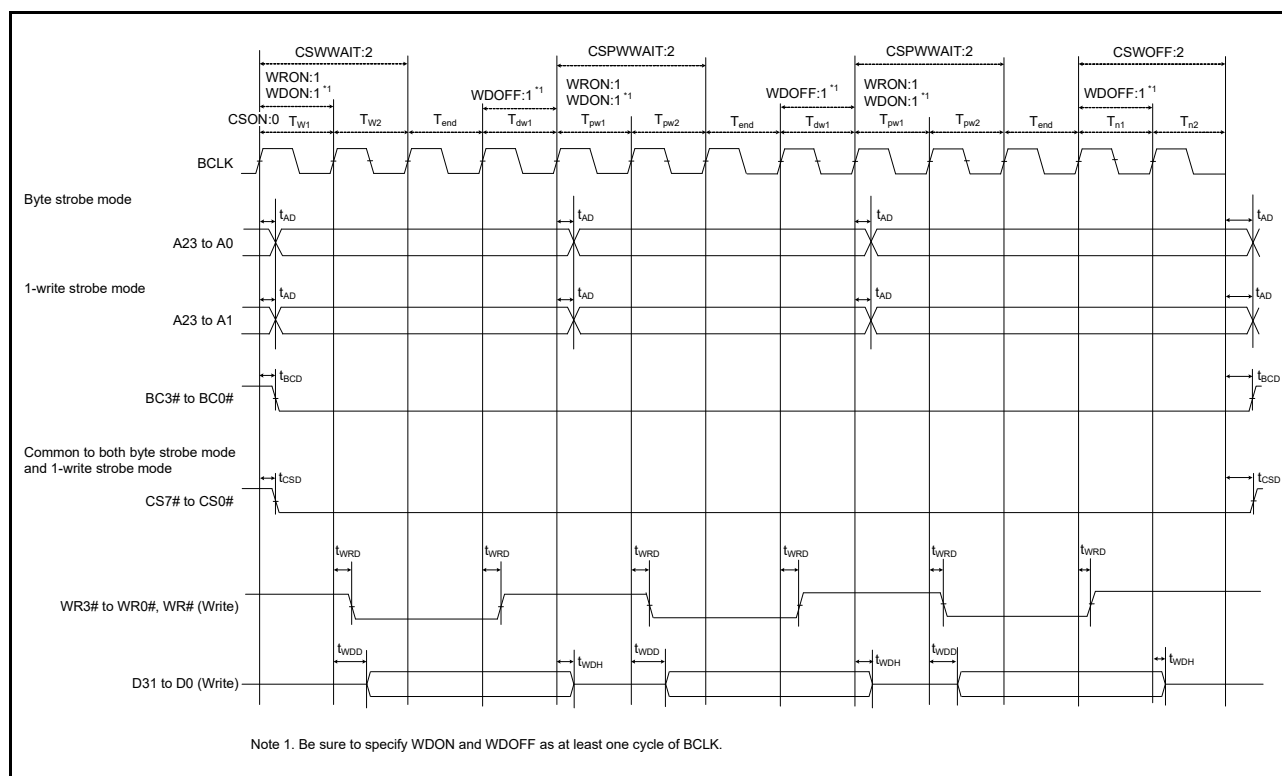


Figure 63.22 External Bus Timing/Page Read Cycle (Bus Clock Synchronized)



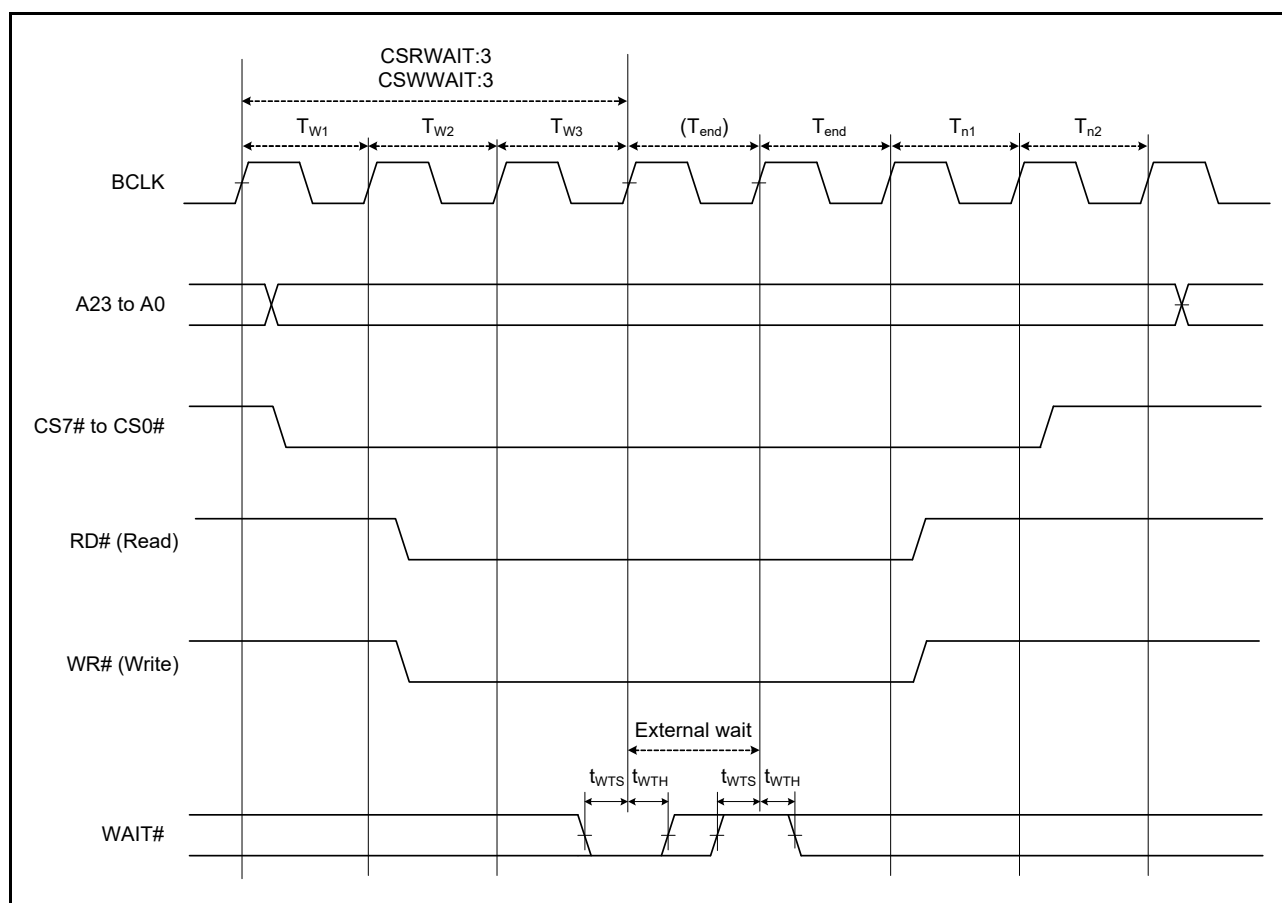


Figure 63.24 External Bus Timing/External Wait Control

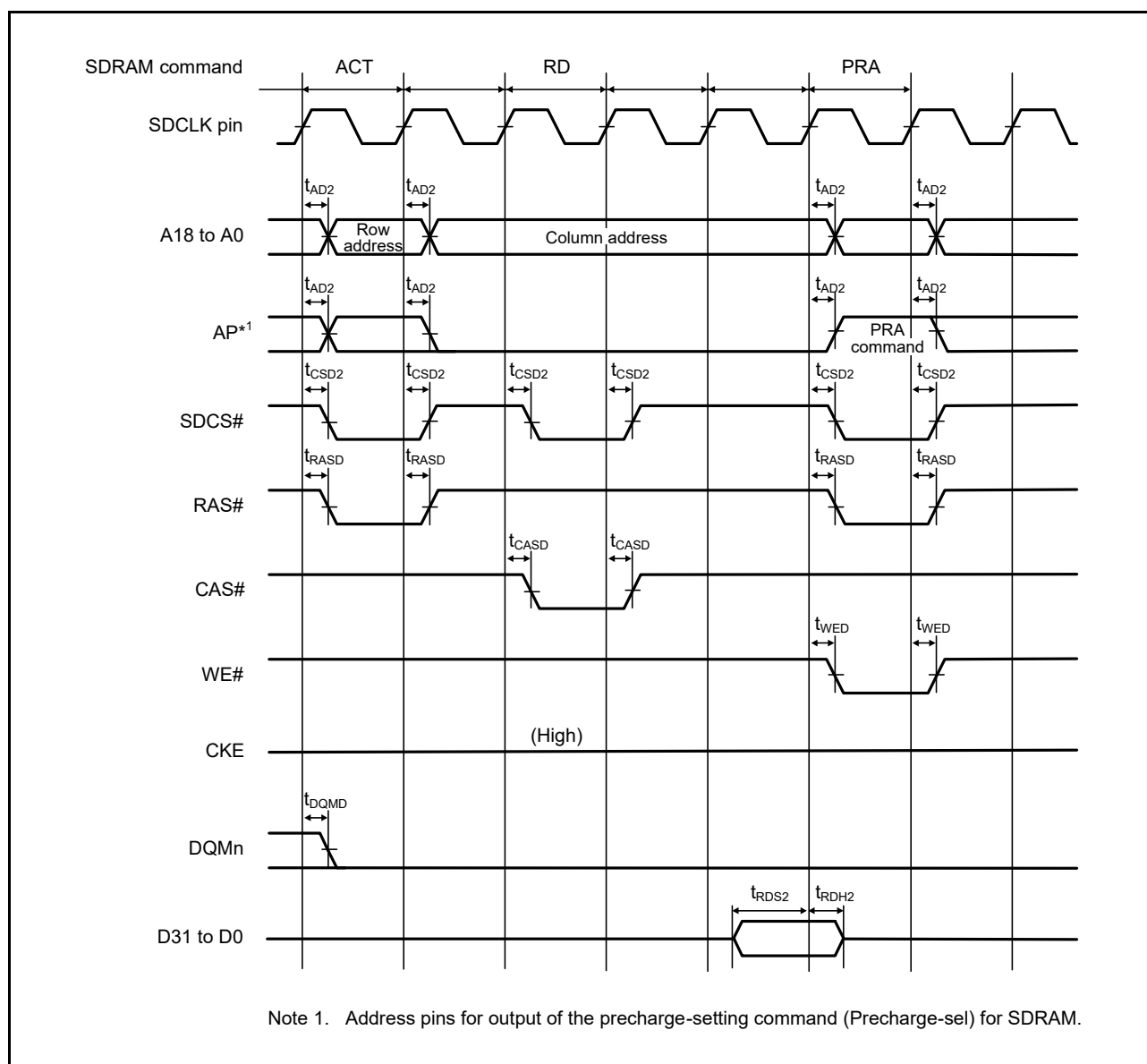


Figure 63.25 SDRAM Space Single Read Bus Timing

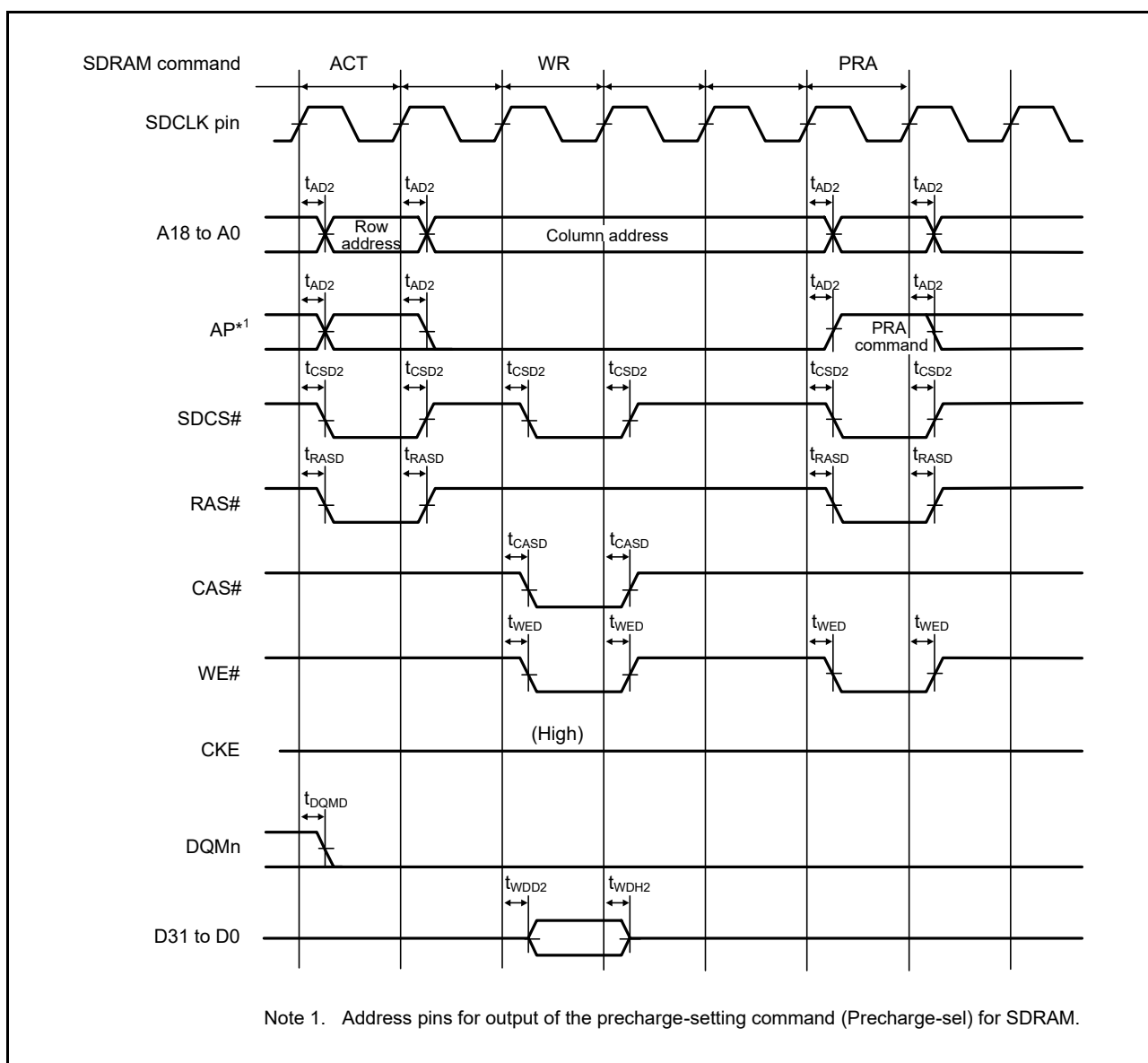
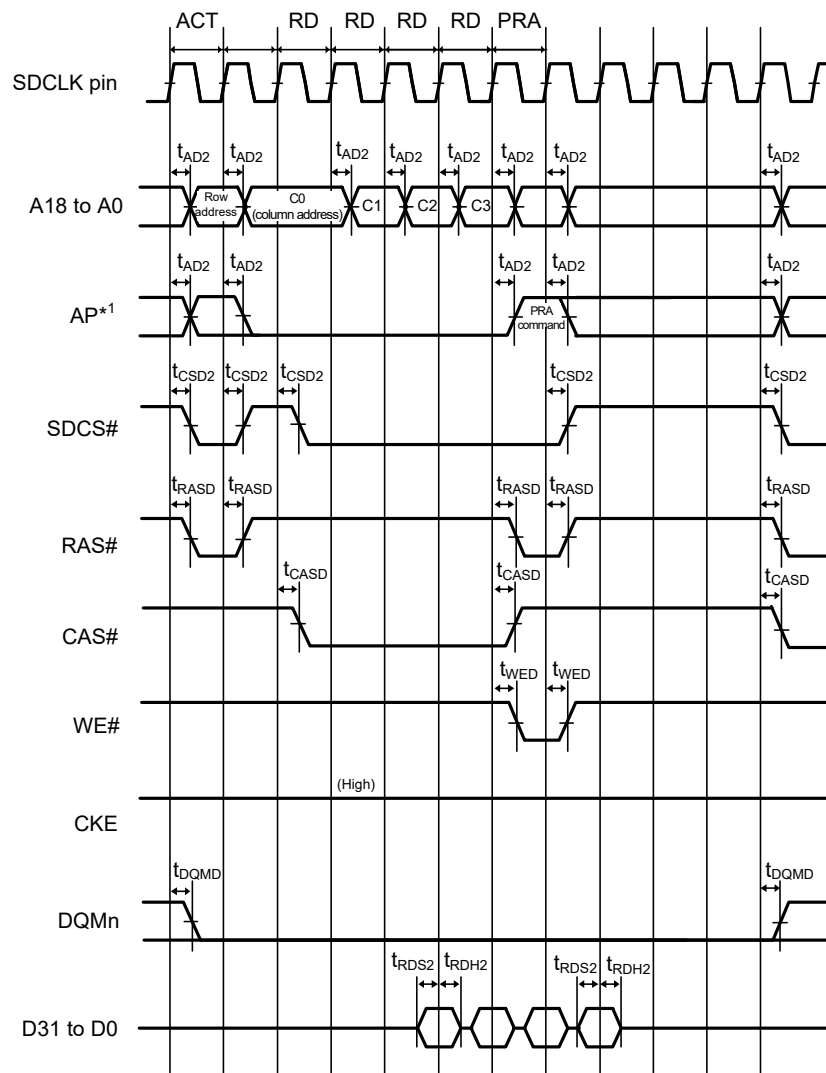
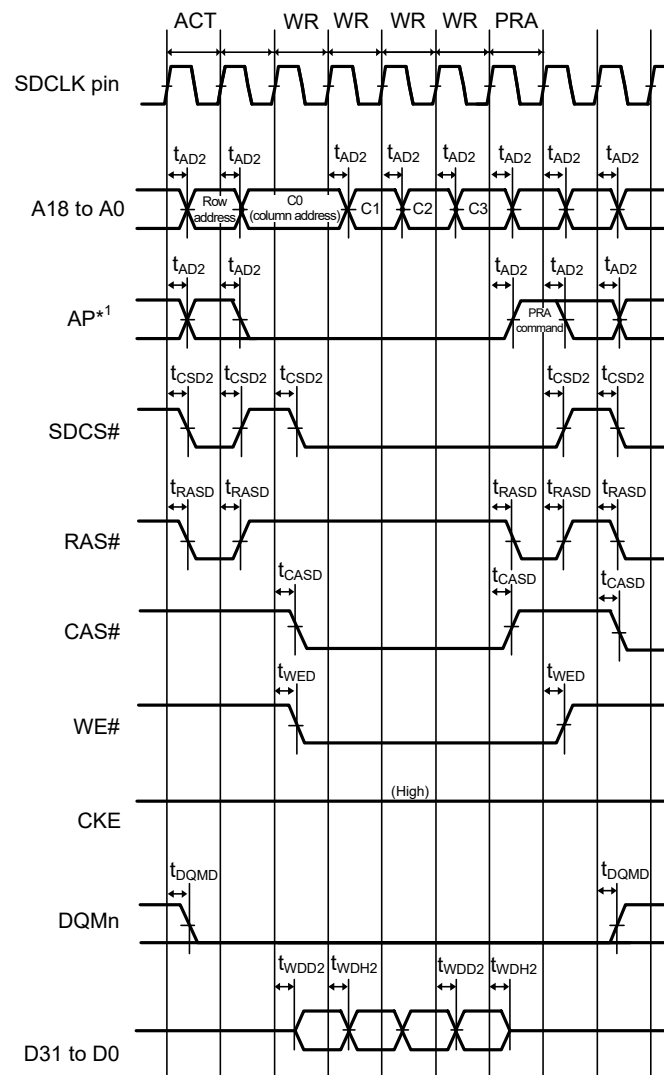


Figure 63.26 SDRAM Space Single Write Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

Figure 63.27 SDRAM Space Multiple Read Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

Figure 63.28 SDRAM Space Multiple Write Bus Timing

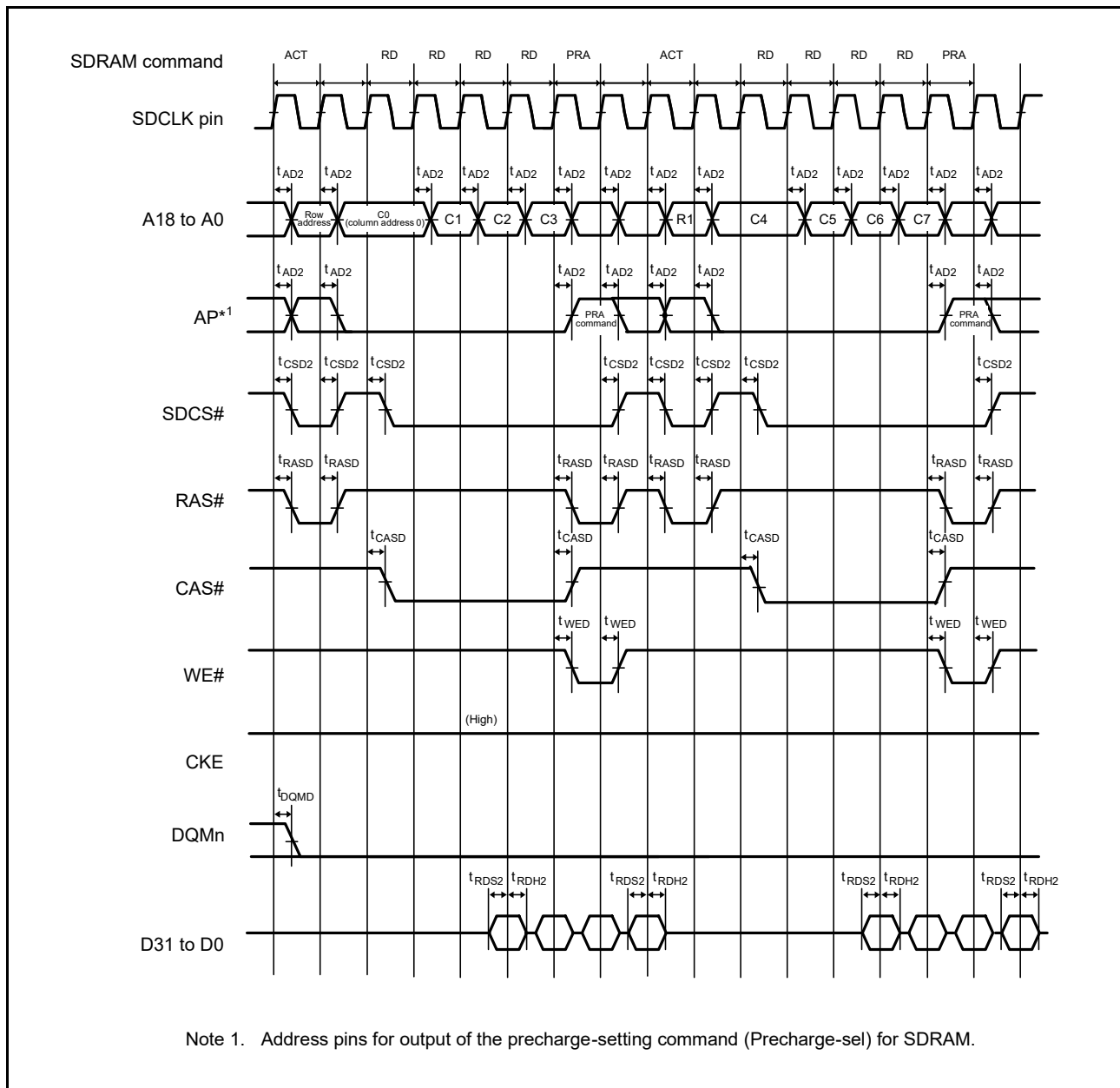
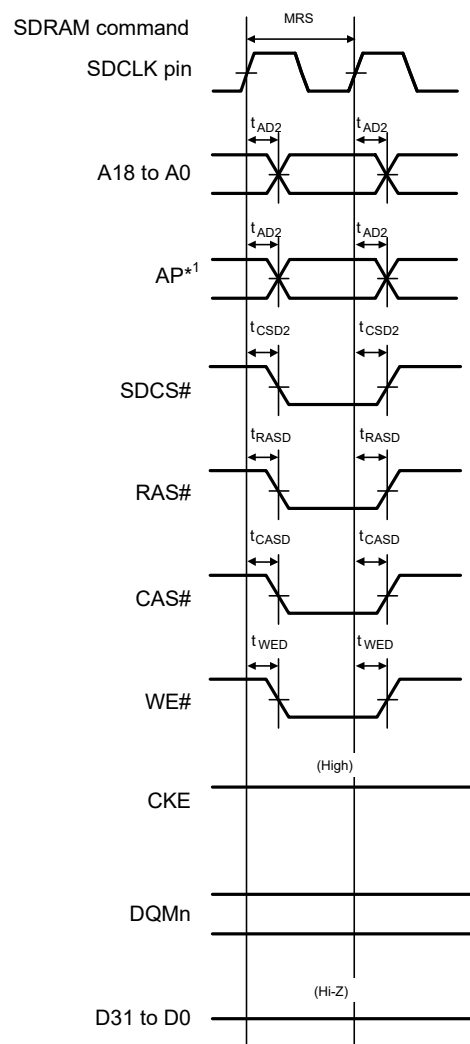
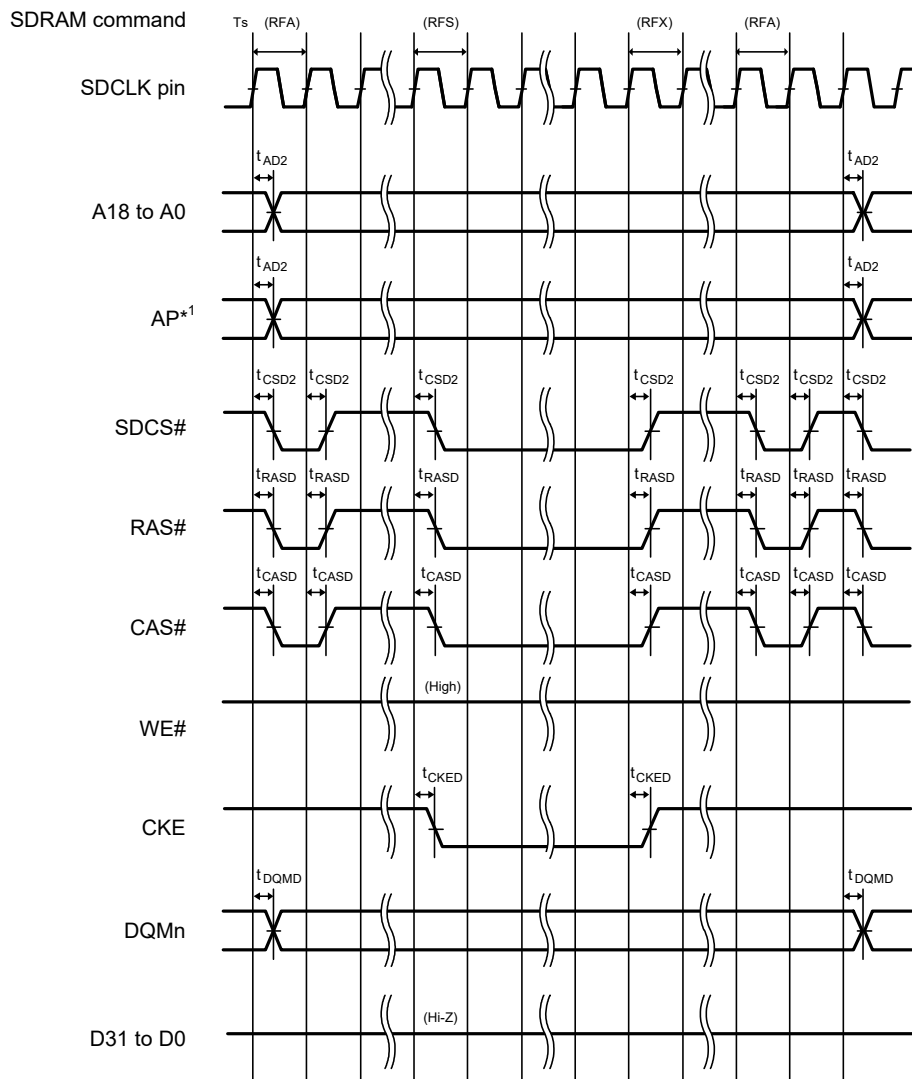


Figure 63.29 SDRAM Space Multiple Read Line Stride Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

Figure 63.30 SDRAM Space Mode Register Set Bus Timing



Note 1. Address pins for output of the precharge-setting command (Precharge-sel) for SDRAM.

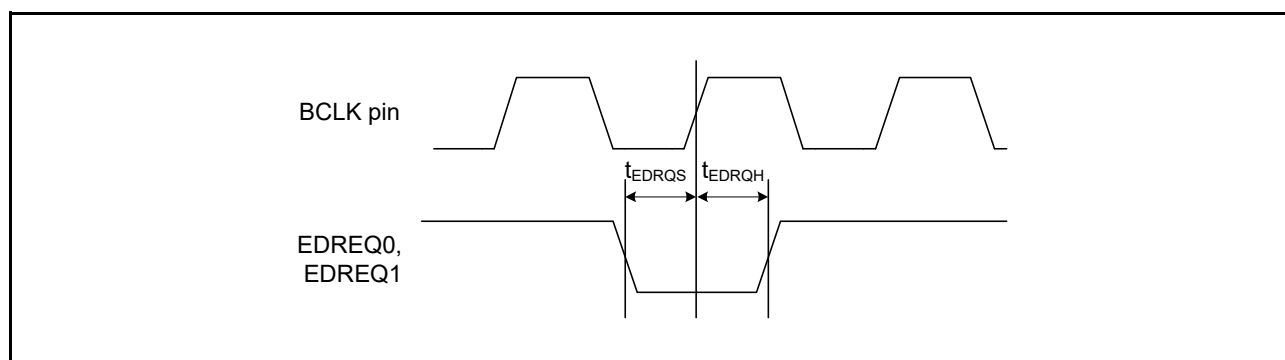
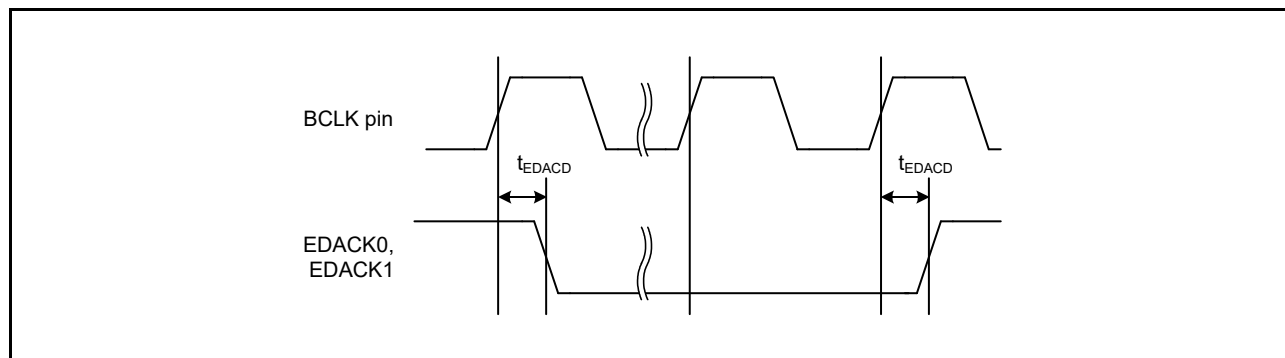
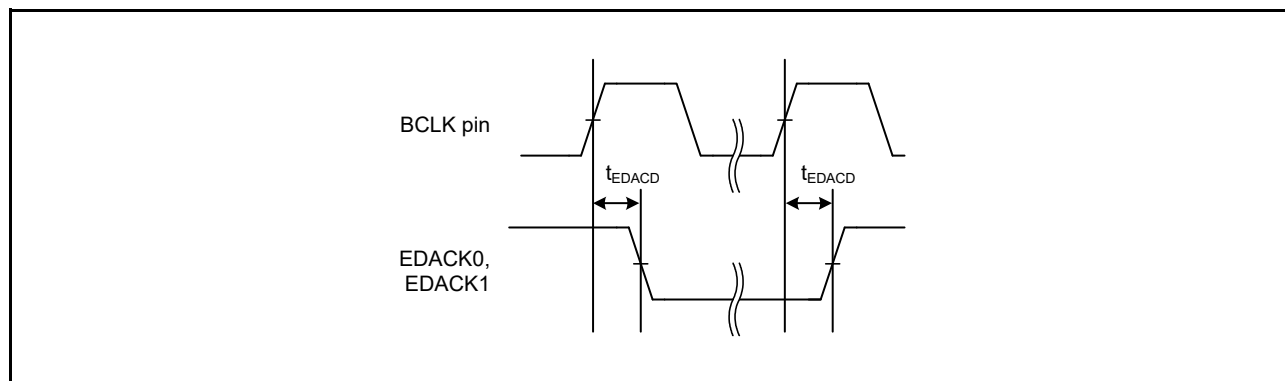
Figure 63.31 SDRAM Space Self-Refresh Bus Timing

63.4.6 EXDMAC Timing

Table 63.27 EXDMAC Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $ICLK = 8$ to 240 MHz, $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $BCLK = SDCLK = 8$ to 80 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

	Item	Symbol	Min.	Max.	Unit	Test Conditions
EXDMAC	EDREQ setup time	t_{EDRQS}	13	—	ns	Figure 63.32
	EDREQ hold time	t_{EDRQH}	2	—	ns	
	EDACK delay time	t_{EDACD}	—	13	ns	Figure 63.33, Figure 63.34

**Figure 63.32 EDREQ0 and EDREQ1 Input Timing****Figure 63.33 EDACK0 and EDACK1 Single-Address Transfer Timing (for a CS Area)****Figure 63.34 EDACK0 and EDACK1 Single-Address Transfer Timing (for SDRAM)**

63.4.7 Timing of On-Chip Peripheral Modules

63.4.7.1 I/O Port

Table 63.28 I/O Port Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq VREFH0 \leq AVCC0$,
 $VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = 0.5 \times VCC$, $V_{OL} = 0.5 \times VCC$, $C = 30$ pF,
High-drive output is selected by the drive capacity control register.

Item		Symbol	Min.	Max.	Unit*1	Test Conditions
I/O ports	Input data pulse width	t_{PRW}	1.5	—	t_{PBcyc}	Figure 63.35

Note 1. t_{PBcyc} : PCLKB cycle

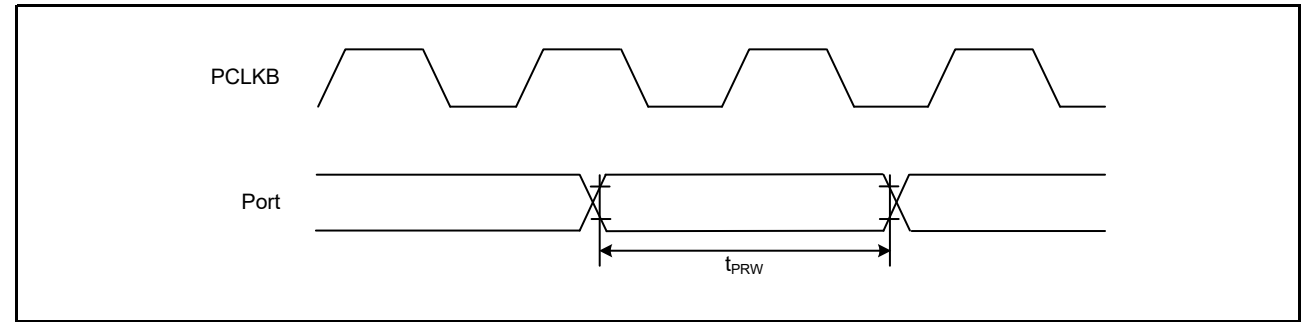


Figure 63.35 I/O Port Input Timing

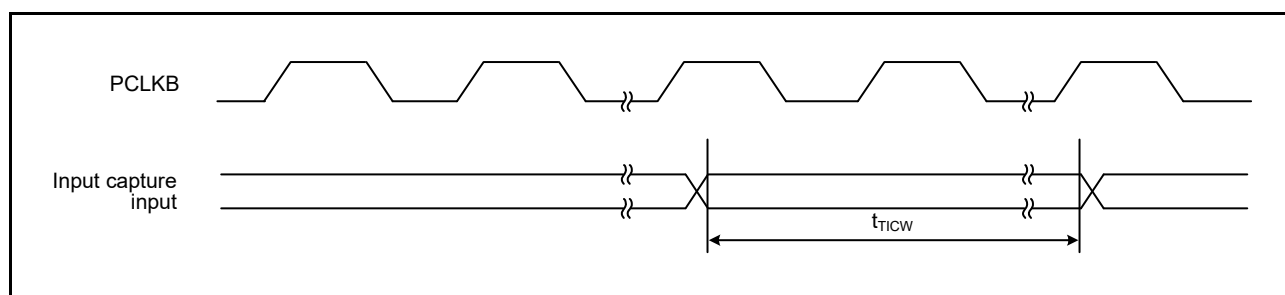
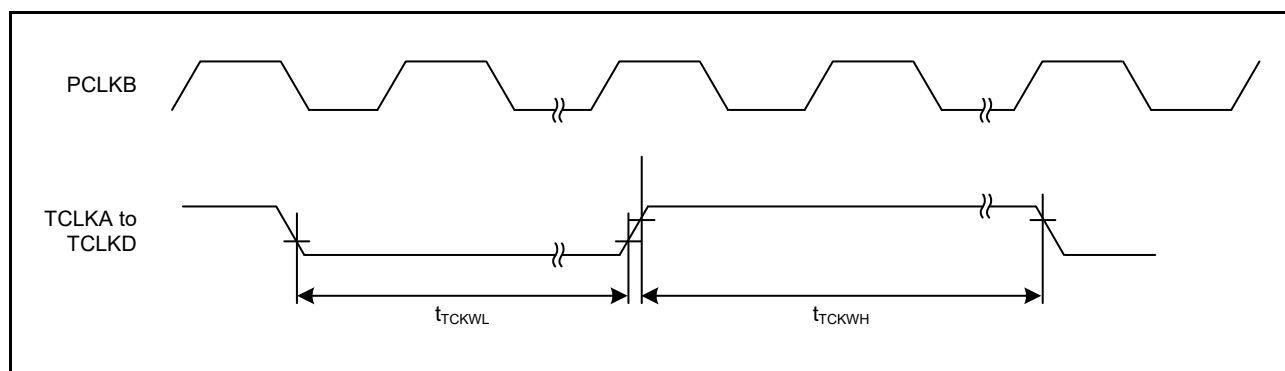
63.4.7.2 TPU

Table 63.29 TPU Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
TPU	Input capture input pulse width	Single-edge setting	t_{TICW}	1.5	—	t_{PBcyc}	Figure 63.36
		Both-edge setting		2.5	—		
	Timer clock pulse width	Single-edge setting	t_{TCKWH} , t_{TCKWL}	1.5	—	t_{PBcyc}	Figure 63.37
		Both-edge setting		2.5	—		
		Phase counting mode		2.5	—		

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.36 TPU Input Capture Input Timing****Figure 63.37 TPU Clock Input Timing**

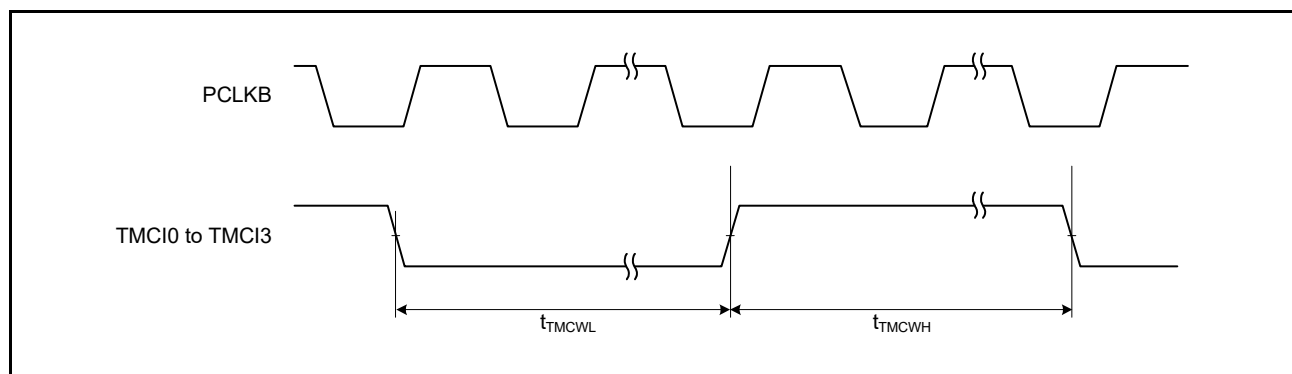
63.4.7.3 TMR

Table 63.30 TMR Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
TMR	Timer clock pulse width	Single-edge setting	t_{TMCWH}	1.5	—	t_{PBcyc}	Figure 63.38
		Both-edge setting	t_{TMCWL}	2.5	—		

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.38 TMR Clock Input Timing**

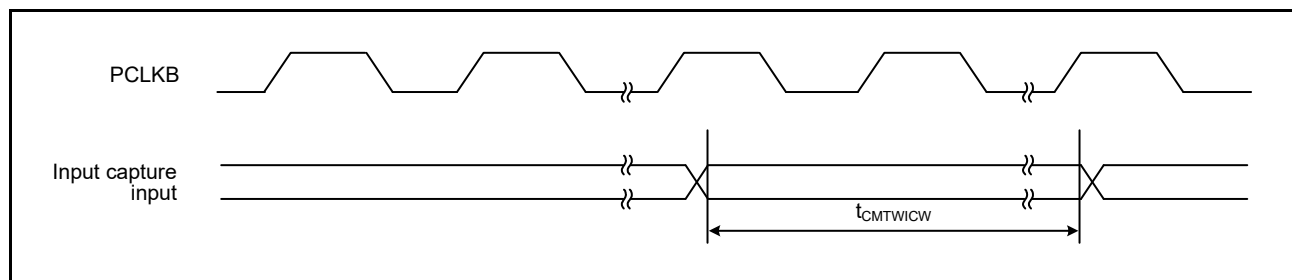
63.4.7.4 CMTW

Table 63.31 CMTW Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
CMTW	Input capture input pulse width	Single-edge setting	$t_{CMTWICW}$	1.5	—	t_{PBcyc}	Figure 63.39
		Both-edge setting		2.5	—		

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.39 CMTW Input Capture Input Timing**

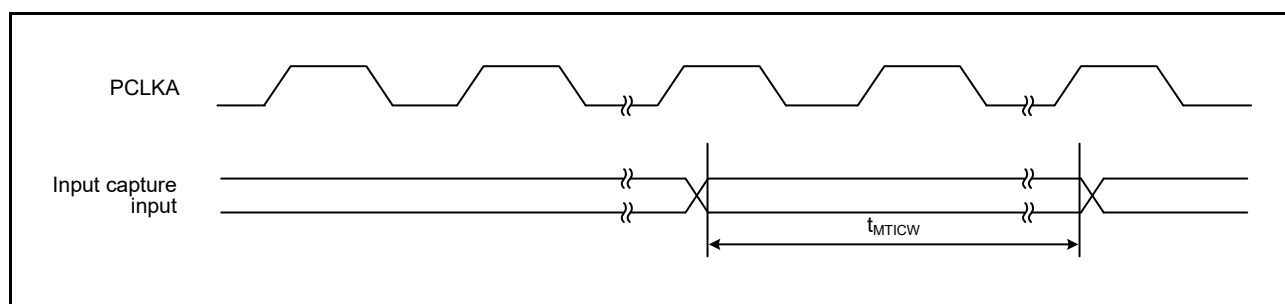
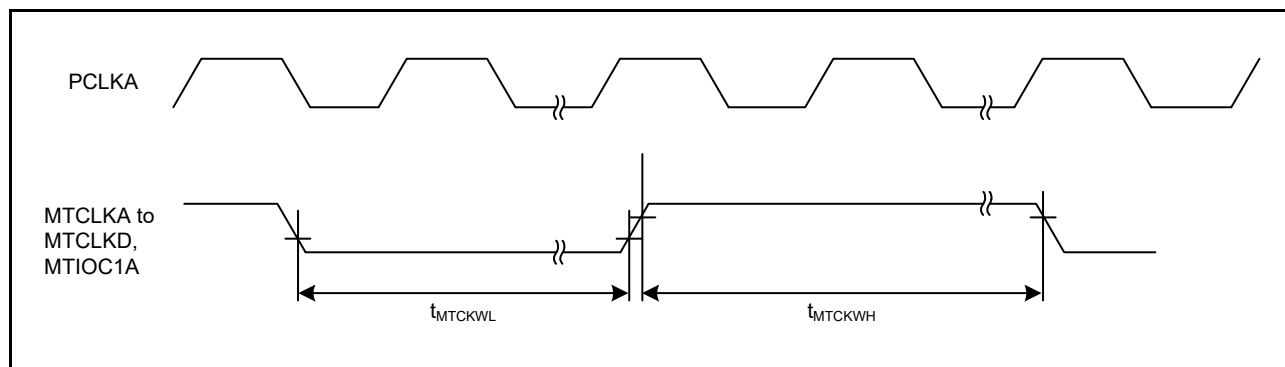
63.4.7.5 MTU

Table 63.32 MTU Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
MTU	Input capture input pulse width	Single-edge setting	t_{MTICW}	1.5	—	t_{PAcyc}	Figure 63.40
		Both-edge setting		2.5	—		
	Timer clock pulse width	Single-edge setting	t_{MTCKWH} , t_{MTCKWL}	1.5	—	t_{PAcyc}	Figure 63.41
		Both-edge setting		2.5	—		
		Phase counting mode		2.5	—		

Note 1. t_{PAcyc} : PCLKA cycle

**Figure 63.40 MTU Input Capture Input Timing****Figure 63.41 MTU Clock Input Timing**

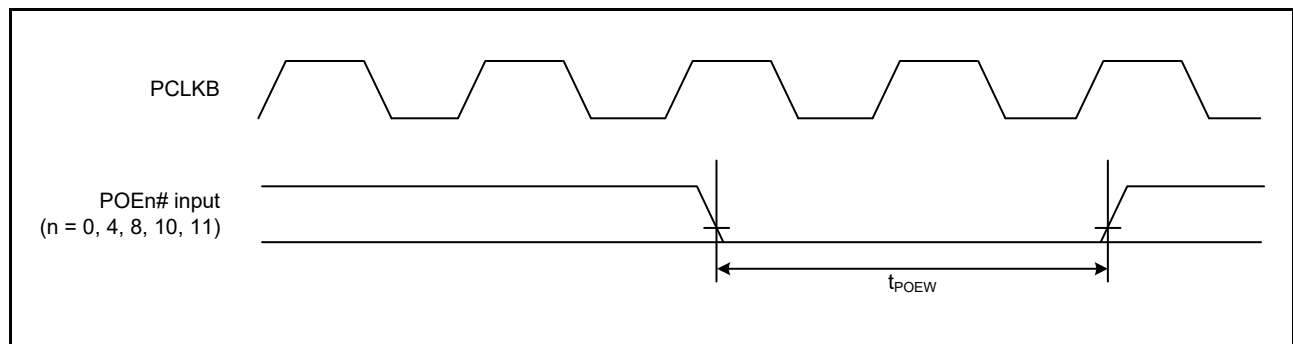
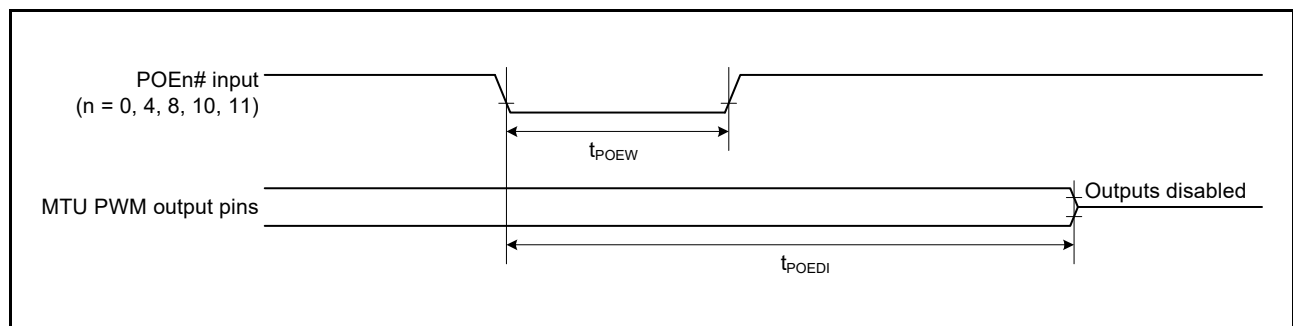
63.4.7.6 POE

Table 63.33 POE Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item	Symbol	Min.	Typ.	Max.	Unit*1	Test Conditions
POE	POEn# input pulse width (n = 0, 4, 8, 10, 11)	t_{POEW}	1.5	—	—	t_{PBcyc} Figure 63.42
Output disable time	Transition of the POEn# signal level	t_{POEDI}	—	—	$5 PCLKB + 0.24$	μs Figure 63.43 When detecting falling edges (ICSRm.POEnM[3:0] = 0000 (m = 1 to 5; n = 0, 4, 8, 10, 11))
	Simultaneous conduction of output pins	t_{POEDO}	—	—	$3 PCLKB + 0.2$	μs Figure 63.44
	Register setting	t_{POEDS}	—	—	$1 PCLKB + 0.2$	μs Figure 63.45 Time for access to the register is not included.
	Oscillation stop detection	t_{POEDOS}	—	—	21	μs Figure 63.46

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.42 POE# Input Timing****Figure 63.43 Output Disable Time for POE in Response to Transition of the POEn# Signal Level**

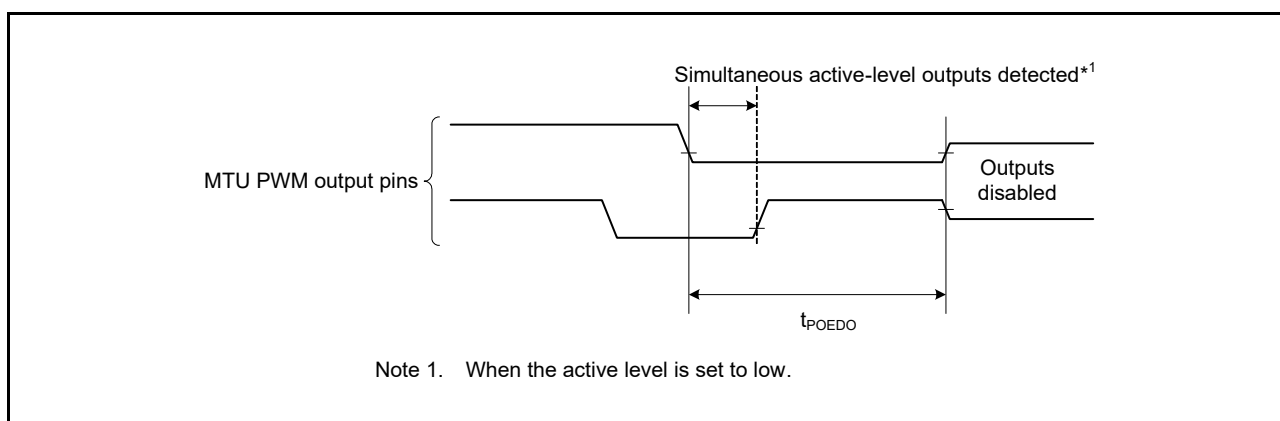


Figure 63.44 Output Disable Time for POE in Response to the Simultaneous Conduction of Output Pins

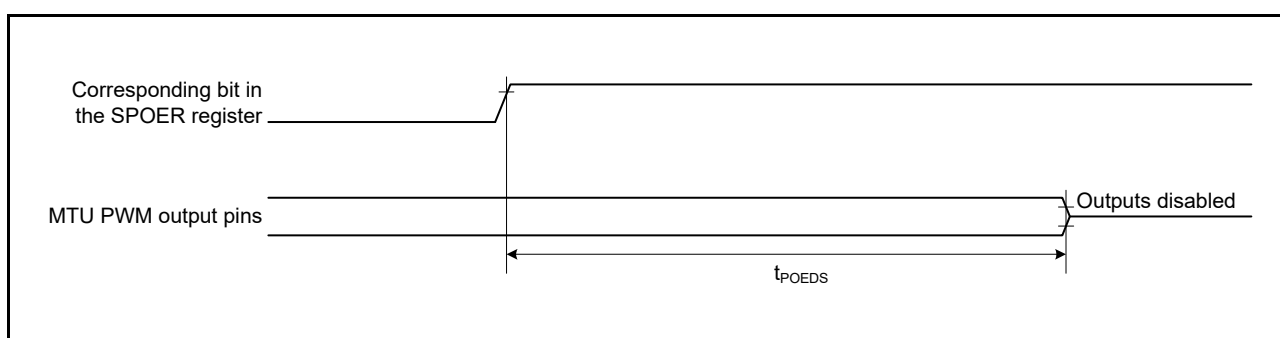


Figure 63.45 Output Disable Time for POE in Response to the Register Setting

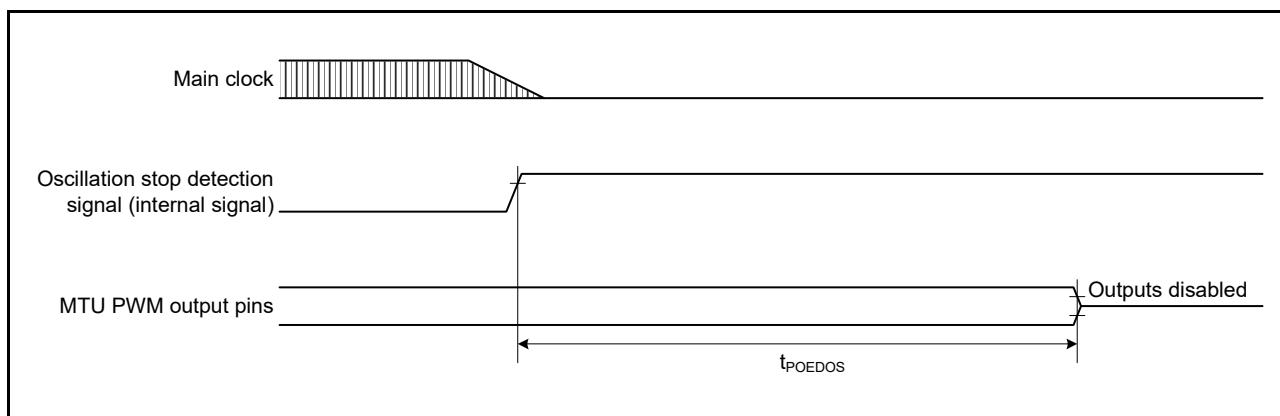


Figure 63.46 Output Disable Time for POE in Response to the Oscillation Stop Detection

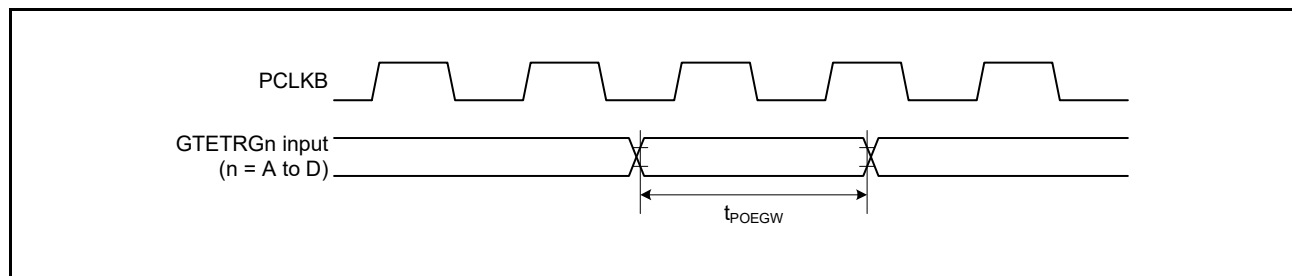
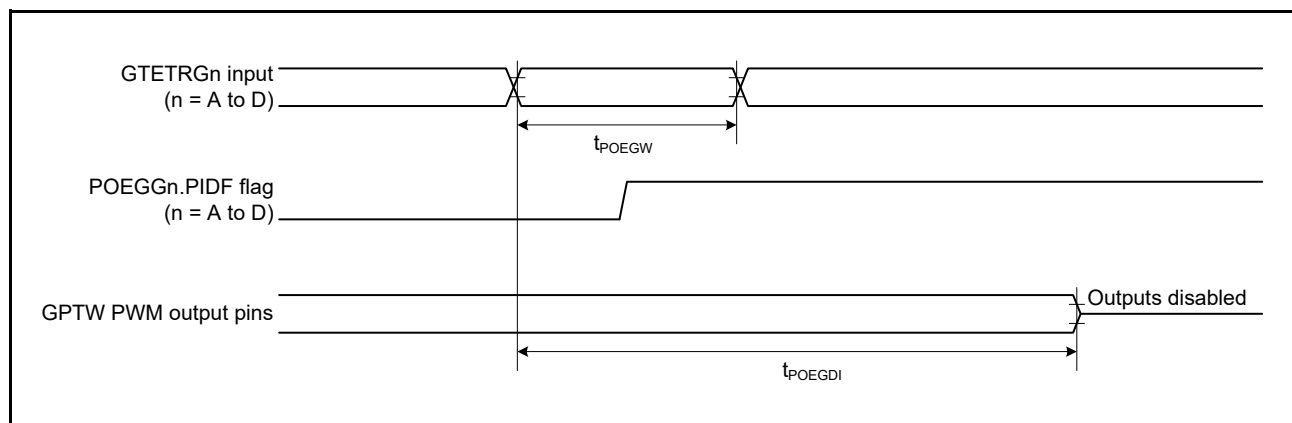
63.4.7.7 POEG

Table 63.34 POEG Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item		Symbol	Min.	Typ.	Max.	Unit*1	Test Conditions
POEG	GTETRn input pulse width (n = A to D)	t_{POEGW}	1.5	—	—	t_{PBcyc}	Figure 63.47
	Output disable time	Input level detection of the GTETRn pin (via flag)	t_{POEGDI}	—	$3 PCLKB + 0.34$	μs	Figure 63.48 When the digital noise filter is not in use (POEGn.NFEN = 0 (n = A to D))
		Detection of the output stopping signal from GPTW (deadtime error, simultaneous high output, or simultaneous low output)	t_{POEGDE}	—	0.5	μs	Figure 63.49
		Register setting	t_{POEGDS}	—	$1 PCLKB + 0.3$	μs	Figure 63.50 Time for access to the register is not included.
		Oscillation stop detection	$t_{POEGDOS}$	—	21	μs	Figure 63.51

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.47 POEG Input Timing****Figure 63.48 Output Disable Time for POEG via Detection Flag in Response to the Input Level Detection of the GTETRn pin**

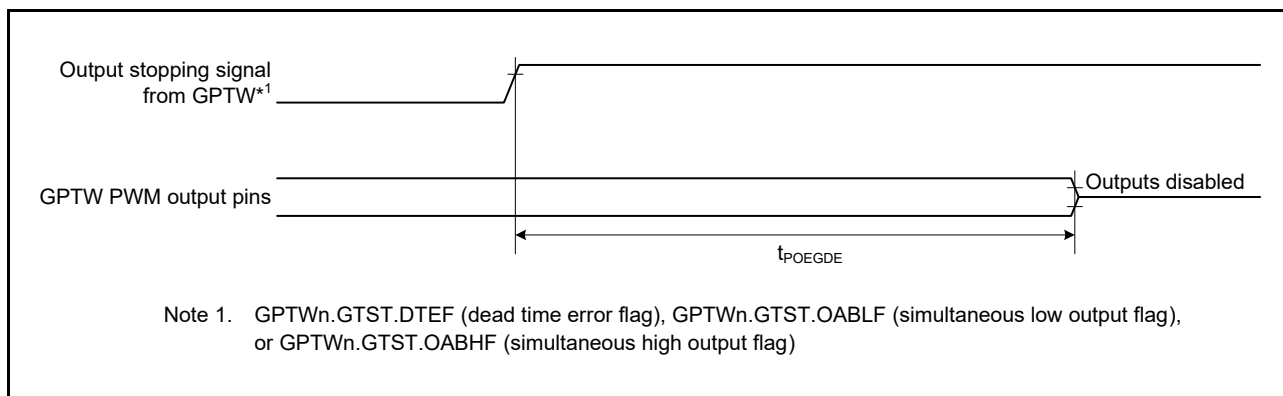


Figure 63.49 Output Disable Time for POEG in Response to Detection of the Output Stopping Signal from GPTW

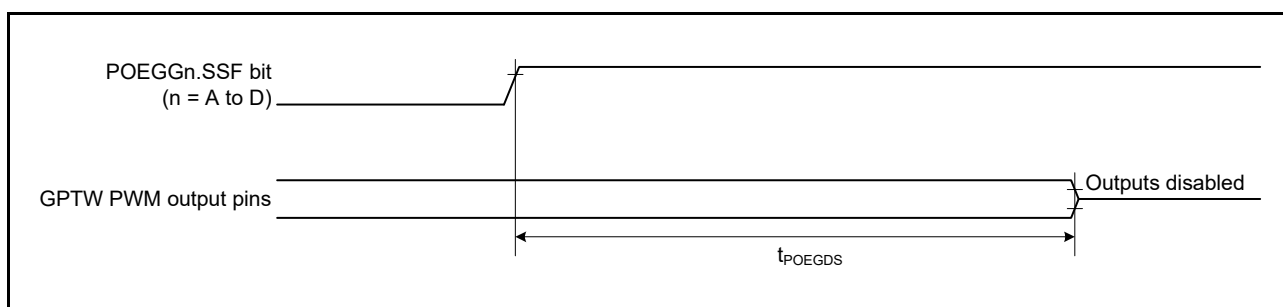


Figure 63.50 Output Disable Time for POEG in Response to the Register Setting

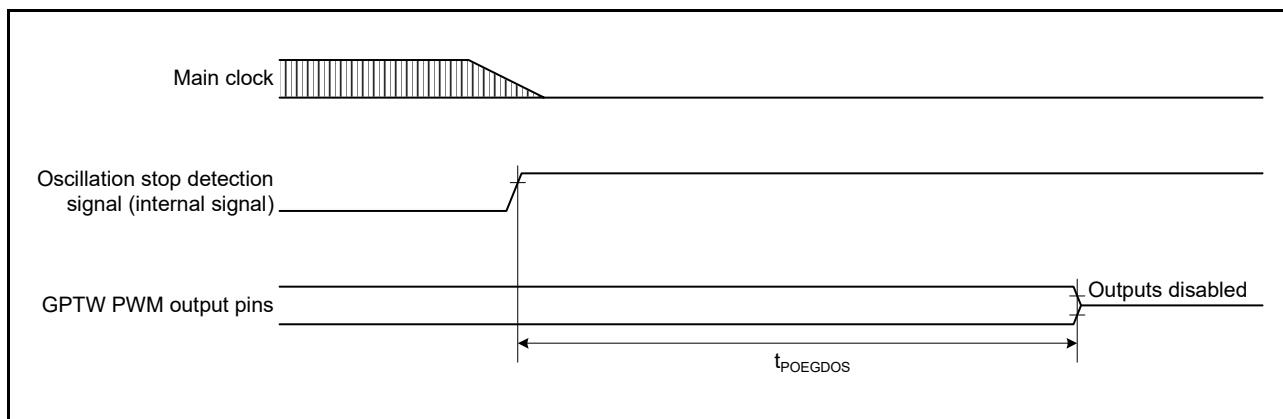


Figure 63.51 Output Disable Time of POEG in Response to the Oscillation Stop Detection

63.4.7.8 GPTW

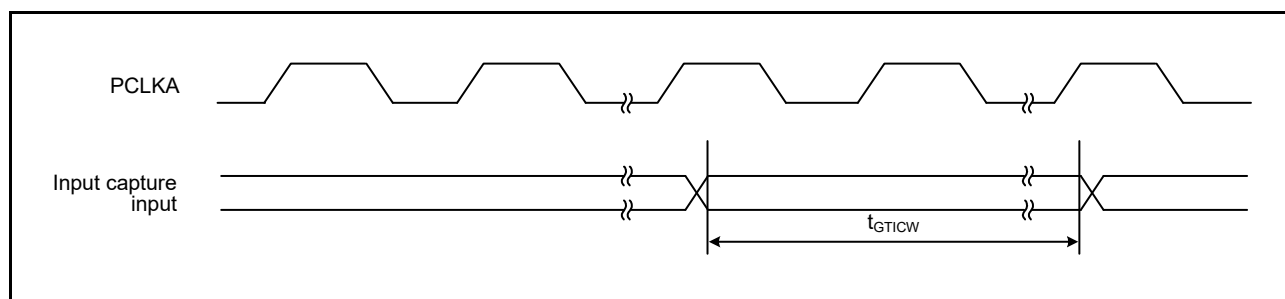
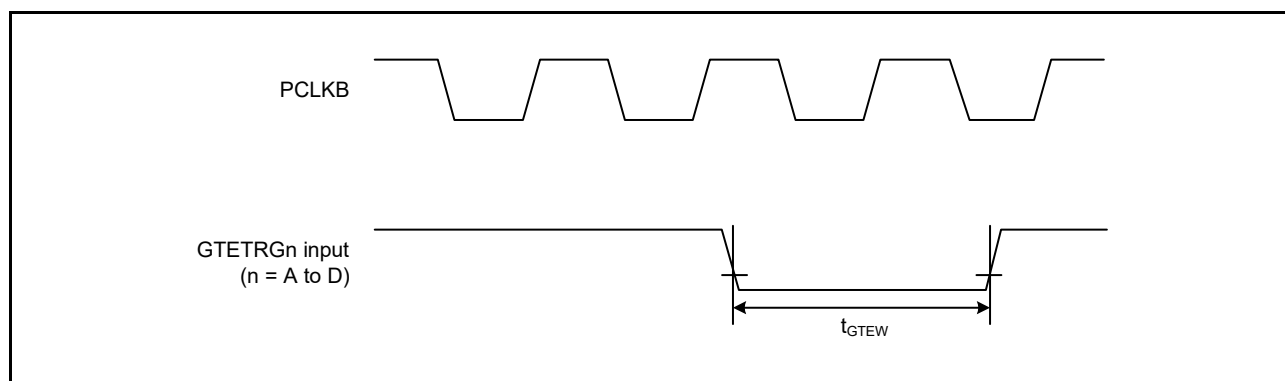
Table 63.35 GPTW Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item			Symbol	Min.	Max.	Unit*1, *2	Test Conditions
GPTW	Input capture input pulse width	Single-edge setting	t_{GTICW}	1.5	—	$t_{P_{Acyc}}$	Figure 63.52
		Both-edge setting		2.5	—		
	External trigger input pulse width	Single-edge setting	t_{GTEW}	1.5	—	$t_{P_{Bcyc}}$	Figure 63.53
		Both-edge setting		2.5	—		

Note 1. $t_{P_{Acyc}}$: PCLKA cycle

Note 2. $t_{P_{Bcyc}}$: PCLKB cycle

**Figure 63.52 GPTW Input Capture Input Timing****Figure 63.53 GPTW External Trigger Input Timing**

63.4.7.9 SCI

Table 63.36 SCI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item			Symbol	Min.	Max.	Unit*1	Test Conditions
SCIh, SCIlj	Input clock cycle	Asynchronous	t _{Scyc}	4	—	t _{PBcyc}	Figure 63.54
		Clock synchronous		6	—		
	Input clock pulse width		t _{SCKW}	0.4	0.6	t _{Scyc}	
	Input clock rise time		t _{SCKr}	—	5	ns	
	Input clock fall time		t _{SCKf}	—	5	ns	
	Output clock cycle	Asynchronous*2	t _{Scyc}	8	—	t _{PBcyc}	
		Clock synchronous		4	—		
	Output clock pulse width		t _{SCKW}	0.4	0.6	t _{Scyc}	
	Output clock rise time		t _{SCKr}	—	5	ns	
	Output clock fall time		t _{SCKf}	—	5	ns	
	Transmit data delay time	Clock synchronous	t _{TXD}	—	28	ns	Figure 63.55
	Receive data setup time	Clock synchronous	t _{RXS}	15	—	ns	
	Receive data hold time	Clock synchronous	t _{RXH}	5	—	ns	
SCli	Input clock cycle	Asynchronous	t _{Scyc}	4	—	t _{PAcyc}	Figure 63.54
		Clock synchronous		12	—		
	Input clock pulse width		t _{SCKW}	0.4	0.6	t _{Scyc}	
	Input clock rise time		t _{SCKr}	—	5	ns	
	Input clock fall time		t _{SCKf}	—	5	ns	
	Output clock cycle	Asynchronous*2	t _{Scyc}	8	—	t _{PAcyc}	
		Clock synchronous		8	—		
	Output clock pulse width		t _{SCKW}	0.4	0.6	t _{Scyc}	
	Output clock rise time		t _{SCKr}	—	5	ns	
	Output clock fall time		t _{SCKf}	—	5	ns	
	Transmit data delay time	Master	t _{TXD}	—	15	ns	Figure 63.55
		Slave		—	28		
	Receive data setup time	Clock synchronous	t _{RXS}	20	—	ns	
	Receive data hold time	Clock synchronous	t _{RXH}	5	—		

Note 1. t_{PBcyc} : PCLKB cycle; t_{PAcyc} : PCLKA cycle

Note 2. When the SEMR.ABCS and SEMR.BGDM bits are set to 1

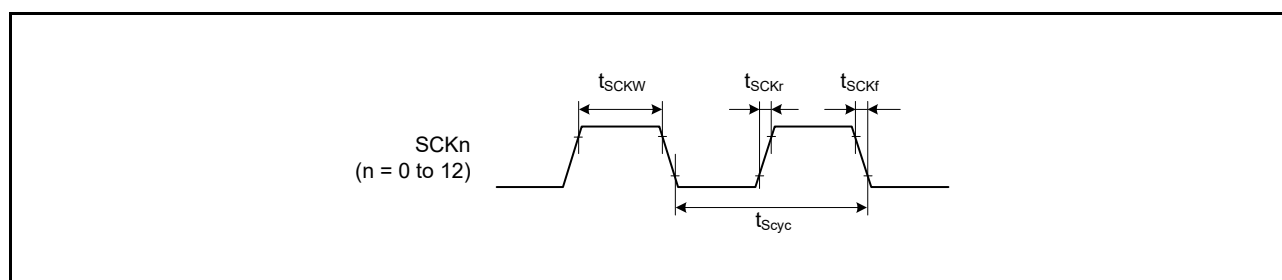
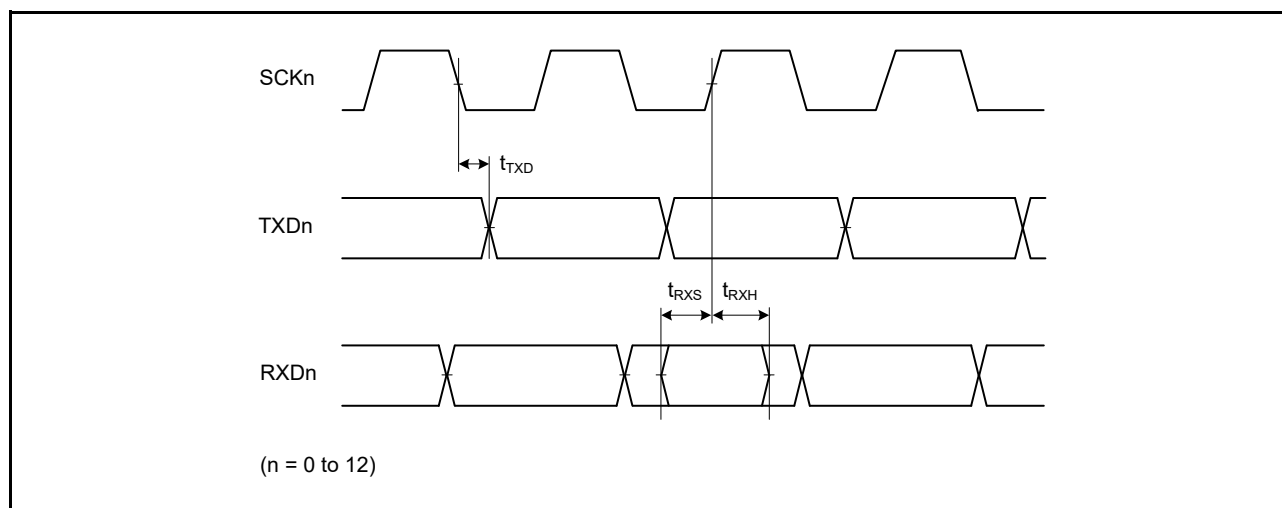
**Figure 63.54 SCK Clock Input Timing****Figure 63.55 SCI Input/Output Timing: Clock Synchronous Mode**

Table 63.37 Simple IIC Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 High-drive output is selected by the drive capacity control register.

Item		Symbol	Min.	Max.	Unit	Test Conditions
Simple IIC (Standard-mode)	SSDA input rise time	t_{Sr}	—	1000	ns	Figure 63.56
	SSDA input fall time	t_{Sf}	—	300	ns	
	SSDA input spike pulse removal time	t_{SP}	0	$4 \times t_{PBcyc}$	ns	
	Data input setup time	t_{SDAS}	250	—	ns	
	Data input hold time	t_{SDAH}	0	—	ns	
	SSCL, SSDA capacitive load	C_b^{*1}	—	400	pF	
Simple IIC (Fast-mode)	SSCL, SSDA input rise time	t_{Sr}	—	300	ns	
	SSCL, SSDA input fall time	t_{Sf}	—	300	ns	
	SSCL, SSDA input spike pulse removal time	t_{SP}	0	$4 \times t_{PBcyc}$	ns	
	Data input setup time	t_{SDAS}	100	—	ns	
	Data input hold time	t_{SDAH}	0	—	ns	
	SSCL, SSDA capacitive load	C_b^{*1}	—	400	pF	

Note: t_{PBcyc} : PCLKB cycle

Note 1. C_b is the total capacitance of the bus lines.

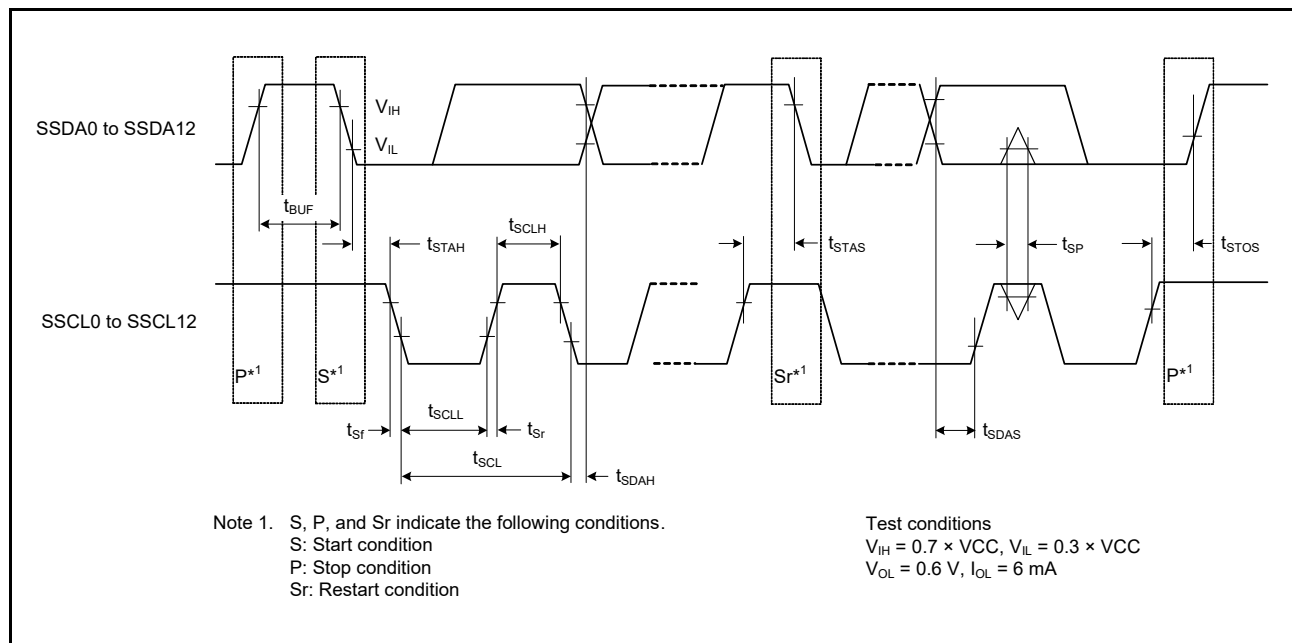
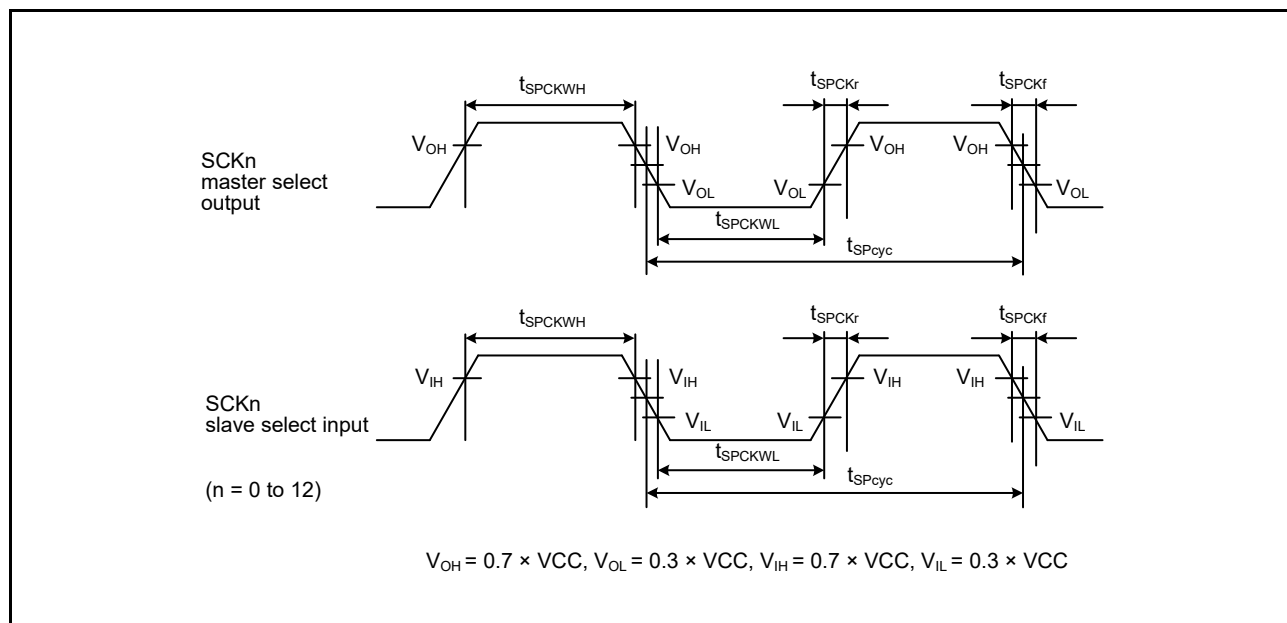
**Figure 63.56 Simple IIC Bus Interface Input/Output Timing**

Table 63.38 Simple SPI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
High-drive output is selected by the drive capacity control register.

	Item	Symbol	Min.	Max.	Unit*1	Test Conditions
Simple SPI	SCK clock cycle output (master)	t_{SPCyc}	4	65536	t_{PAcyc}	Figure 63.57
	SCK clock cycle input (slave)		8	65536		
	SCK clock high pulse width	t_{SPCKWH}	0.4	0.6	t_{SPCyc}	
	SCK clock low pulse width	t_{SPCKWL}	0.4	0.6	t_{SPCyc}	
	SCK clock rise/fall time	t_{SPCKr} , t_{SPCKf}	—	20	ns	
	Data input setup time	t_{SU}	33.3	—	ns	Figure 63.58 to Figure 63.61
	Data input hold time	t_H	33.3	—	ns	
	SS input setup time	t_{LEAD}	1	—	t_{SPCyc}	
	SS input hold time	t_{LAG}	1	—	t_{SPCyc}	
	Data output delay time	t_{OD}	—	33.3	ns	
	Data output hold time	t_{OH}	−10	—	ns	Figure 63.60, Figure 63.61
	Data rise/fall time	t_{Dr} , t_{Df}	—	16.6	ns	
	SS input rise/fall time	t_{SSLr} , t_{SSLf}	—	16.6	ns	
	Slave access time	t_{SA}	—	5	t_{PBcyc}	
	Slave output release time	t_{REL}	—	5	t_{PBcyc}	

Note 1. t_{PAcyc} : PCLKA cycle, t_{PBcyc} : PCLKB cycle

**Figure 63.57 Simple SPI Clock Timing**

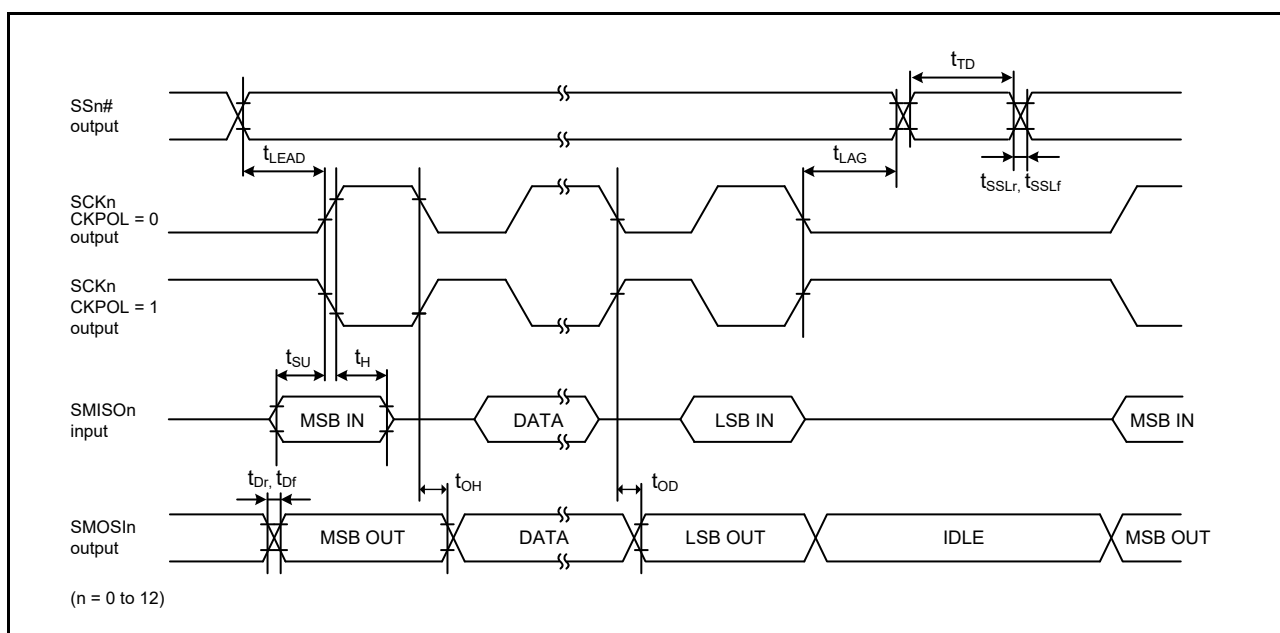


Figure 63.58 Simple SPI Timing (Master, CKPH = 1)

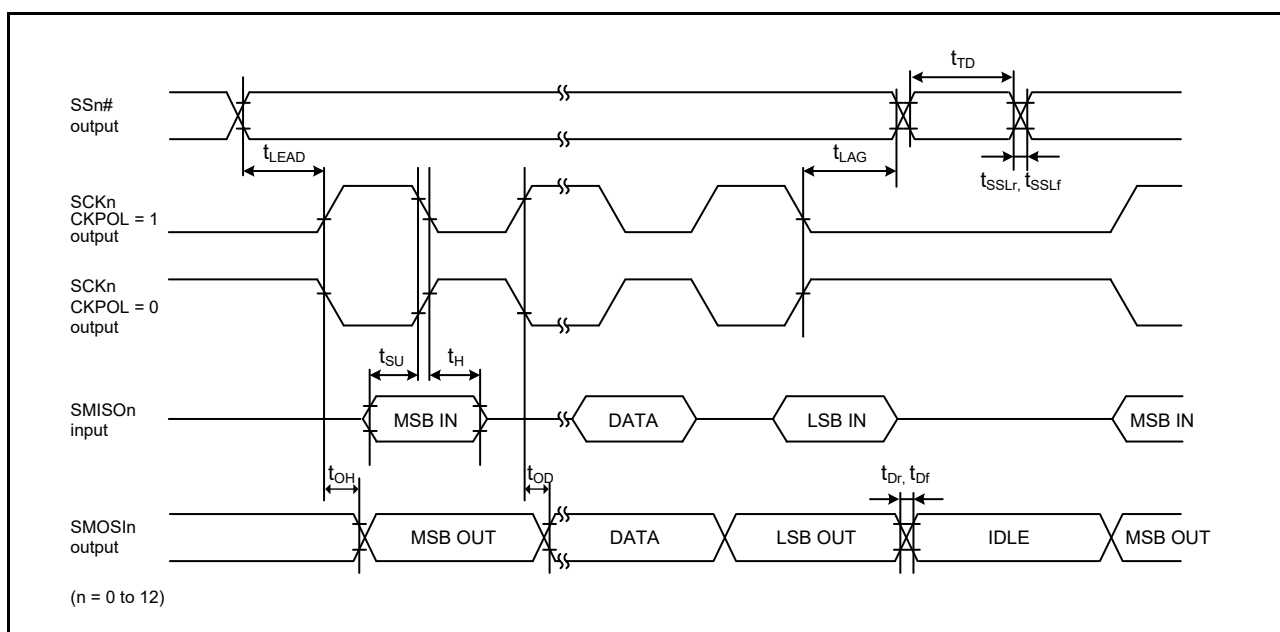


Figure 63.59 Simple SPI Timing (Master, CKPH = 0)

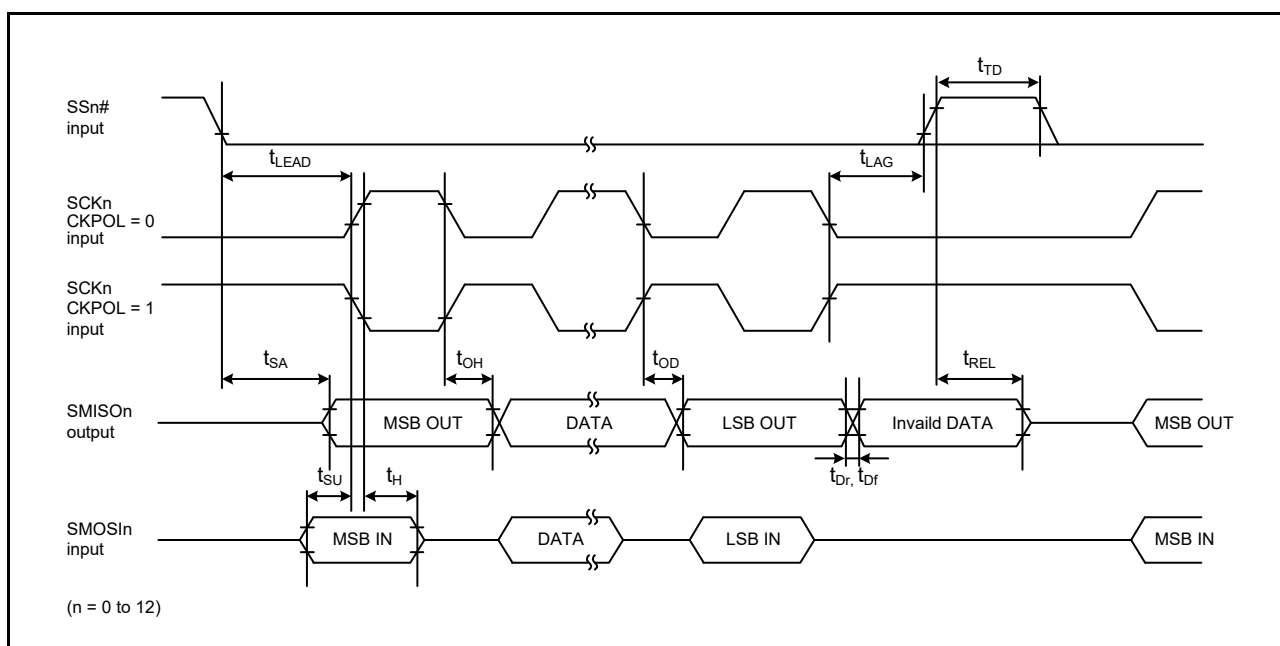


Figure 63.60 Simple SPI Timing (Slave, CKPH = 1)

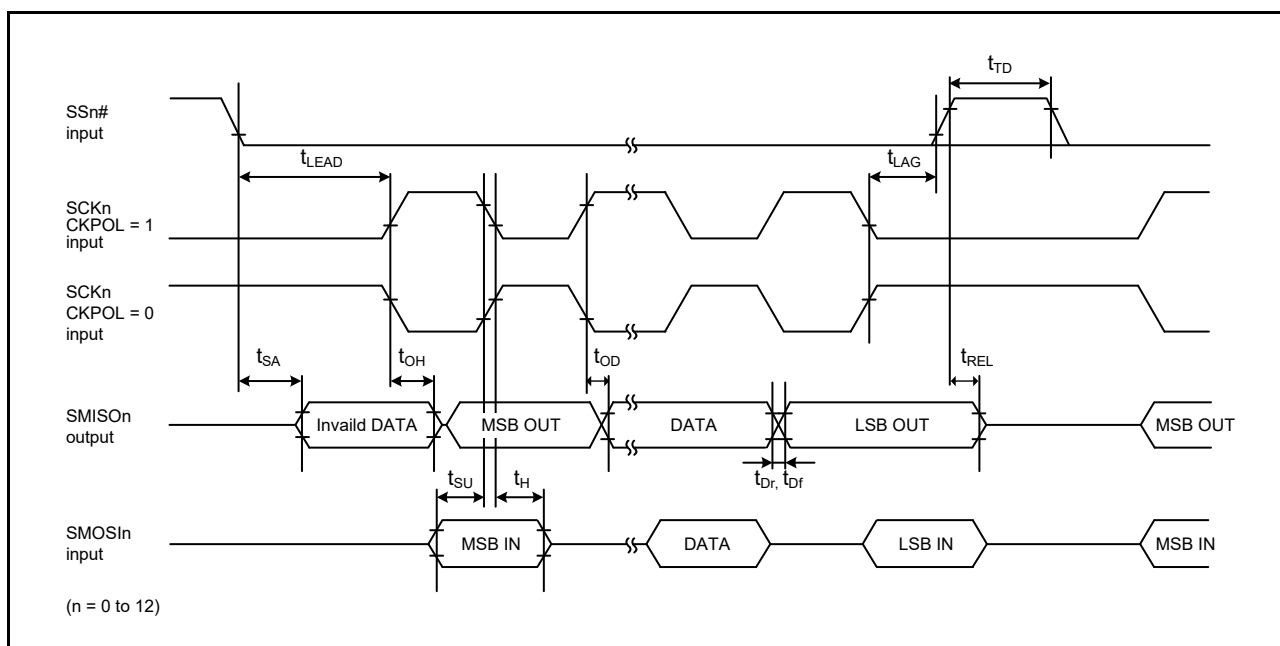


Figure 63.61 Simple SPI Timing (Slave, CKPH = 0)

63.4.7.10 RIIC

Table 63.39 RIIC Timing

Conditions: VCC = AVCC0 = AVCC1 = VCC_USB = V_{BATT} = 2.7 to 3.6 V, 2.7 V ≤ VREFH0 ≤ AVCC0,
VSS = AVSS0 = AVSS1 = VREFL0 = VSS_USB = 0 V,
PCLKA = 8 to 120 MHz, PCLKB = 8 to 60 MHz, T_a = T_{opr},
High-drive output is selected by the drive capacity control register.

Item		Symbol	Min.*1	Max.	Unit	Test Conditions
RIIC (Standard-mode, SMBus) ICFER.FMPE = 0	SCL input cycle time	t _{SCL}	6(12) × t _{II} Ccyc + 1300	—	ns	Figure 63.62
	SCL input high pulse width	t _{SCLH}	3(6) × t _{II} Ccyc + 300	—	ns	
	SCL input low pulse width	t _{SCLL}	3(6) × t _{II} Ccyc + 300	—	ns	
	SCL, SDA input rise time	t _{Sr}	—	1000	ns	
	SCL, SDA input fall time	t _{Sf}	—	300	ns	
	SCL, SDA input spike pulse removal time	t _{SP}	0	1(4) × t _{II} Ccyc	ns	
	SDA input bus free time	t _{BUF}	3(6) × t _{II} Ccyc + 300	—	ns	
	Start condition input hold time	t _{STAH}	t _{II} Ccyc + 300	—	ns	
	Restart condition input setup time	t _{STAS}	1000	—	ns	
	Stop condition input setup time	t _{STOS}	1000	—	ns	
	Data input setup time	t _{SDAS}	t _{II} Ccyc + 50	—	ns	
	Data input hold time	t _{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C _b *2	—	400	pF	
RIIC (Fast-mode) ICFER.FMPE = 0	SCL input cycle time	t _{SCL}	6(12) × t _{II} Ccyc + 600	—	ns	
	SCL input high pulse width	t _{SCLH}	3(6) × t _{II} Ccyc + 300	—	ns	
	SCL input low pulse width	t _{SCLL}	3(6) × t _{II} Ccyc + 300	—	ns	
	SCL, SDA input rise time	t _{Sr}	20 × (External pull-up voltage/5.5V)	300	ns	
	SCL, SDA input fall time	t _{Sf}	20 × (External pull-up voltage/5.5V)	300	ns	
	SCL, SDA input spike pulse removal time	t _{SP}	0	1(4) × t _{II} Ccyc	ns	
	SDA input bus free time	t _{BUF}	3(6) × t _{II} Ccyc + 300	—	ns	
	Start condition input hold time	t _{STAH}	t _{II} Ccyc + 300	—	ns	
	Restart condition input setup time	t _{STAS}	300	—	ns	
	Stop condition input setup time	t _{STOS}	300	—	ns	
	Data input setup time	t _{SDAS}	t _{II} Ccyc + 50	—	ns	
	Data input hold time	t _{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C _b *2	—	400	pF	
RIIC (Fast-mode+) ICFER.FMPE = 1	SCL input cycle time	t _{SCL}	6(12) × t _{II} Ccyc + 240	—	ns	
	SCL input high pulse width	t _{SCLH}	3(6) × t _{II} Ccyc + 120	—	ns	
	SCL input low pulse width	t _{SCLL}	3(6) × t _{II} Ccyc + 120	—	ns	
	SCL, SDA input rise time	t _{Sr}	—	120	ns	
	SCL, SDA input fall time	t _{Sf}	—	120	ns	
	SCL, SDA input spike pulse removal time	t _{SP}	0	1(4) × t _{II} Ccyc	ns	
	SDA input bus free time	t _{BUF}	3(6) × t _{II} Ccyc + 120	—	ns	
	Start condition input hold time	t _{STAH}	t _{II} Ccyc + 120	—	ns	
	Restart condition input setup time	t _{STAS}	120	—	ns	
	Stop condition input setup time	t _{STOS}	120	—	ns	
	Data input setup time	t _{SDAS}	t _{II} Ccyc + 20	—	ns	
	Data input hold time	t _{SDAH}	0	—	ns	
	SCL, SDA capacitive load	C _b *2	—	550	pF	

Note: t_{IICcyC} : RIIC internal reference clock (IIC ϕ) cycle

Note 1. The value within parentheses is applicable when the value of the ICMR3.NF[1:0] bits is 11b while the digital filter is enabled by the setting ICFER.NFE = 1.

Note 2. Cb is the total capacitance of the bus lines.

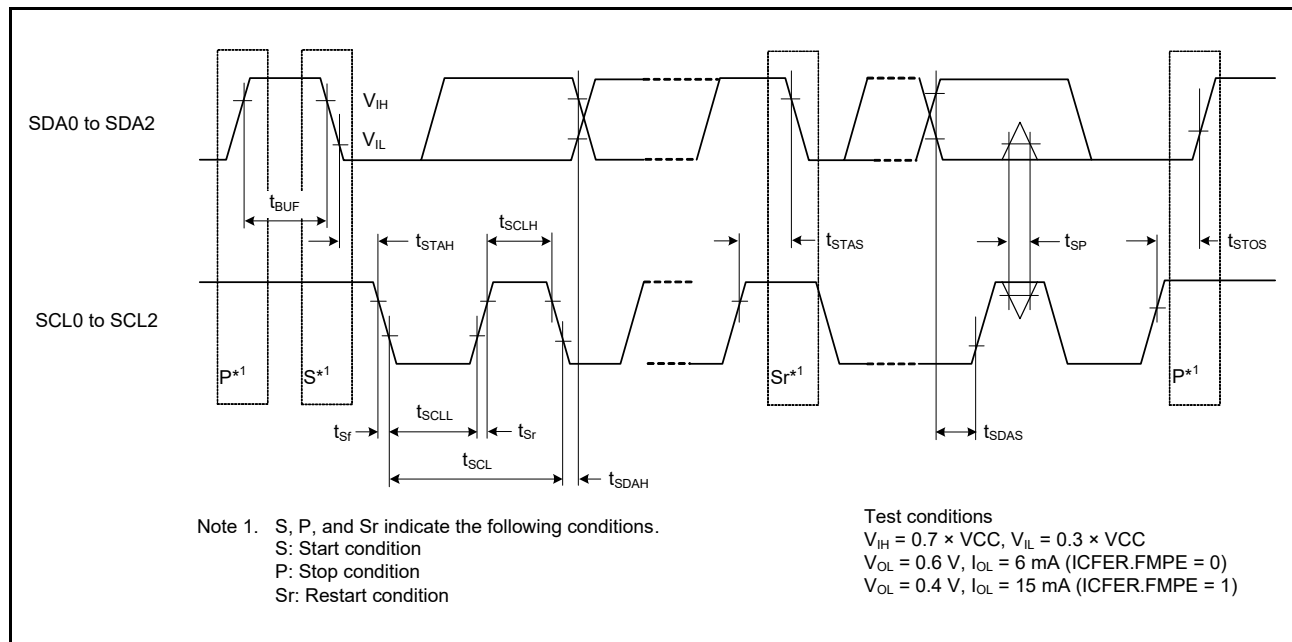


Figure 63.62 RIIC Bus Interface Input/Output Timing

63.4.7.11 RSPI

Table 63.40 RSPI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item				Symbol	Min.*1	Max.*1	Unit*1	Test Conditions*2	
RSPI	RSPCK clock cycle	Master		t_{SPcyc}	2	—	t_{PAcyc}	Figure 63.63	
		Slave			4	—			
	RSPCK clock high pulse width	Master		t_{SPCKWH}	$(t_{SPcyc} - t_{SPCKr} - t_{SPCKf}) / 2 - 3$	—	ns		
		Slave			$(t_{SPcyc} - t_{SPCKr} - t_{SPCKf}) / 2$	—			
	RSPCK clock low pulse width	Master		t_{SPCKWL}	$(t_{SPcyc} - t_{SPCKr} - t_{SPCKf}) / 2 - 3$	—	ns		
		Slave			$(t_{SPcyc} - t_{SPCKr} - t_{SPCKf}) / 2$	—			
	RSPCK clock rise/fall time	Output		t_{SPCKr}	—	5	ns		
		Input		t_{SPCKf}	—	1	μs		
	Data input setup time	Master		t_{SU}	6	—	ns	Figure 63.64 to Figure 63.69	
		Slave			8.3	—			
	Data input hold time	Master	PCLKA divi- sion ratio set to 1/2	t_{HF}	0	—	ns		
			PCLKA divi- sion ratio set to a value other than 1/2	t_H	t_{PAcyc}	—			
		Slave			8.3	—			
	SSL setup time	Master		t_{LEAD}	1	8	t_{SPcyc}		
		Slave			6	—	t_{PAcyc}		
	SSL hold time	Master		t_{LAG}	1	8	t_{SPcyc}		
		Slave			6	—	t_{PAcyc}		
	Data output delay time	Master		t_{OD}	—	6.3	ns		
		Slave			—	28			
	Data output hold time	Master		t_{OH}	0	—	ns		
		Slave			0	—			
	Successive transmission delay time	Master		t_{TD}	$t_{SPcyc} + 2 \times t_{PAcyc}$	$8 \times t_{SPcyc} + 2 \times t_{PAcyc}$	ns		
		Slave			$6 \times t_{PAcyc}$	—			
	MOSI and MISO rise/fall time	Output		t_{Dr}, t_{Df}	—	5	ns		
		Input			—	1	μs		
	SSL rise/fall time	Output		t_{SSLr}	—	5	ns		
		Input		t_{SSLf}	—	1	μs		
	Slave access time				t_{SA}	—	$2 \times t_{PAcyc} + 28$	ns	Figure 63.68, Figure 63.69
	Slave output release time				t_{REL}	—	$2 \times t_{PAcyc} + 28$	ns	

Note 1. t_{PACyc} : PCLKA cycle

Note 2. We recommend using pins that have a letter ("A", "B", etc.) to indicate group membership appended to their names as groups. For the RSPI interface, the AC portion of the electrical characteristics is measured for each group.

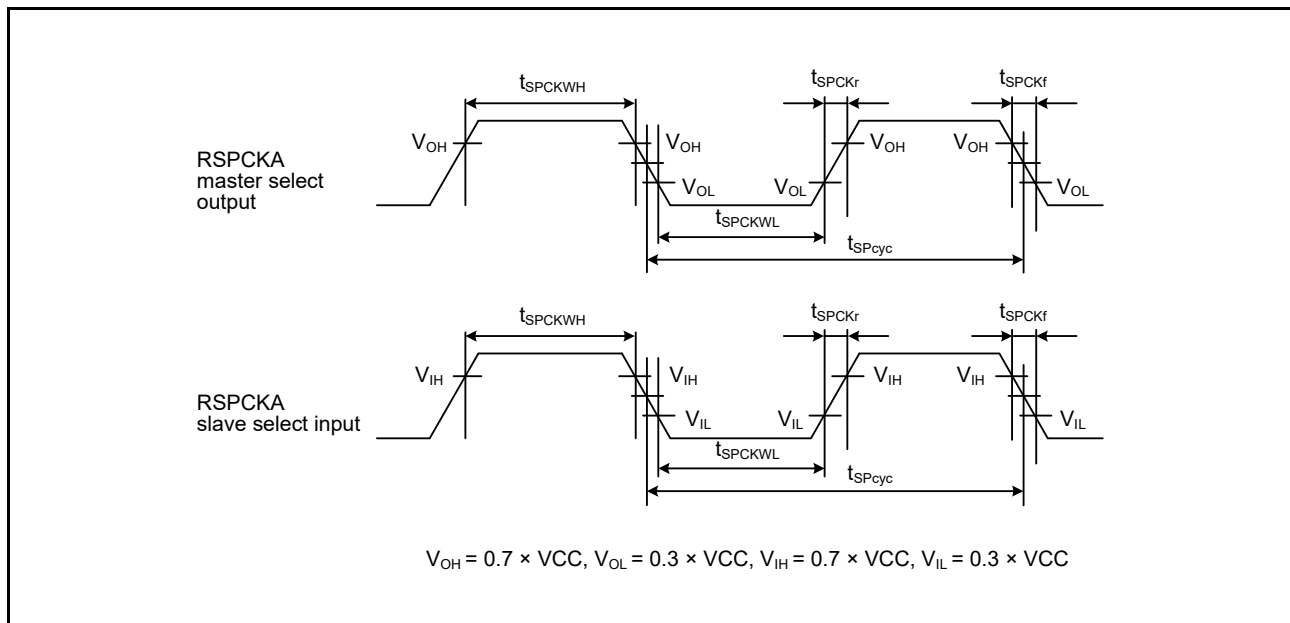


Figure 63.63 RSPI Clock Timing

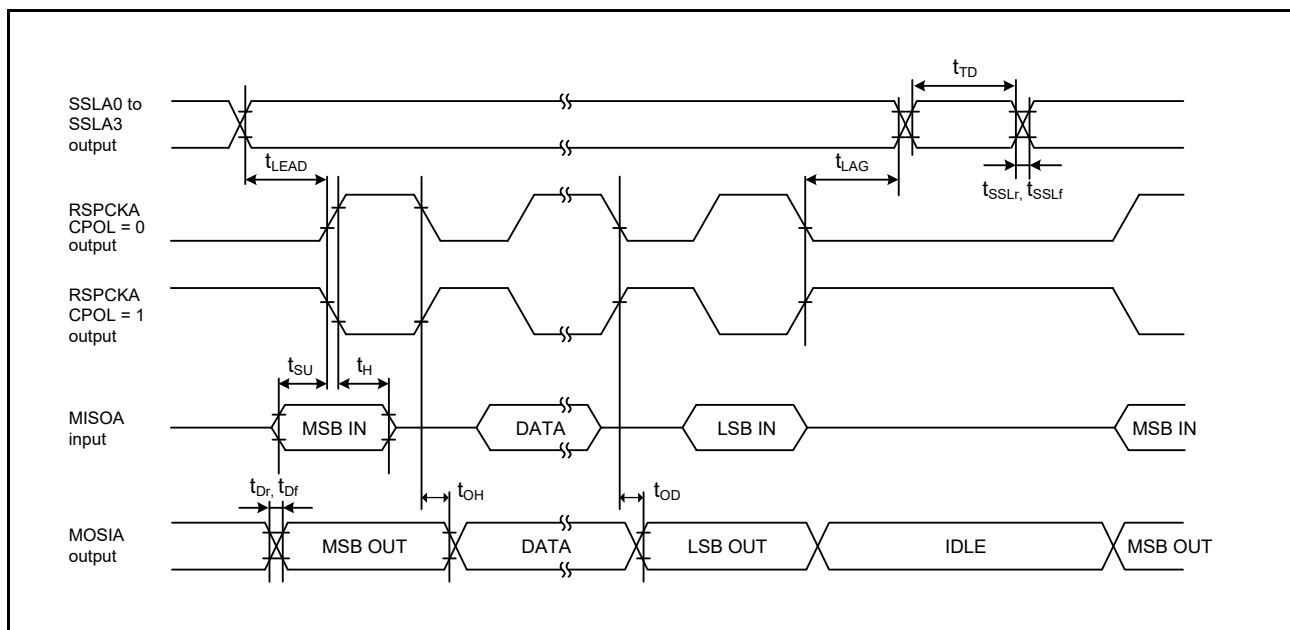


Figure 63.64 RSPI Timing (Master, CPHA = 0) (Bit Rate: PCLKA Division Ratio Set to a Value Other Than 1/2)

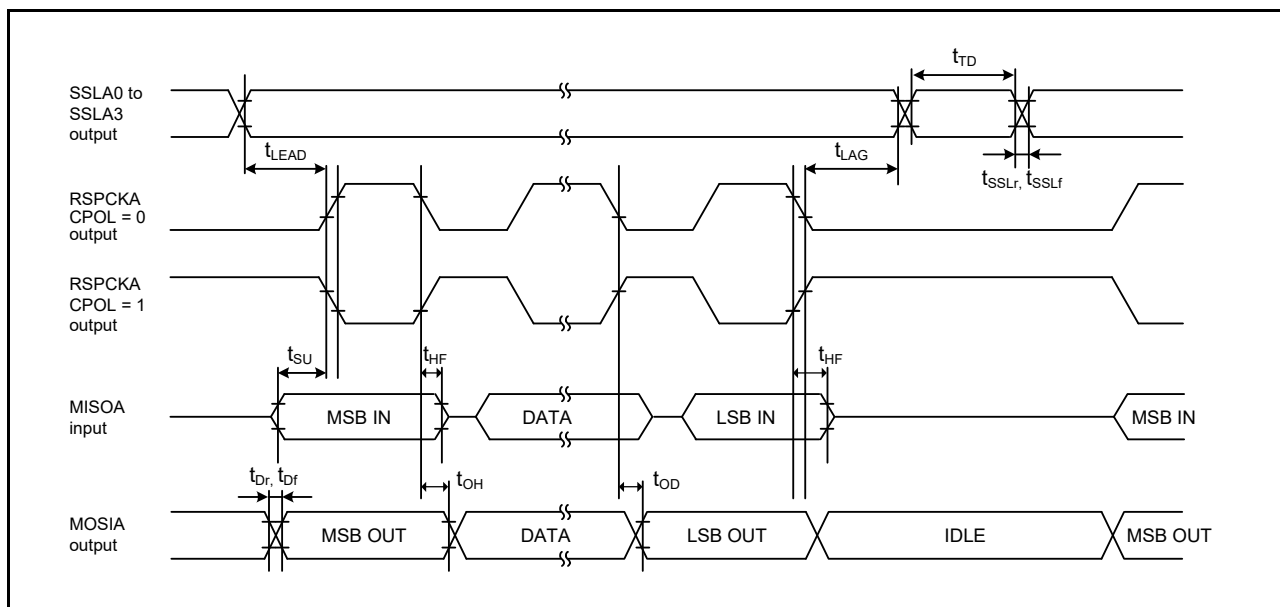


Figure 63.65 RSPI Timing (Master, CPHA = 0) (Bit Rate: PCLKA Division Ratio Set to 1/2)

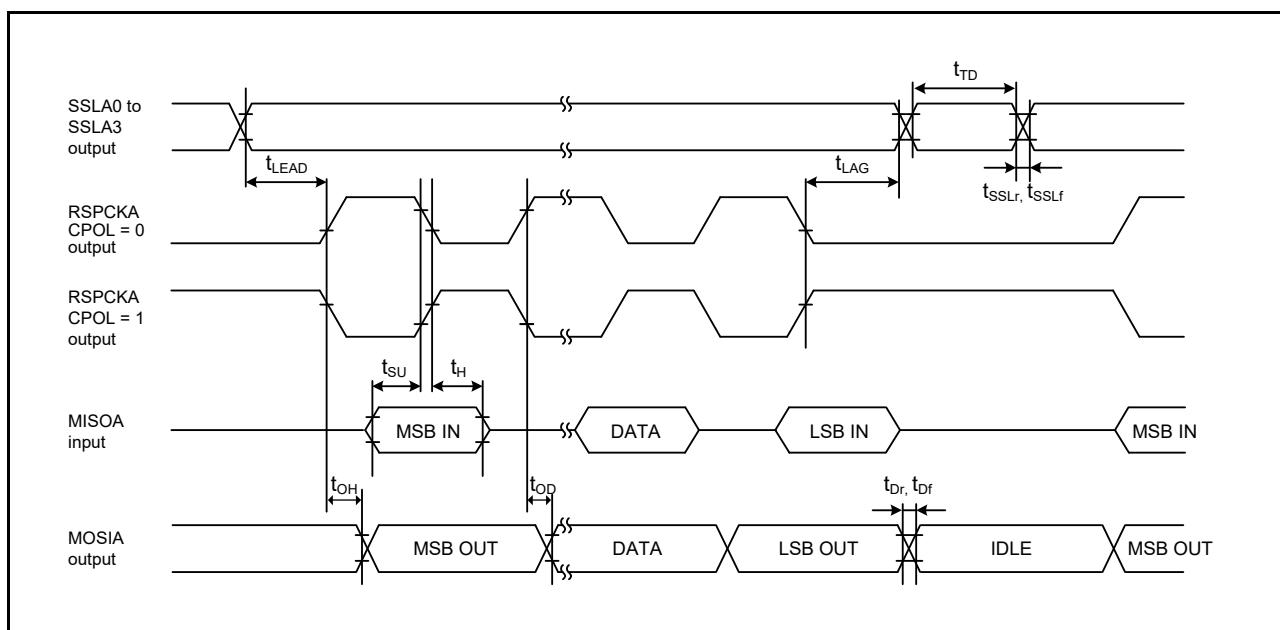


Figure 63.66 RSPI Timing (Master, CPHA = 1) (Bit Rate: PCLKA Division Ratio Set to a Value Other Than 1/2)

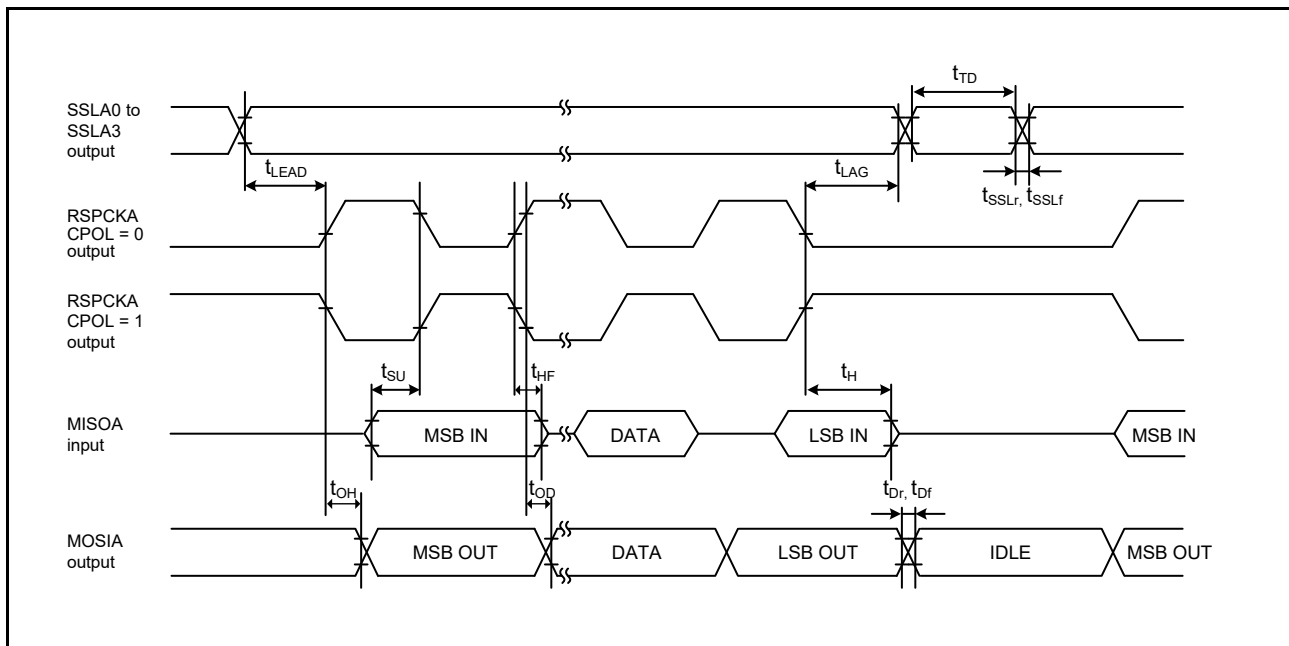


Figure 63.67 RSPI Timing (Master, CPHA = 1) (Bit Rate: PCLKA Division Ratio Set to 1/2)

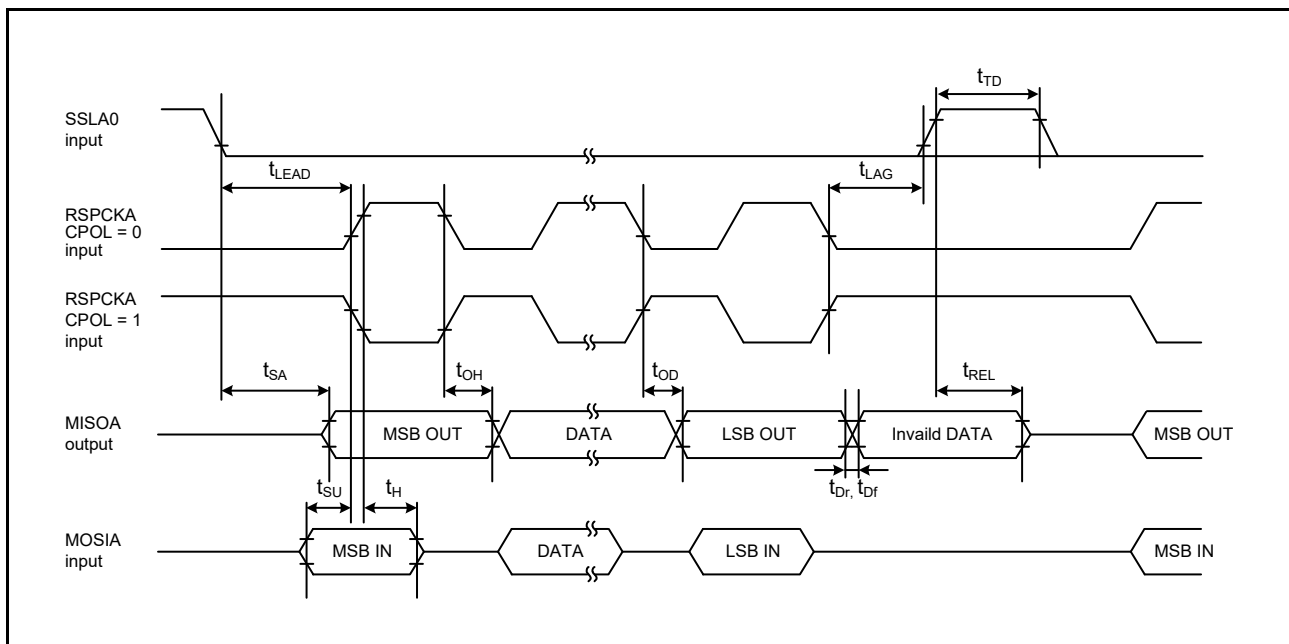


Figure 63.68 RSPI Timing (Slave, CPHA = 0)

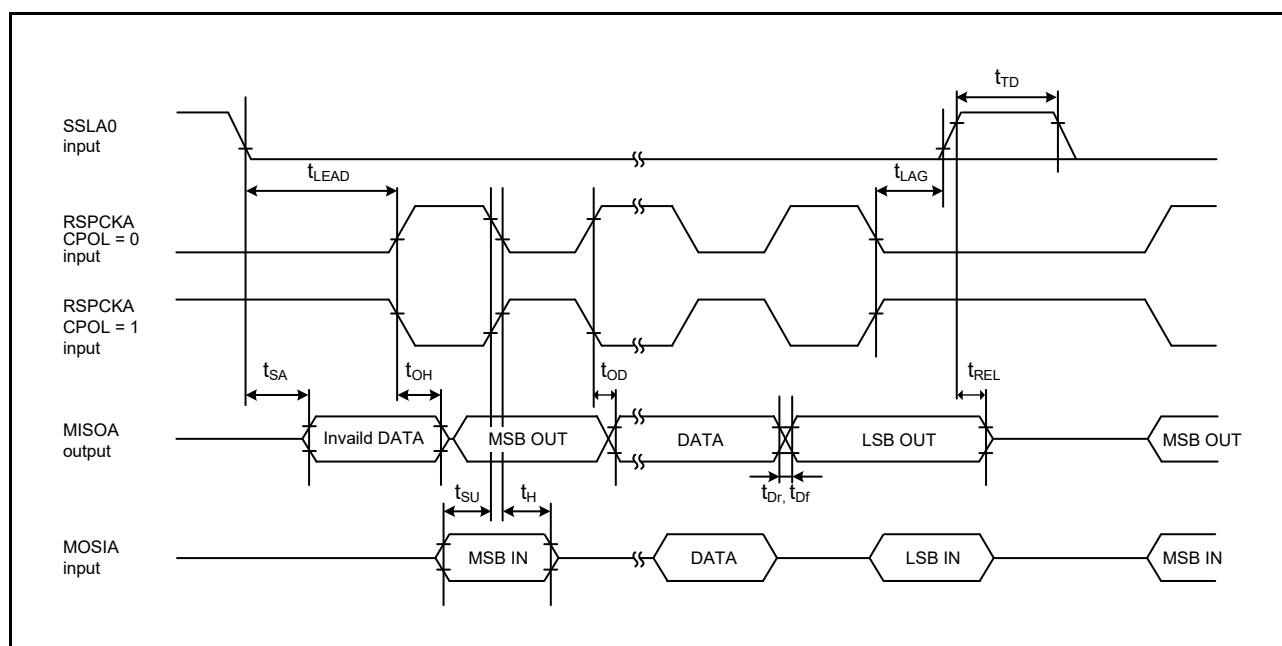


Figure 63.69 RSPI Timing (Slave, CPHA = 1)

63.4.7.12 QSPI

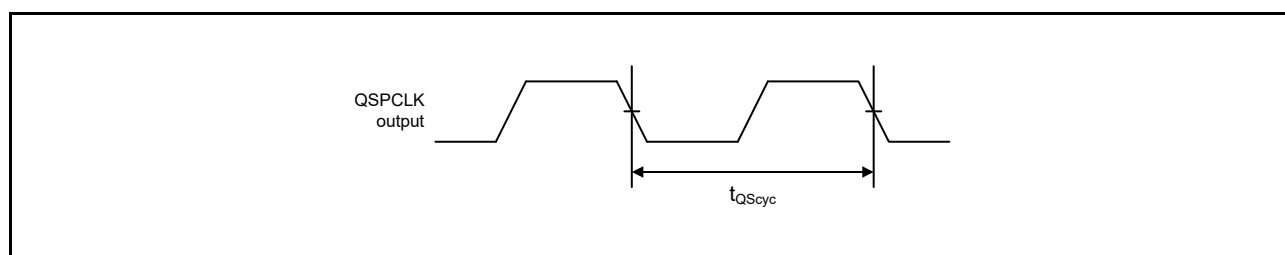
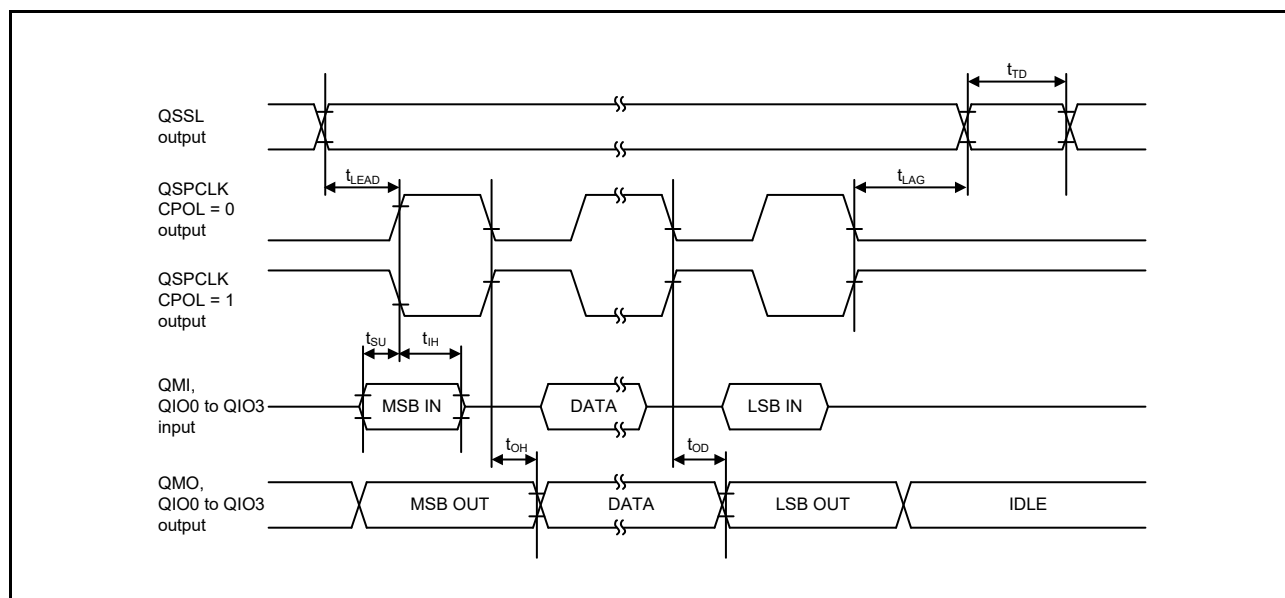
Table 63.41 QSPI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

	Item	Symbol	Min.	Max.	Unit*1	Test Conditions*2
QSPI	QSPCLK clock cycle	t_{QScyc}	2	4080	t_{PBcyc}	Figure 63.70
	Data input setup time	t_{Su}	6.5	—	ns	Figure 63.71, Figure 63.72
	Data input hold time	t_{IH}	5	—	ns	
	SS setup time	t_{LEAD}	1.5	8.5	t_{QScyc}	
	SS hold time	t_{LAG}	1	8	t_{QScyc}	
	Data output delay time	t_{OD}	—	10.0	ns	
	Data output hold time	t_{OH}	—5	—	ns	
	Successive transmission delay time	t_{TD}	1	8	t_{QScyc}	

Note 1. t_{PBcyc} : PCLKB cycle

Note 2. We recommend using pins that have a letter ("A", "B", etc.) to indicate group membership appended to their names as groups.
 For the QSPI interface, the AC portion of the electrical characteristics is measured for each group.

**Figure 63.70 QSPI Clock Timing****Figure 63.71 Transmit/Receive Timing (CPHA = 0)**

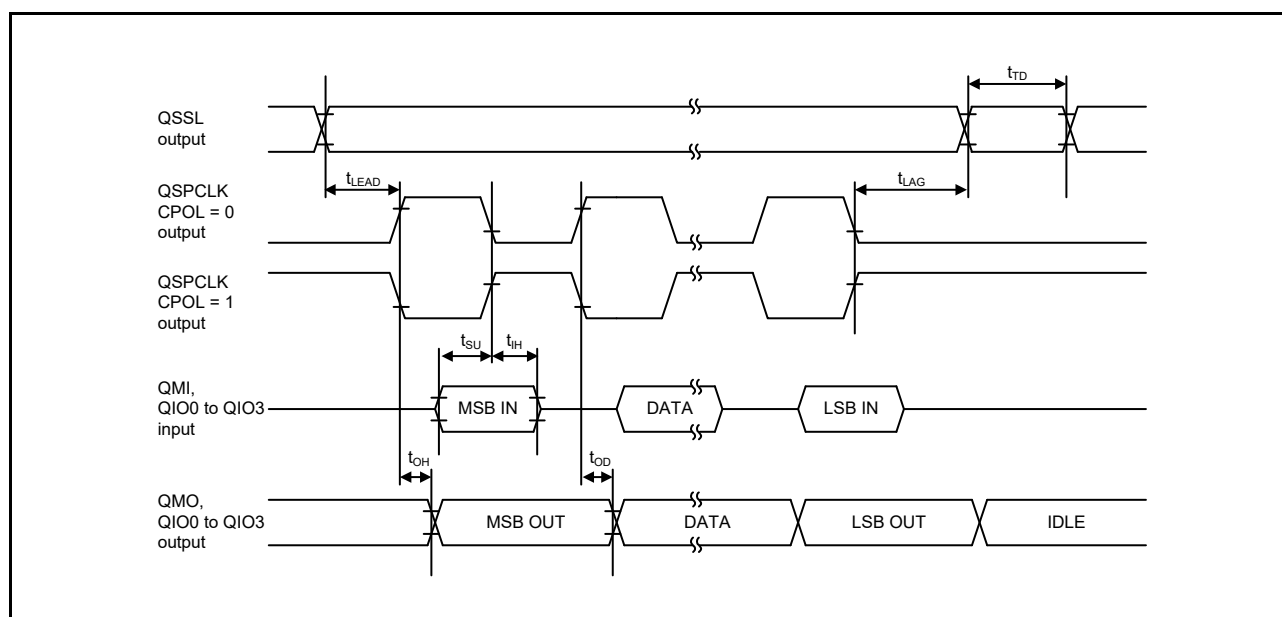


Figure 63.72 Transmit/Receive Timing (CPHA = 1)

63.4.7.13 SSIE

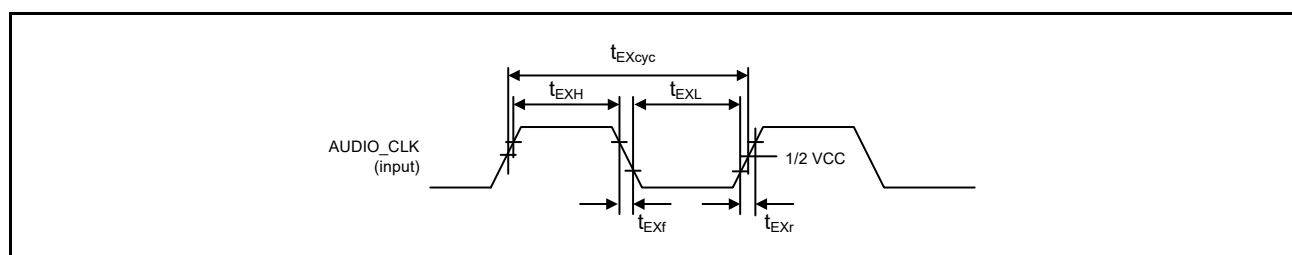
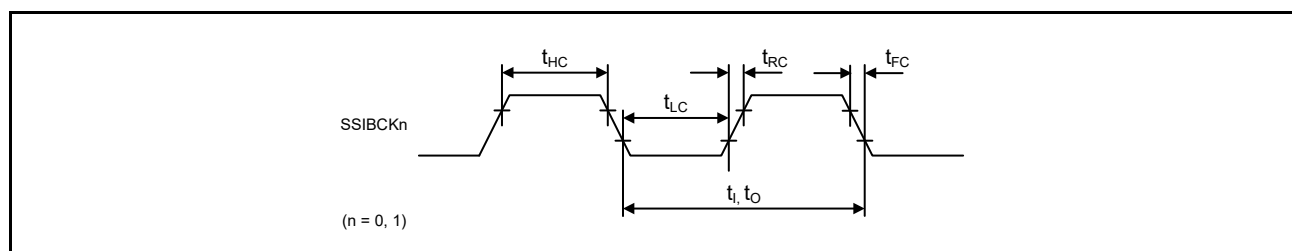
Table 63.42 Expansion Serial Sound Interface Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, $V_{REFH0} = 2.7$ V to AV_{CC0} ,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item			Symbol	Min.	Max.	Unit	Test Conditions
AUDIO_CLK	Cycle		t_{EXcyc}	20	—	ns	Figure 63.73
	High/low level		t_{EXL}/t_{EXH}	0.4	0.6	t_{EXcyc}	
SSIBCKn	Cycle	Master	t_O	80	—	ns	Figure 63.74
		Slave	t_I	80	—	ns	
	Output clock high level	Master	t_{HC}	0.35	—	t_O	
	Output clock low level		t_{LC}	0.35	—	t_O	
	Input clock high level	Slave	t_{HC}	0.35	—	t_I	
	Input clock low level		t_{LC}	0.35	—	t_I	
	Output clock rise time	Master	t_{RC}	—	0.15	t_O	
	Output clock fall time		t_{FC}	—	0.15	t_O	
	Input clock rise time	Slave	t_{RC}	—	0.15	t_I	
	Input clock fall time		t_{FC}	—	0.15	t_I	
SSILRCKn, SSITXD0, SSIRXD0, SSIDATA1	Input setup time	Master	t_{SR}	12	—	ns	Figure 63.75, Figure 63.76
		Slave		12	—	ns	
	Input hold time	Master	t_{HR}	8	—	ns	
		Slave		15	—	ns	
	Output delay time	Master	t_{DTR}	−10	5	ns	
		Slave		0	20	ns	
	Output delay time from when an SSILRCK0 signal is changed*1	Slave	t_{DTRW}	—	20	ns	Figure 63.77

n = 0, 1

Note 1. The SSIE has a single path for transmission in slave mode. To generate the data for transmission, the signals input through the SSILRCKn pin through the abovementioned path are used. After that, the data for transmission proceed to be used as the logical outputs to the SSITXD0 or SSIDATA1 pin.

**Figure 63.73 Clock Input Timing****Figure 63.74 SSIE Clock Input/Output Timing**

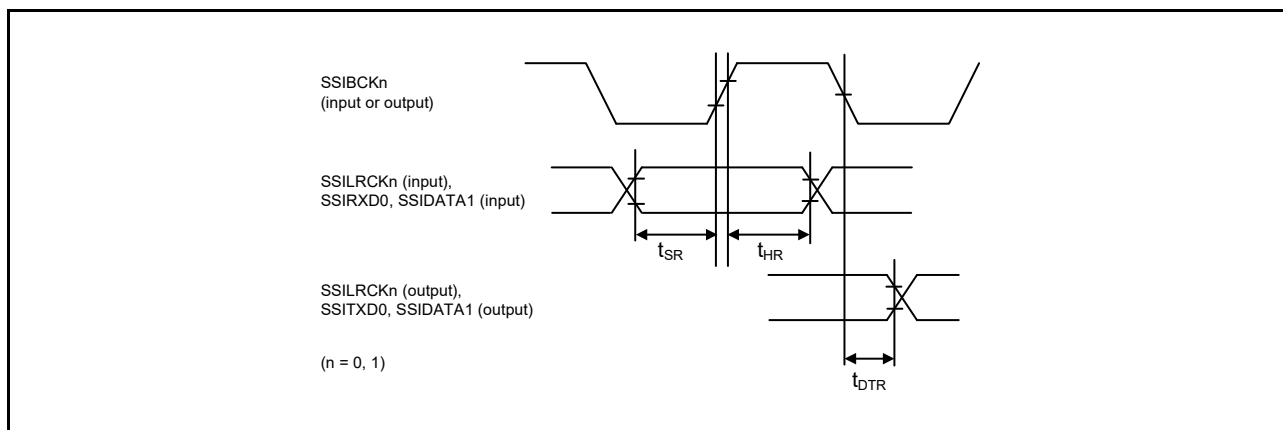


Figure 63.75 Transmission and Reception Timing for the SSIE Data When the SSICR.BCKP Bit is 0

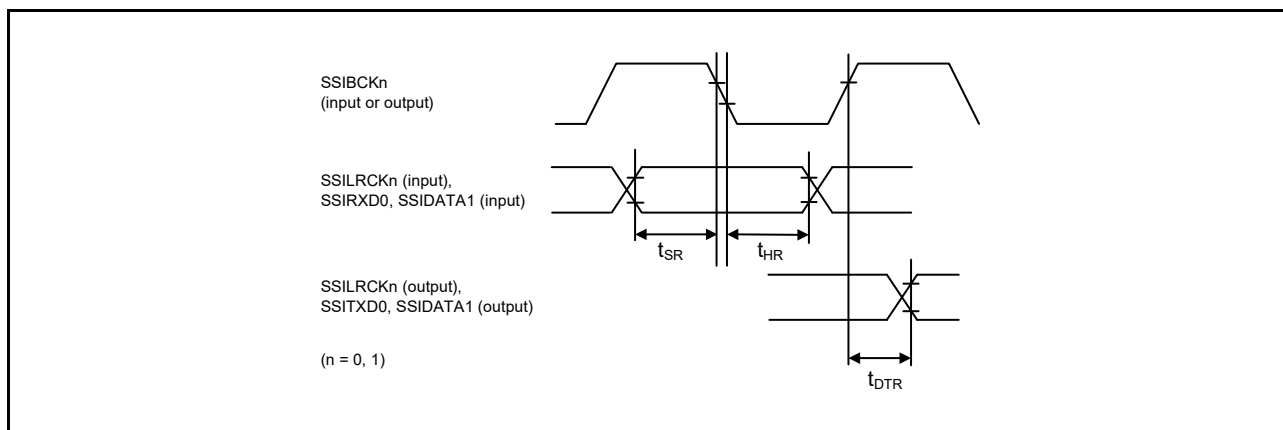


Figure 63.76 Transmission and Reception Timing for the SSIE Data When the SSICR.BCKP Bit is 1

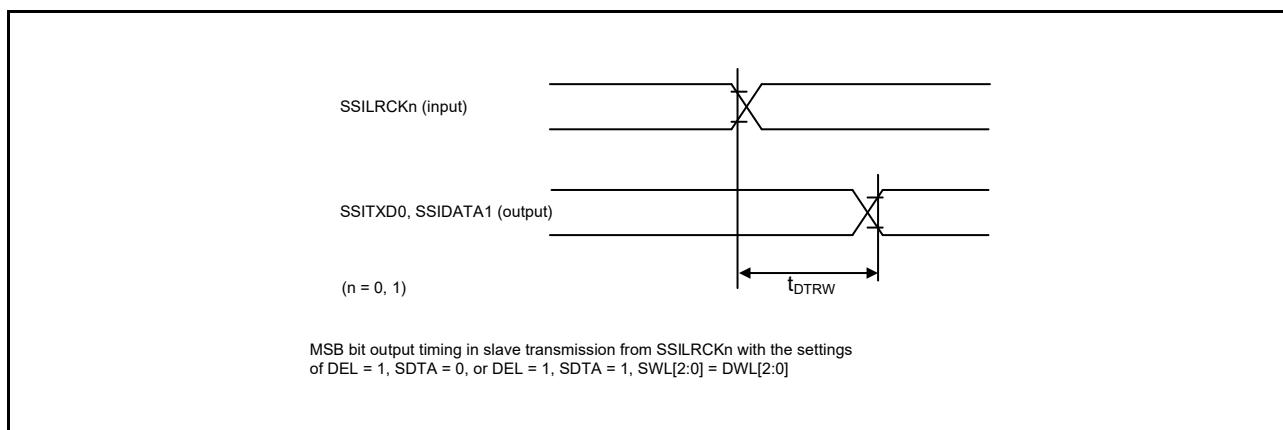


Figure 63.77 Output Delay of the SSIE Data from When an SSILRCKn Signal is Changed

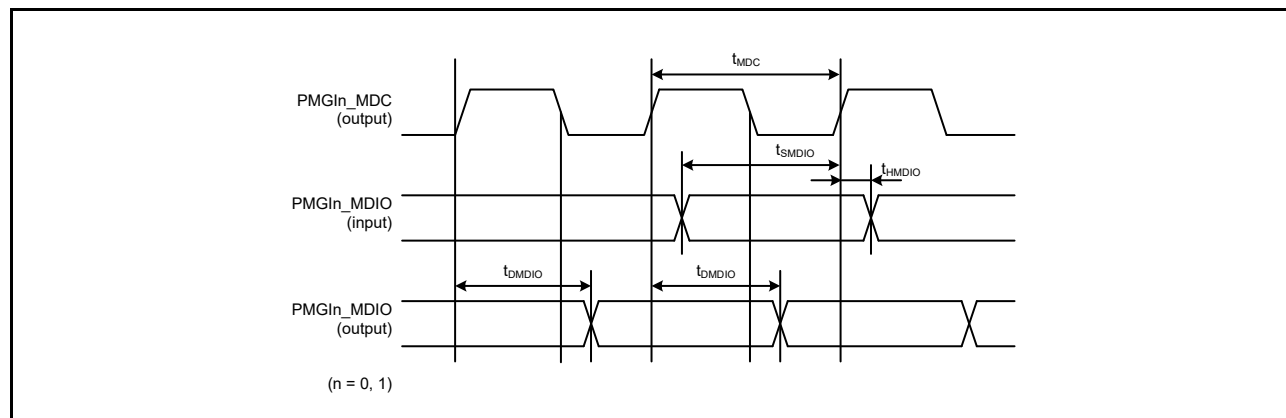
63.4.7.14 PMGI

Table 63.43 PMGI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $ICLK = PCLKA = 8$ to 120 MHz, $PCLKB = BCLK = SDCLK = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item		Symbol	Min.	Max.	Unit	Test Conditions
PMGI	PMGIn_MDC output cycle	t_{MDC}	80	—	ns	Figure 63.78
	PMGIn_MDIO setup time (relative to PMGIn_MDC $\uparrow$)	t_{SMDIO}	20	—	ns	
	PMGIn_MDIO hold time (relative to PMGIn_MDC $\uparrow$)	t_{HMDIO}	0	—	ns	
	PMGIn_MDIO output delay time (relative to PMGIn_MDC $\uparrow$)	t_{DMDIO}	0	20	ns	

$n = 0, 1$

**Figure 63.78 Timing of Serial Management Access**

63.4.7.15 MMC

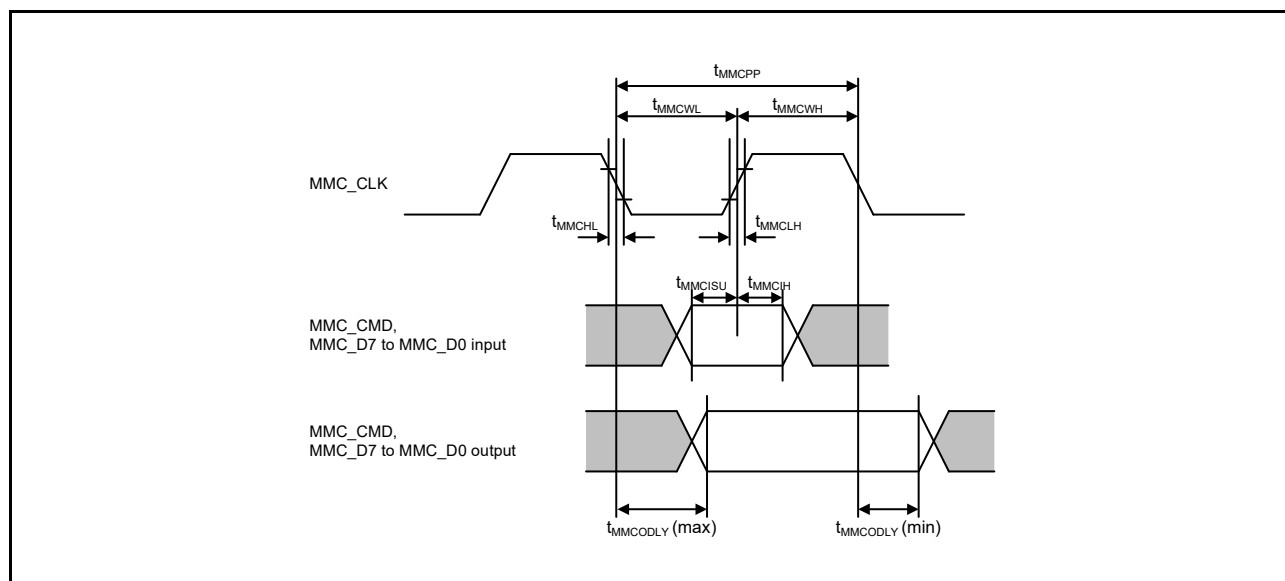
Table 63.44 MMC Host Interface Timing

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item	Symbol	Min.*1	Max.	Unit	Test Conditions*2
MMCIF	MMC_CLK clock cycle	t_{MMCPP}	$2 \times t_{PBcyc}$	ns	Figure 63.79
	MMC_CLK clock high level width	t_{MMCWH}	6.5	ns	
	MMC_CLK clock low level width	t_{MMCWL}	6.5	ns	
	MMC_CLK clock rising time	t_{MMCLH}	3	ns	
	MMC_CLK clock falling time	t_{MMCHL}	3	ns	
	MMC_CMD, MMC_D7 to MMC_D0 output data delay (data transfer mode)	t_{MMCODY}	-6.6	ns	
	MMC_CMD, MMC_D7 to MMC_D0 input data setup	t_{MMCISU}	8	ns	
	MMC_CMD, MMC_D7 to MMC_D0 input data hold	t_{MMCIH}	2.5	ns	

Note 1. t_{PBcyc} : PCLKB cycle

Note 2. We recommend using pins that have a letter ("A", "B", etc.) to indicate group membership appended to their names as groups.
 For the MMC interface, the AC portion of the electrical characteristics is measured for each group.

**Figure 63.79 MMC Interface**

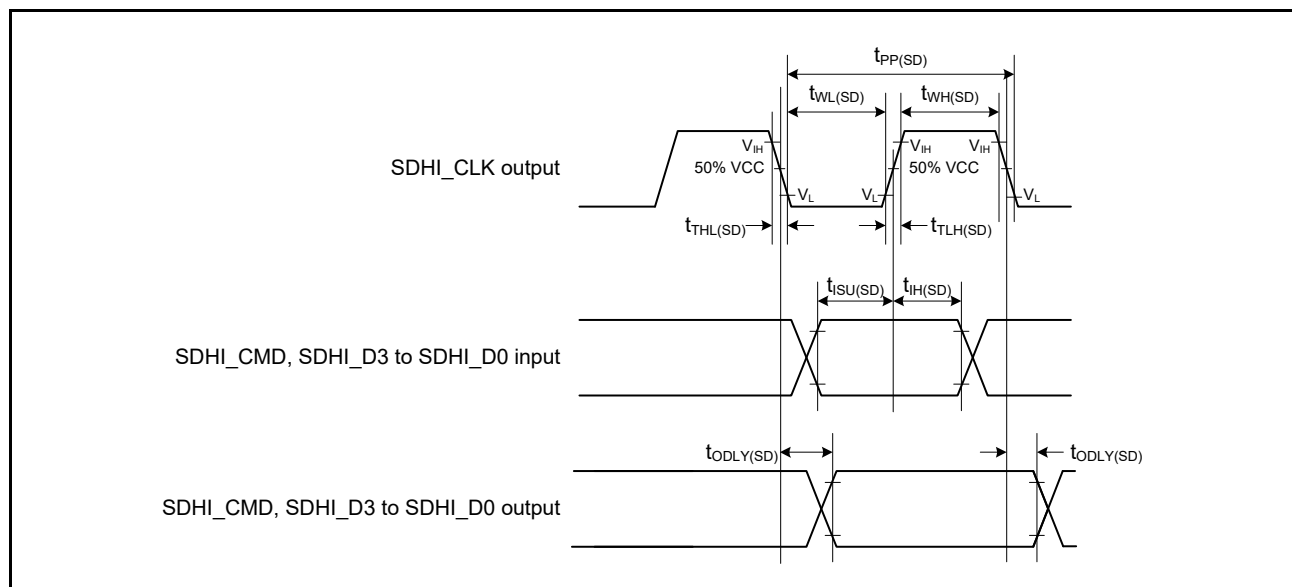
63.4.7.16 SDHI

Table 63.45 SDHI Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to $3.6V$, $2.7V \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0V$,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30pF$
 High-drive output is selected by the drive capacity control register.

	Item	Symbol	Min.	Max.	Unit	Test Conditions*1
SDHI	SDHI_CLK output cycle time	$t_{PP(SD)}$	20	—	ns	Figure 63.80
	SDHI_CLK output width at high level	$t_{WH(SD)}$	$0.4 \times t_{PP(SD)}$	—	ns	
	SDHI_CLK output width at low level	$t_{WL(SD)}$	$0.4 \times t_{PP(SD)}$	—	ns	
	SDHI_CLK output rising time	$t_{TLH(SD)}$	—	3	ns	
	SDHI_CLK output falling time	$t_{THL(SD)}$	—	3	ns	
	SDHI_CMD, SDHI_D3 to SDHI_D0 output data delay (data transfer mode)	$t_{ODLY(SD)}$	-6.5	4	ns	
	SDHI_CMD, SDHI_D3 to SDHI_D0 input data setup time	$t_{SU(SD)}$	6	—	ns	
	SDHI_CMD, SDHI_D3 to SDHI_D0 input data hold time	$t_{IH(SD)}$	2	—	ns	

Note 1. We recommend using pin names that have a letter ("A", "B", etc.) to indicate group membership per group in the test.
 For the SDHI, the AC portion of the electrical characteristics is measured per group.

**Figure 63.80 SD Host Interface Input/Output Signal Timing**

63.4.7.17 ETHERC

Table 63.46 ETHERC Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
RMII: High-drive output for the high-speed interface is selected in the drive capacity control register.
MII: High-drive output is selected by the drive capacity control register.

	Item	Symbol	Min.	Max.	Unit	Test Conditions
ETHERC (RMII)	REF50CK cycle time	T_{ck}	20	—	ns	Figure 63.81 to Figure 63.84
	REF50CK frequency Typ. 50 MHz	—	—	50 + 100 ppm	MHz	
	REF50CK duty	—	35	65	%	
	REF50CK rise/fall time	$T_{ckr/ckf}$	0.5	3.5	ns	
	RMII _n _xxxx*1 output delay time	T_{co}	2.5	15.0	ns	
	RMII _n _xxxx*2 setup time	T_{su}	3	—	ns	
	RMII _n _xxxx*2 hold time	T_{hd}	1	—	ns	
	RMII _n _xxxx*1, *2 rise/fall time	T_r/T_f	—	5	ns	
	ETn_WOL output delay time	t_{WOLd}	1	23.5	ns	Figure 63.85
ETHERC (MII)	ETn_TX_CLK cycle time	t_{Tcyc}	40	—	ns	—
	ETn_TX_EN output delay time	t_{TEND}	1	20	ns	Figure 63.86
	ETn_ETXD0 to ETn_ETXD3 output delay time	t_{MTDd}	1	20	ns	
	ETn_CRS setup time	t_{CRSs}	10	—	ns	
	ETn_CRS hold time	t_{CRSh}	10	—	ns	
	ETn_COL setup time	t_{COLs}	10	—	ns	Figure 63.87
	ETn_COL hold time	t_{COLh}	10	—	ns	
	ETn_RX_CLK cycle time	t_{TRcyc}	40	—	ns	—
	ETn_RX_DV setup time	t_{RDVs}	10	—	ns	Figure 63.88
	ETn_RX_DV hold time	t_{RDVh}	10	—	ns	
	ETn_ERXD0 to ETn_ERXD3 setup time	t_{MRDs}	10	—	ns	
	ETn_ERXD0 to ETn_ERXD3 hold time	t_{MRDh}	10	—	ns	
	ETn_RX_ER setup time	t_{RERs}	10	—	ns	Figure 63.89
	ETn_RX_ER hold time	t_{RERh}	10	—	ns	
	ETn_WOL output delay time	t_{WOLd}	1	23.5	ns	Figure 63.90

n = 0, 1

Note 1. RMII_n_TXD_EN, RMII_n_TXD1, RMII_n_TXD0

Note 2. RMII_n_CRS_DV, RMII_n_RXD1, RMII_n_RXD0, RMII_n_RX_ER

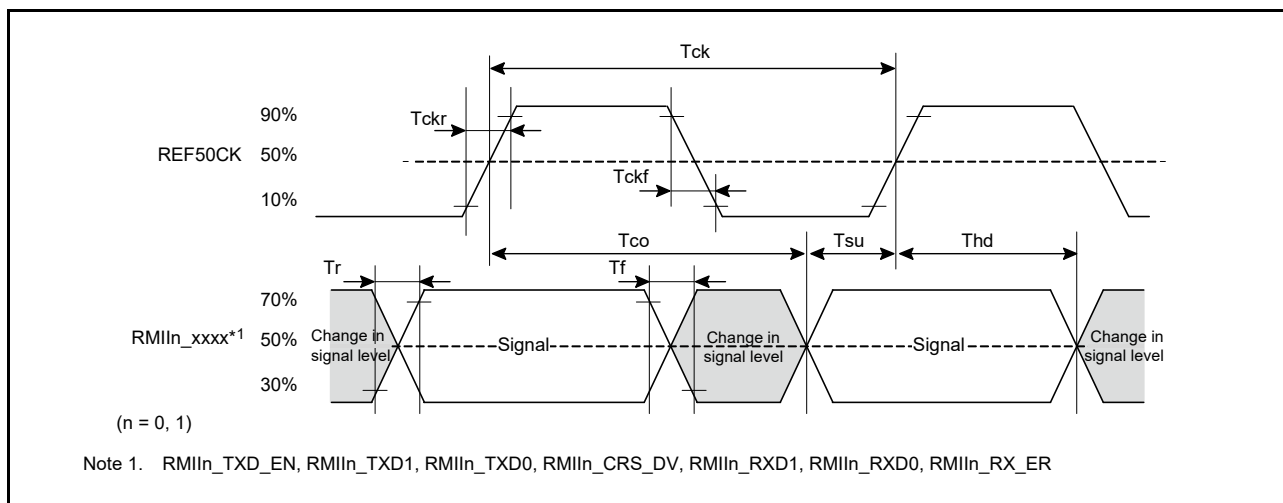


Figure 63.81 Timing with the REF50CK and RMII Signals

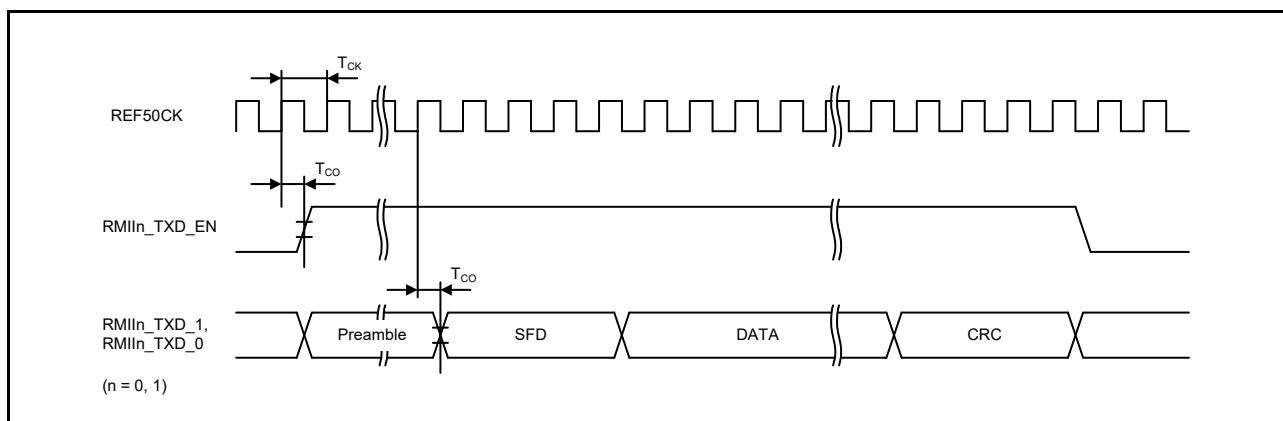


Figure 63.82 RMII Transmission Timing

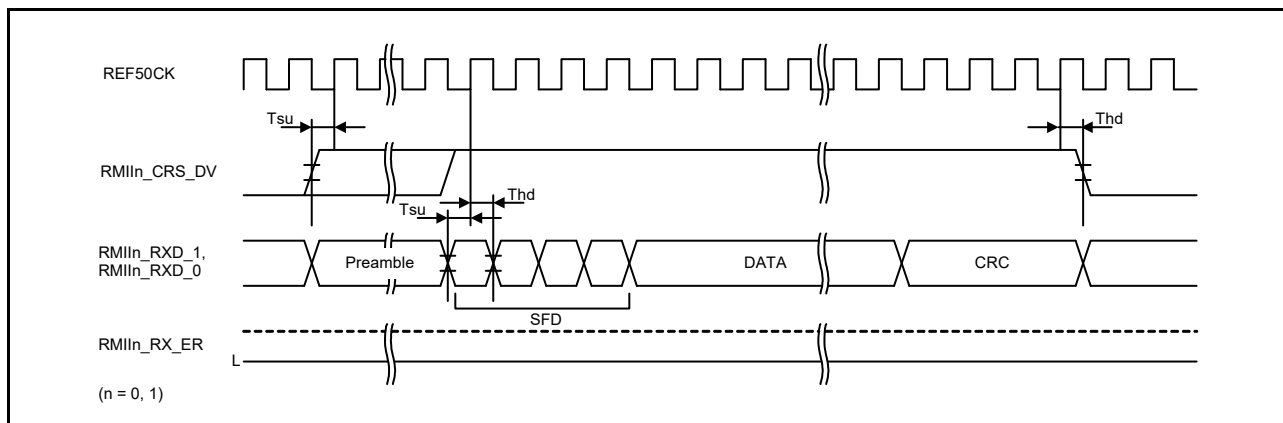


Figure 63.83 RMII Reception Timing (Normal Operation)

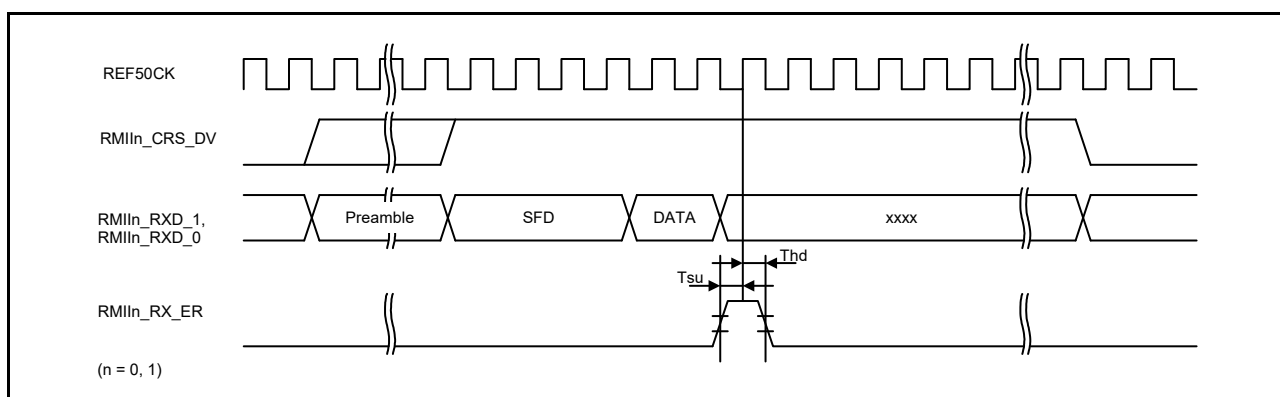


Figure 63.84 RMII Reception Timing (Error Occurrence)

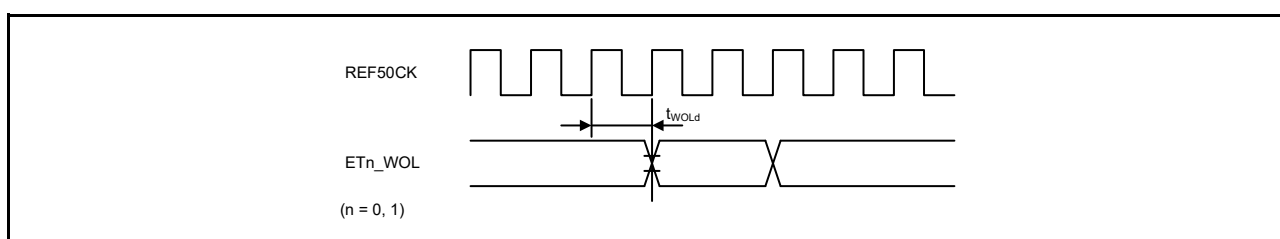


Figure 63.85 WOL Output Timing (RMII)

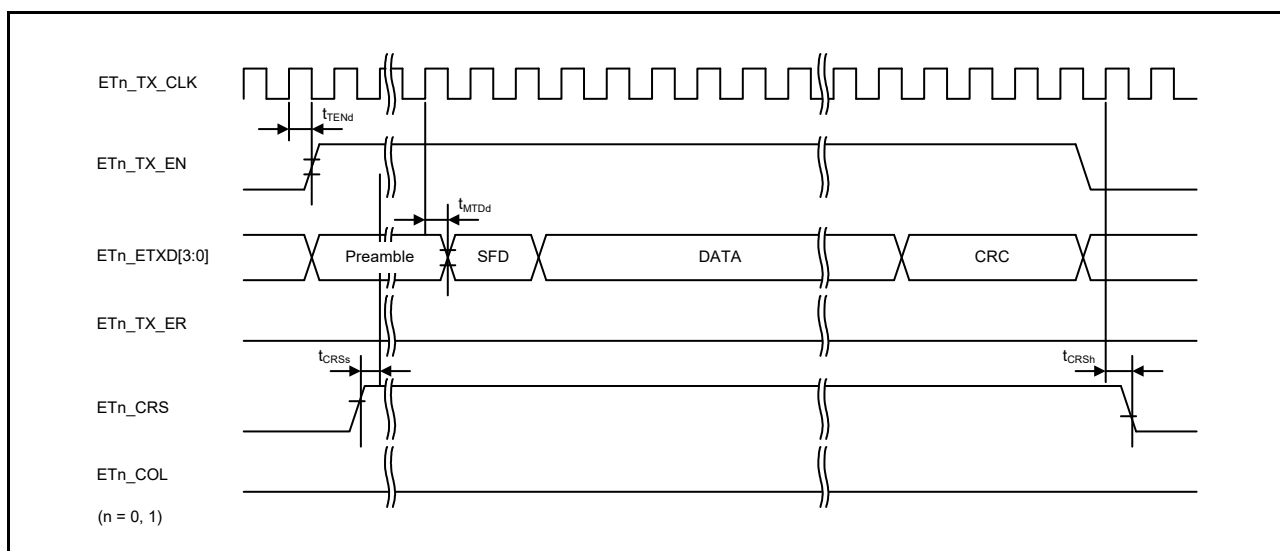


Figure 63.86 MII Transmission Timing (Normal Operation)

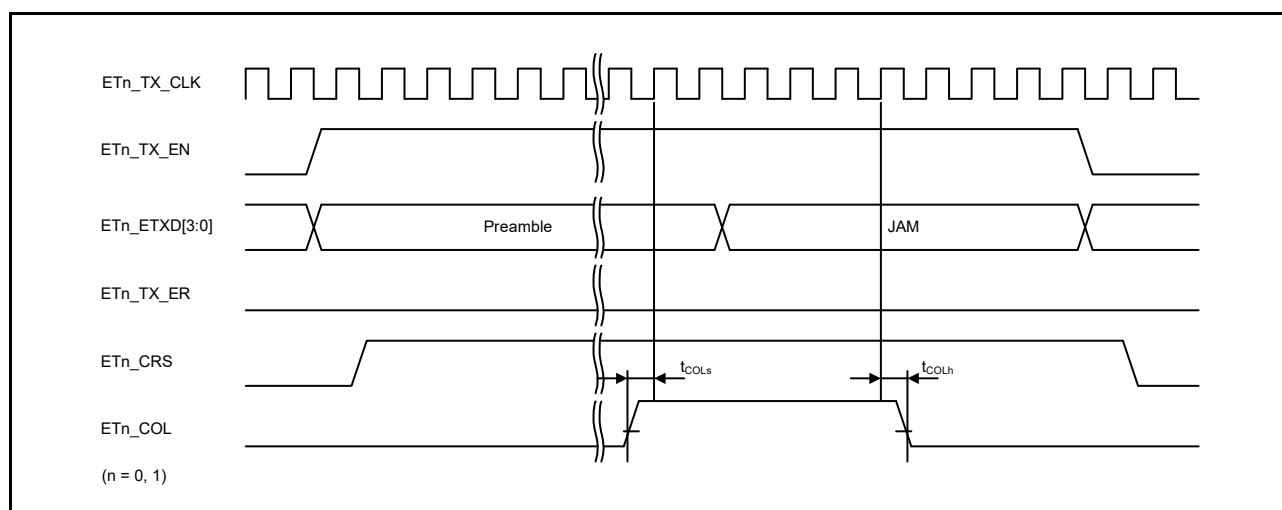


Figure 63.87 MII Transmission Timing (Conflict Occurrence)

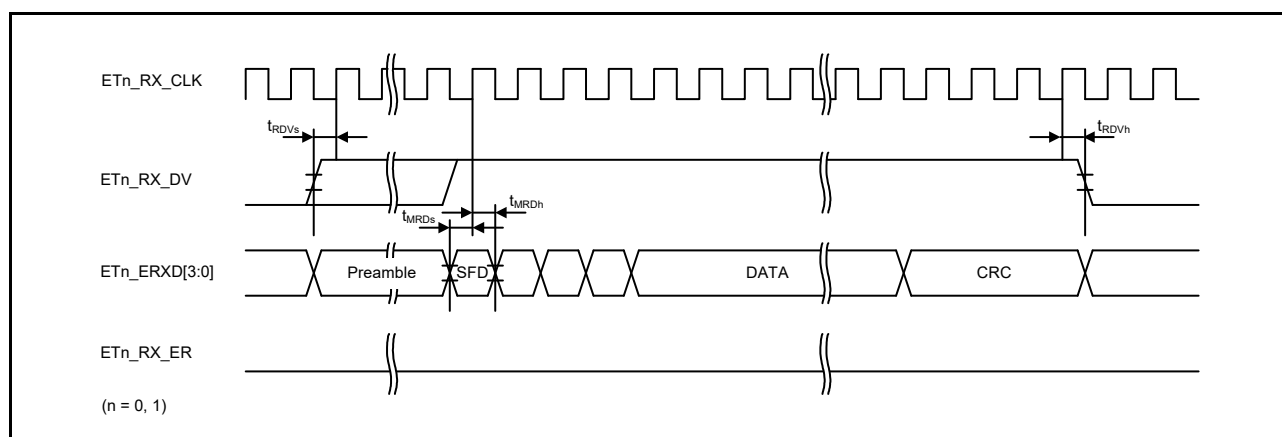


Figure 63.88 MII Reception Timing (Normal Operation)

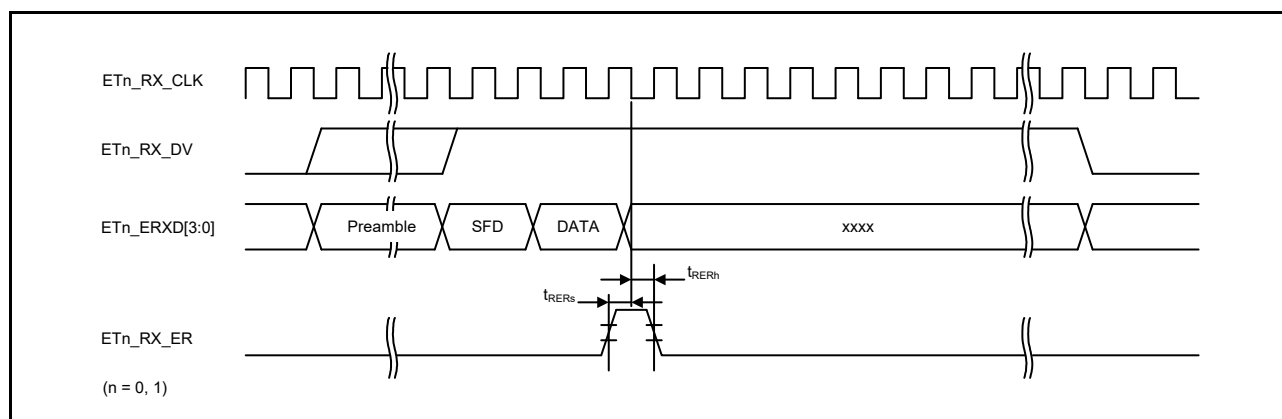
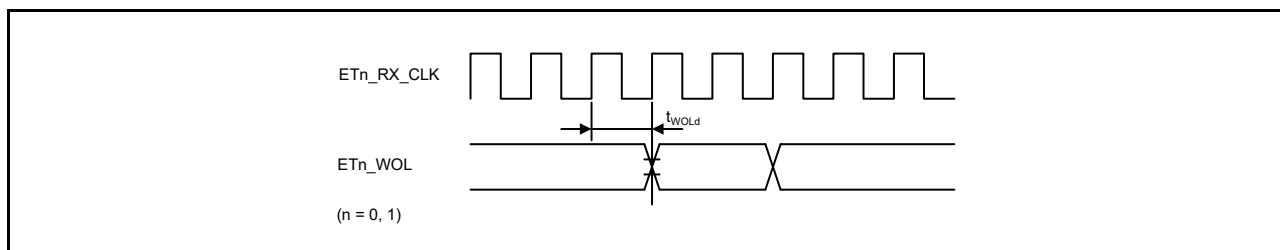


Figure 63.89 MII Reception Timing (Error Occurrence)

**Figure 63.90 WOL Output Timing (MII)**

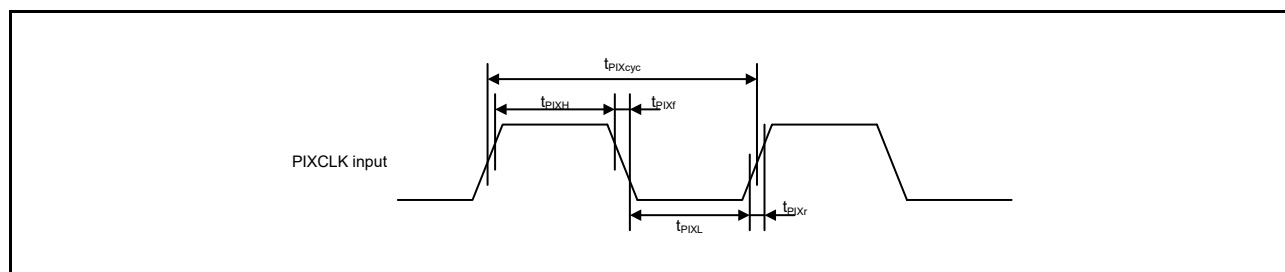
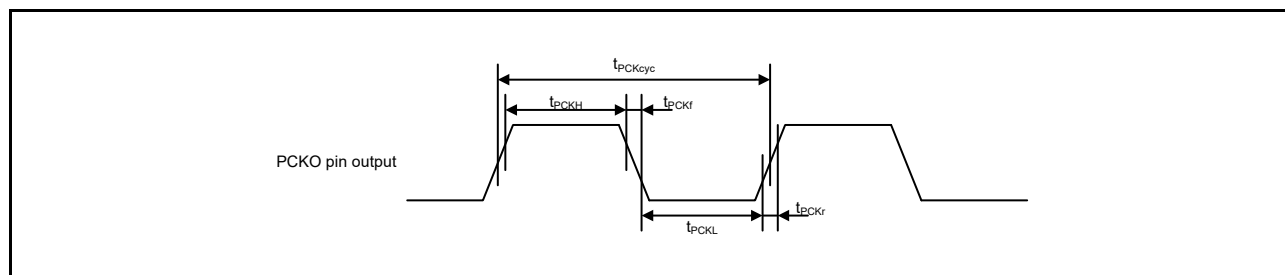
63.4.7.18 PDC

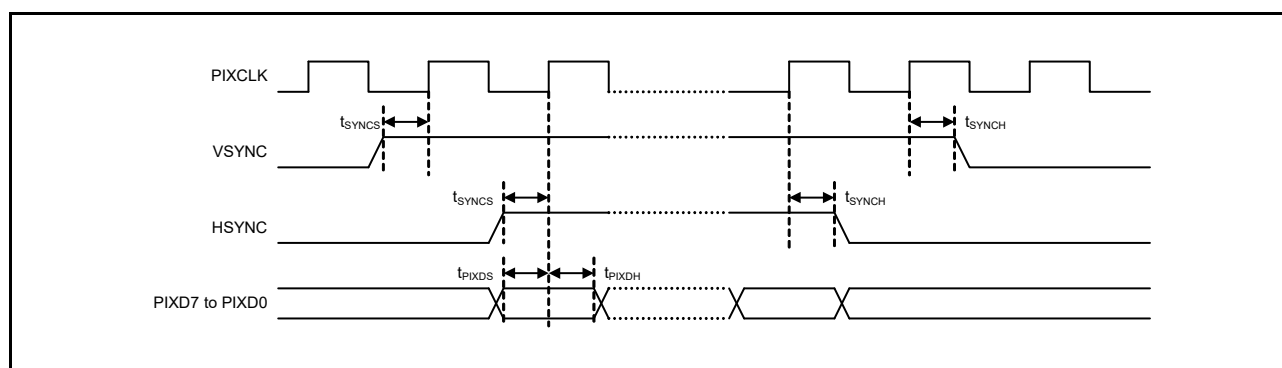
Table 63.47 PDC Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item	Symbol	Min.*1	Max.	Unit	Test Conditions
PDC	PIXCLK input cycle time	t_{PIXcyc}	37	—	Figure 63.91
	PIXCLK input high pulse width	t_{PIXH}	10	—	
	PIXCLK input low pulse width	t_{PIXL}	10	—	
	PIXCLK rising time	t_{PIXr}	5	ns	
	PIXCLK falling time	t_{PIXf}	5	ns	
	PCKO output cycle time	t_{PCKcyc}	$2 \times t_{PBcyc}$	—	Figure 63.92
	PCKO output high pulse width	t_{PCKH}	$(t_{PCKcyc} - t_{PCKr} - t_{PCKf})/2 - 3$	—	
	PCKO output low pulse width	t_{PCKL}	$(t_{PCKcyc} - t_{PCKr} - t_{PCKf})/2 - 3$	—	
	PCKO rising time	t_{PCKr}	5	ns	
	PCKO falling time	t_{PCKf}	5	ns	
	VSYNC/HSYNC input setup time	t_{SYNCS}	10	—	Figure 63.93
	VSYNC/HSYNC input hold time	t_{SYNCH}	5	—	
	PIXD input setup time	t_{PIXDS}	10	—	
	PIXD input hold time	t_{PIXDH}	5	—	

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.91 PDC Input Clock Timing****Figure 63.92 PDC Output Clock Timing**

**Figure 63.93 PDC AC Timing**

63.4.7.19 GLCDC

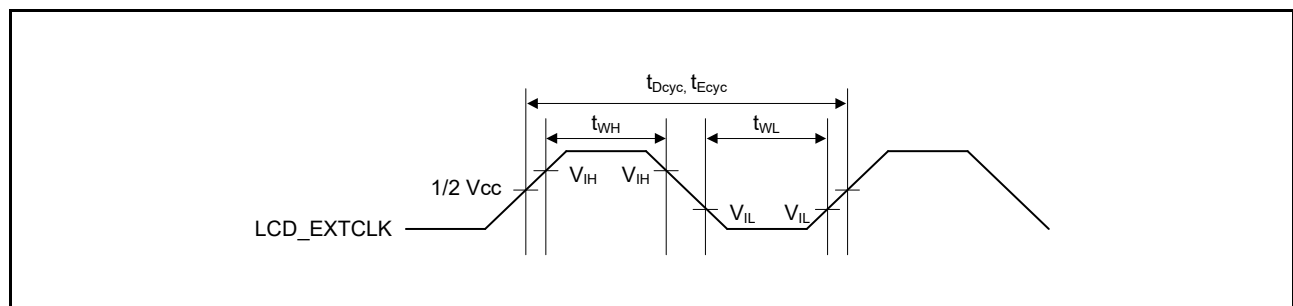
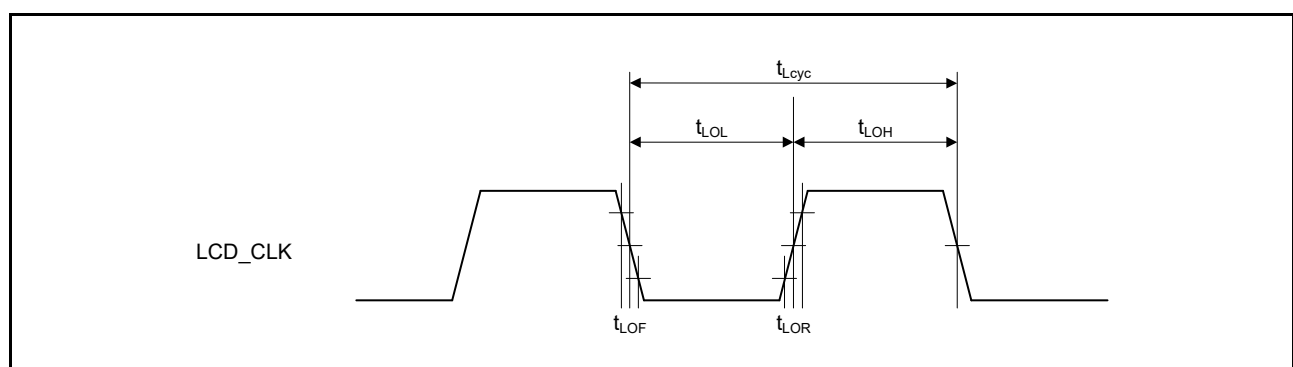
Table 63.48 GLCDC Timing

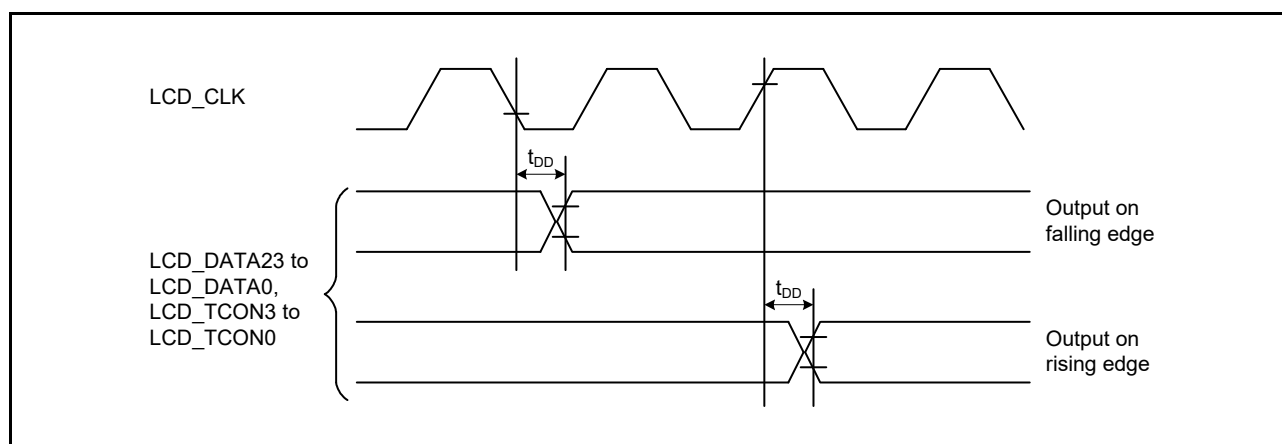
Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
LCD_EXTCLK Input clock frequency	$t_{E_{cyc}}$	—	—	30*1	MHz	Figure 63.94
LCD_EXTCLK Input clock Low pulse width	t_{WL}	0.45	—	0.55	$t_{E_{cyc}}$	
LCD_EXTCLK Input clock High pulse width	t_{WH}	0.45	—	0.55	$t_{E_{cyc}}$	
LCD_CLK Output clock frequency	$t_{L_{cyc}}$	—	—	30*1	MHz	Figure 63.95
LCD_CLK Output clock Low pulse width	t_{LOL}	0.4	—	0.6	$t_{L_{cyc}}$	
LCD_CLK Output clock High pulse width	t_{LOH}	0.4	—	0.6	$t_{L_{cyc}}$	
LCD_CLK Output clock rise time	t_{LOR}	—	—	5	ns	
LCD_CLK Output clock fall time	t_{LOF}	—	—	5	ns	
LCD data output Delay timing	t_{DD}	-3.5^*2	—	4^*2	ns	Figure 63.96

Note 1. Parallel RGB888,666,565: Max. 27 MHz
 Serial RGB888: Max. 30 MHz (4x speed)

Note 2. We recommend using pins that have a letter ("A", "B", etc) to indicate group membership appended to their names as groups.
 For the GLCDC interface, the AC portion of the electrical characteristics is measured for each group.
 If we use group "A" and "B" combination, "LCD data output Delay timing (t_{DD})" is Min = -5.0 ns, Max = 5.5 ns.

**Figure 63.94 LCD_EXTCLK Clock Input Timing****Figure 63.95 LCD_CLK Clock Output Timing**

**Figure 63.96** LCD Output Data Timing

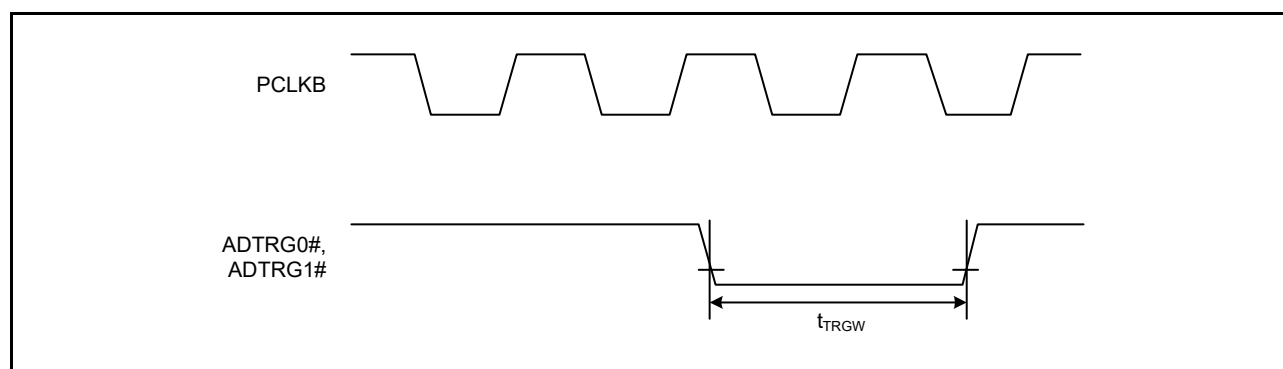
63.4.7.20 A/D Converter Trigger

Table 63.49 A/D Converter Trigger Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item		Symbol	Min.	Max.	Unit*1	Test Conditions
A/D converter	A/D converter trigger input pulse width	t_{TRGW}	1.5	—	t_{PBcyc}	Figure 63.97

Note 1. t_{PBcyc} : PCLKB cycle

**Figure 63.97 A/D Converter Trigger Input Timing**

63.4.7.21 CAC

Table 63.50 CAC Timing

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKA = 8$ to 120 MHz, $PCLKB = 8$ to 60 MHz, $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item*1, *2			Symbol	Min.*1, *2	Max.	Unit	Test Conditions
CAC	CACREF input pulse width	$t_{PBcyc} \leq t_{cac}$	t_{CACREF}	$4.5 t_{cac} + 3 t_{PBcyc}$	—	ns	
		$t_{PBcyc} > t_{cac}$		$5 t_{cac} + 6.5 t_{PBcyc}$	—		

Note 1. t_{PBcyc} : PCLKB cycle

Note 2. t_{cac} : CAC count clock source cycle

63.5 USB Characteristics

Table 63.51 On-Chip USB Low Speed (Host Only) Characteristics (DP and DM Pin Characteristics)

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 3.0$ to 3.6 V, 3.0 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $U_{CLK} = 48$ MHz, $P_{CLKA} = 8$ to 120 MHz,
 $P_{CLKB} = 8$ to 60 MHz, $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Input characteristics	Input high level voltage	V_{IH}	2.0	—	—	V
	Input low level voltage	V_{IL}	—	—	0.8	V
	Differential input sensitivity	V_{DI}	0.2	—	—	V DP – DM
	Differential common mode range	V_{CM}	0.8	—	2.5	V
Output characteristics	Output high level voltage	V_{OH}	2.8	—	3.6	V $I_{OH} = -200$ μ A
	Output low level voltage	V_{OL}	0.0	—	0.3	V $I_{OL} = 2$ mA
	Cross-over voltage	V_{CRS}	1.3	—	2.0	V Figure 63.98
	Rise time	t_{LR}	75	—	300	ns
	Fall time	t_{LF}	75	—	300	ns
	Rise/fall time ratio	t_{LR} / t_{LF}	80	—	125	% t_{LR} / t_{LF}
Pull-down characteristics	DP/DM pull-down resistance (when the host controller function is selected)	R_{pd}	14.25	—	24.80	k Ω

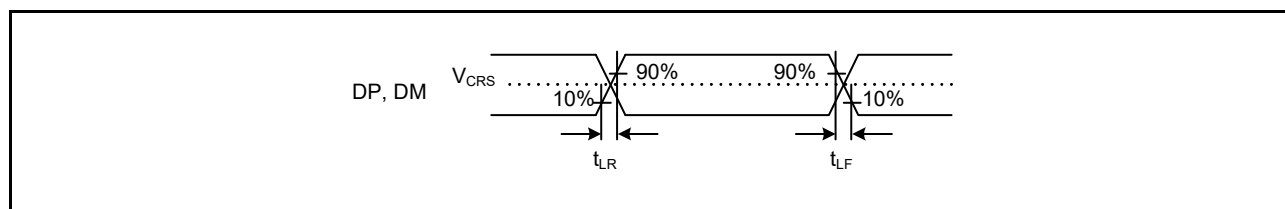
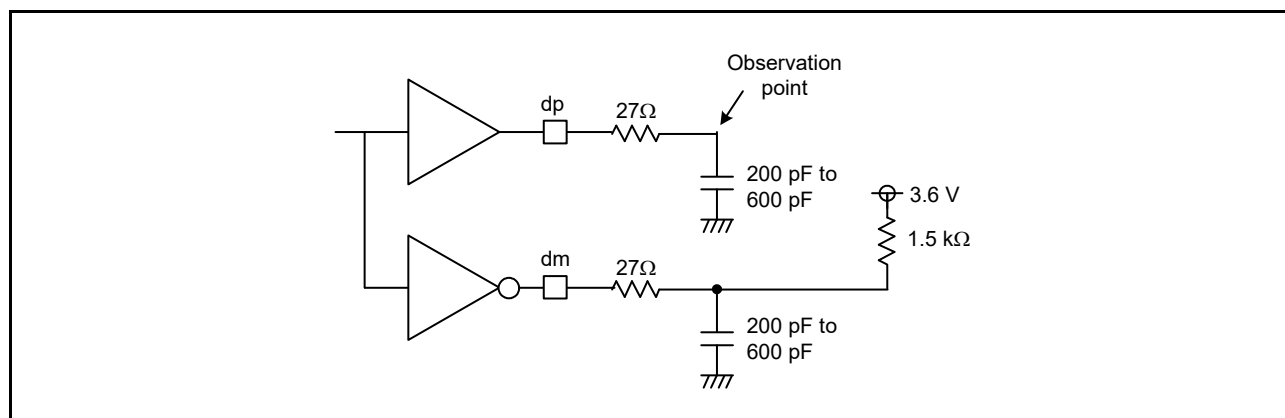
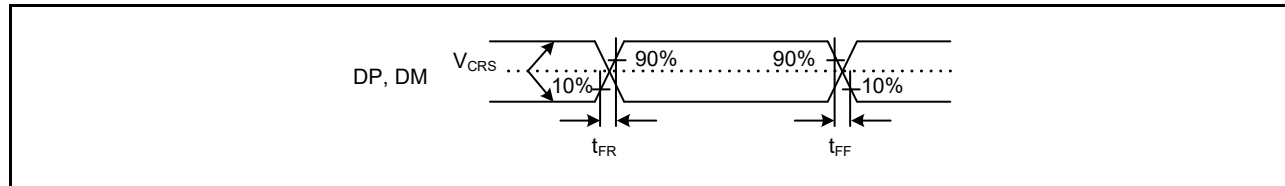
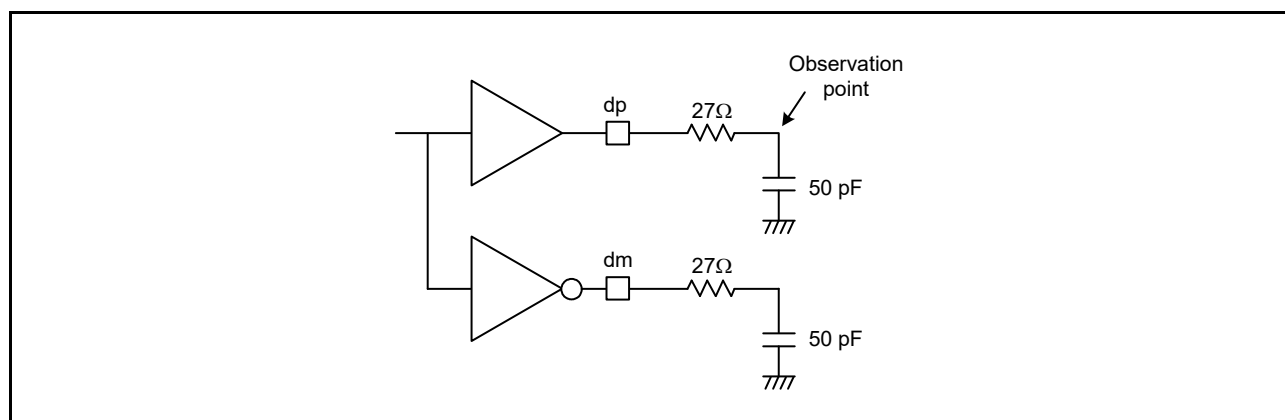
**Figure 63.98 DP and DM Output Timing (Low Speed)****Figure 63.99 Test Circuit (Low Speed)**

Table 63.52 On-Chip USB Full-Speed Characteristics (DP and DM Pin Characteristics)

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 3.0$ to 3.6 V, 3.0 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $U_{CLK} = 48$ MHz, $P_{CLKA} = 8$ to 120 MHz,
 $P_{CLKB} = 8$ to 60 MHz, $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Input characteristics	Input high level voltage	V_{IH}	2.0	—	—	V	
	Input low level voltage	V_{IL}	—	—	0.8	V	
	Differential input sensitivity	V_{DI}	0.2	—	—	V	DP – DM
	Differential common mode range	V_{CM}	0.8	—	2.5	V	
Output characteristics	Output high level voltage	V_{OH}	2.8	—	3.6	V	$I_{OH} = -200$ μ A
	Output low level voltage	V_{OL}	0.0	—	0.3	V	$I_{OL} = 2$ mA
	Cross-over voltage	V_{CRS}	1.3	—	2.0	V	Figure 63.100
	Rise time	t_{FR}	4	—	20	ns	
	Fall time	t_{FF}	4	—	20	ns	
	Rise/fall time ratio	t_{FR} / t_{FF}	90	—	111.11	%	t_{FR} / t_{FF}
	Output resistance	Z_{DRV}	28	—	44	Ω	$R_s = 27$ Ω included
Pull-up and pull-down characteristics	DP pull-up resistance (when the function controller function is selected)	R_{pu}	0.900	—	1.575	k Ω	Idle state
			1.425	—	3.090	k Ω	At transmission and reception
	DP/DM pull-down resistance (when the host controller function is selected)	R_{pd}	14.25	—	24.80	k Ω	

**Figure 63.100 DP and DM Output Timing (Full-Speed)****Figure 63.101 Test Circuit (Full-Speed)**

63.6 A/D Conversion Characteristics

Table 63.53 12-Bit A/D (Unit 0) Conversion Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = PCLKC = 1$ MHz to 60 MHz, $T_a = T_{opr}$, Source impedance = 1.0 k Ω

Item		Min.	Typ.	Max.	Unit	Test Conditions
Resolution		8	—	12	Bit	
Analog input capacitance		—	—	30	pF	
Channel-dedicated sample-and-hold circuits in use (AN000 to AN002)	Conversion time*1 (Operation at PCLKC = 60 MHz)	1.06 (0.4 + 0.25) *2	—	—	μ s	- Sampling of channel-dedicated sample-and-hold circuits in 24 states - Sampling in 15 states
	Offset error	—	± 1.5	± 3.5	LSB	AN000 to AN002 = 0.25 V
	Full-scale error	—	± 1.5	± 3.5	LSB	AN000 to AN002 = $V_{REFH0} - 0.25$ V
	Quantization error	—	± 0.5	—	LSB	
	Absolute accuracy	—	± 3.0	± 5.5	LSB	
	DNL differential nonlinearity error	—	± 1.0	± 2.0	LSB	
	INL integral nonlinearity error	—	± 1.5	± 3.0	LSB	
	Holding characteristics of sample-and-hold circuits	—	—	20	μ s	
	Dynamic range	0.25	—	$V_{REFH0} - 0.25$	V	
Channel-dedicated sample-and-hold circuits not in use (AN000 to AN007)	Conversion time*1 (Operation at PCLKC = 60 MHz)	0.48 (0.267)*2	—	—	μ s	Sampling in 16 states
	Offset error	—	± 1.0	± 2.5	LSB	
	Full-scale error	—	± 1.0	± 2.5	LSB	
	Quantization error	—	± 0.5	—	LSB	
	Absolute accuracy	—	± 2.5	± 4.5	LSB	
	DNL differential nonlinearity error	—	± 0.5	± 1.5	LSB	
	INL integral nonlinearity error	—	± 1.0	± 2.5	LSB	

Note: The above specification values apply when there is no access to the external bus during A/D conversion. If access proceeds during A/D conversion, values may not fall within the above ranges.

Note 1. The conversion time includes the sampling time and the comparison time. As the test conditions, the number of sampling states is indicated.

Note 2. The value in parentheses indicates the sampling time.

Table 63.54 12-Bit A/D (Unit 1) Conversion Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = PCLKD = 1$ MHz to 60 MHz, $T_a = T_{opr}$, Source impedance = 1.0 k Ω

Item	Min.	Typ.	Max.	Unit	Test Conditions
Resolution	8	—	12	Bit	
Conversion time*1 (Operation at PCLKD = 60 MHz)	0.88 (0.633)*2	—	—	μ s	Sampling in 38 states (ADSAM.SAM = 1)
Conversion time*1 (Operation at PCLKD = 30 MHz)	1 (0.500)*2	—	—	μ s	Sampling in 15 states (ADSAM.SAM = 1)
Analog input capacitance	—	—	30	pF	
Offset error	—	± 2.0	± 3.5	LSB	
Full-scale error	—	± 2.0	± 3.5	LSB	
Quantization error	—	± 0.5	—	LSB	
Absolute accuracy	—	± 4.0	± 6.0	LSB	
DNL differential nonlinearity error (Operation at PCLKD = 60 MHz)	—	± 1.5	± 4.0	LSB	
DNL differential nonlinearity error (Operation at PCLKD = 30 MHz)	—	± 1.5	± 2.5	LSB	
INL integral nonlinearity error (Operation at PCLKD = 60 MHz)	—	± 2.0	± 4.0	LSB	
INL integral nonlinearity error (Operation at PCLKD = 30 MHz)	—	± 2.0	± 3.5	LSB	

Note: The above specification values apply when there is no access to the external bus during A/D conversion. If access proceeds during A/D conversion, values may not fall within the above ranges.

Note 1. The conversion time includes the sampling time and the comparison time. As the test conditions, the number of sampling states is indicated.

Note 2. The value in parentheses indicates the sampling time.

Table 63.55 A/D Internal Reference Voltage Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $PCLKB = PCLKD = 60$ MHz, $T_a = T_{opr}$

Item	Min.	Typ.	Max.	Unit	Test Conditions
A/D internal reference voltage	1.13	1.18	1.23	V	

63.7 D/A Conversion Characteristics

Table 63.56 D/A Conversion Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item		Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Resolution		—	12	12	12	Bit	
Unbuffered output	Absolute accuracy	—	—	—	± 6.0	LSB	2-M Ω resistive load 10-bit conversion
	Differential nonlinearity error	DNL	—	± 1.0	± 2.0	LSB	2-M Ω resistive load
	Output resistance	R_O	—	8.6	—	k Ω	
	Setting time	t_S	—	—	3	μ s	20-pF capacitive load
Buffered output	Load resistance	R_L	5	—	—	k Ω	
	Load capacitance	C_L	—	—	50	pF	
	Output voltage	V_O	0.2	—	$AVCC1 - 0.2$	V	
	Differential nonlinearity error	DNL	—	± 1.0	± 2.0	LSB	
	Integral nonlinearity error	INL	—	± 2.0	± 4.0	LSB	
	Setting time	t_S	—	—	4	μ s	

63.8 Temperature Sensor Characteristics

Table 63.57 Temperature Sensor Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

Item	Min.	Typ.	Max.	Unit	Test Conditions
Relative accuracy	—	± 1	—	$^{\circ}$ C	
Temperature slope	—	4	—	mV/ $^{\circ}$ C	
Output voltage	—	1.21	—	V	$T_a = 25^{\circ}$ C
Temperature sensor start time	—	—	30	μ s	
Sampling time*1	4.15	—	—	μ s	

Note 1. Set the S12AD1.ADSSTRT register such that the sampling time of the 12-bit A/D converter satisfies this specification.

63.9 Power-on Reset Circuit and Voltage Detection Circuit Characteristics

Table 63.58 Power-on Reset Circuit and Voltage Detection Circuit Characteristics

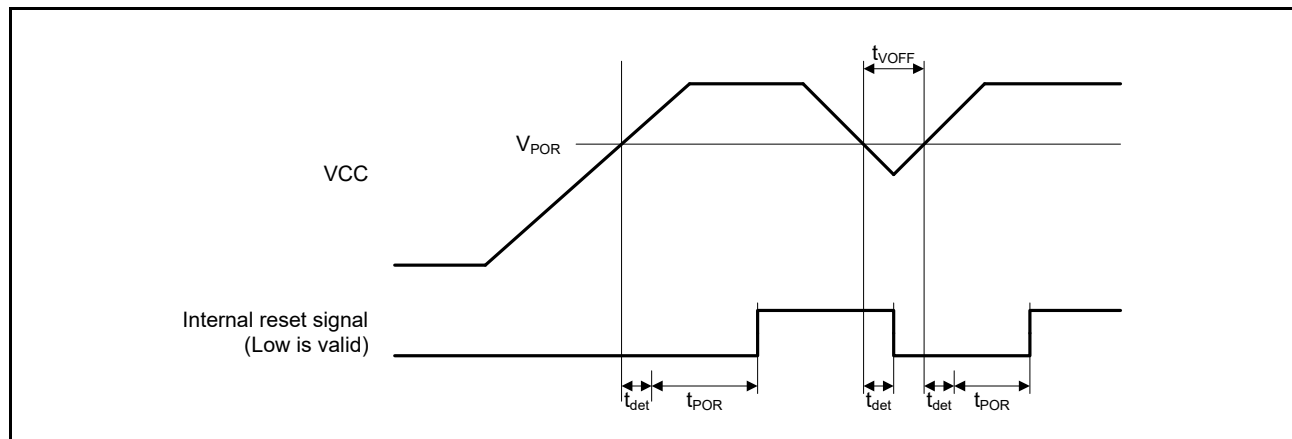
Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$

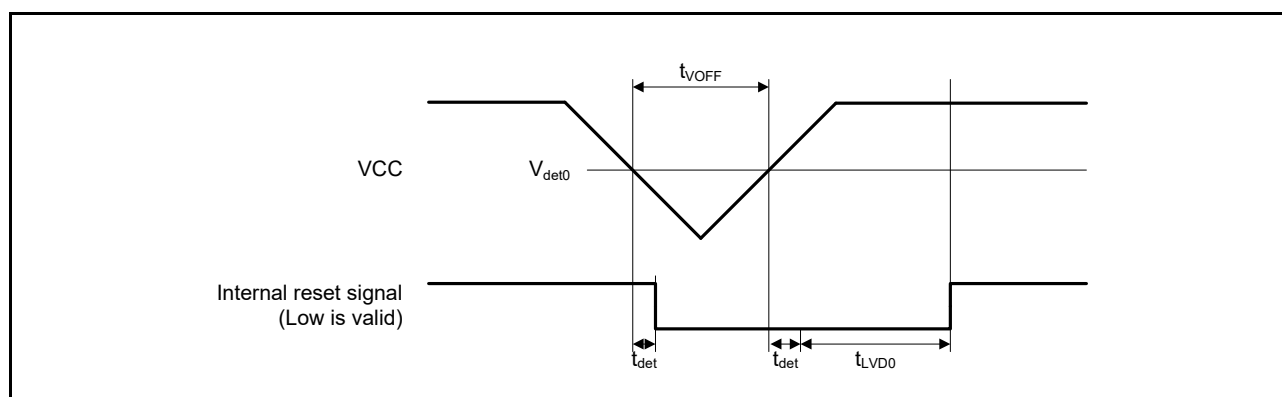
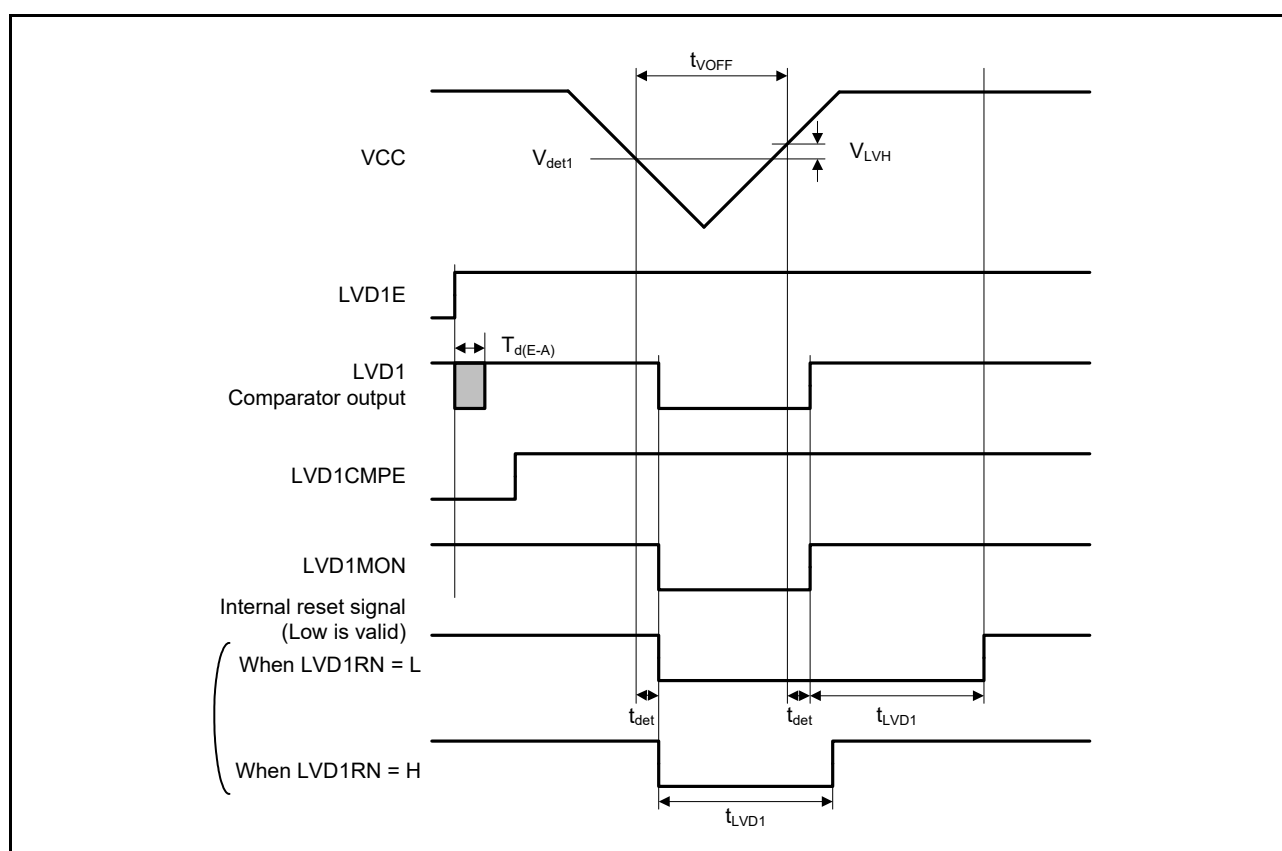
Item			Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Voltage detection level	Power-on reset (POR)	Low power consumption function disabled*1	V _{POR}	2.5	2.6	2.7	V	Figure 63.102
		Low power consumption function enabled*2		1.8	2.25	2.7		
	Voltage detection circuit (LVD0)		V _{det0_1}	2.84	2.94	3.04		Figure 63.103
			V _{det0_2}	2.77	2.87	2.97		
			V _{det0_3}	2.70	2.80	2.90		
	Voltage detection circuit (LVD1)		V _{det1_1}	2.89	2.99	3.09		Figure 63.104
			V _{det1_2}	2.82	2.92	3.02		
			V _{det1_3}	2.75	2.85	2.95		
	Voltage detection circuit (LVD2)		V _{det2_1}	2.89	2.99	3.09		Figure 63.105
			V _{det2_2}	2.82	2.92	3.02		
			V _{det2_3}	2.75	2.85	2.95		
Internal reset time	Power-on reset time		t _{POR}	—	4.6	—	ms	Figure 63.102
	LVD0 reset time		t _{LVD0}	—	0.70	—		Figure 63.103
	LVD1 reset time		t _{LVD1}	—	0.57	—		Figure 63.104
	LVD2 reset time		t _{LVD2}	—	0.57	—		Figure 63.105
Minimum VCC down time			t _{VOFF}	200	—	—	μs	Figure 63.102, Figure 63.103
Response delay time			t _{det}	—	—	200	μs	Figure 63.102 to Figure 63.105
LVD operation stabilization time (after LVD is enabled)			t _{d(E-A)}	—	—	10	μs	Figure 63.104, Figure 63.105
Hysteresis width (LVD1 and LVD2)			V _{LVH}	—	70	—	mV	

Note: The minimum VCC down time indicates the time when VCC is below the minimum value of voltage detection levels V_{POR} , V_{det1} , and V_{det2} for the POR/ LVD.

Note 1. The low power consumption function is disabled and DEEPCUT[1:0] = 00b or 01b.

Note 2. The low power consumption function is enabled and DEEPCUT[1:0] = 11b.

**Figure 63.102 Power-on Reset Timing**

Figure 63.103 Voltage Detection Circuit Timing (V_{det0})Figure 63.104 Voltage Detection Circuit Timing (V_{det1})

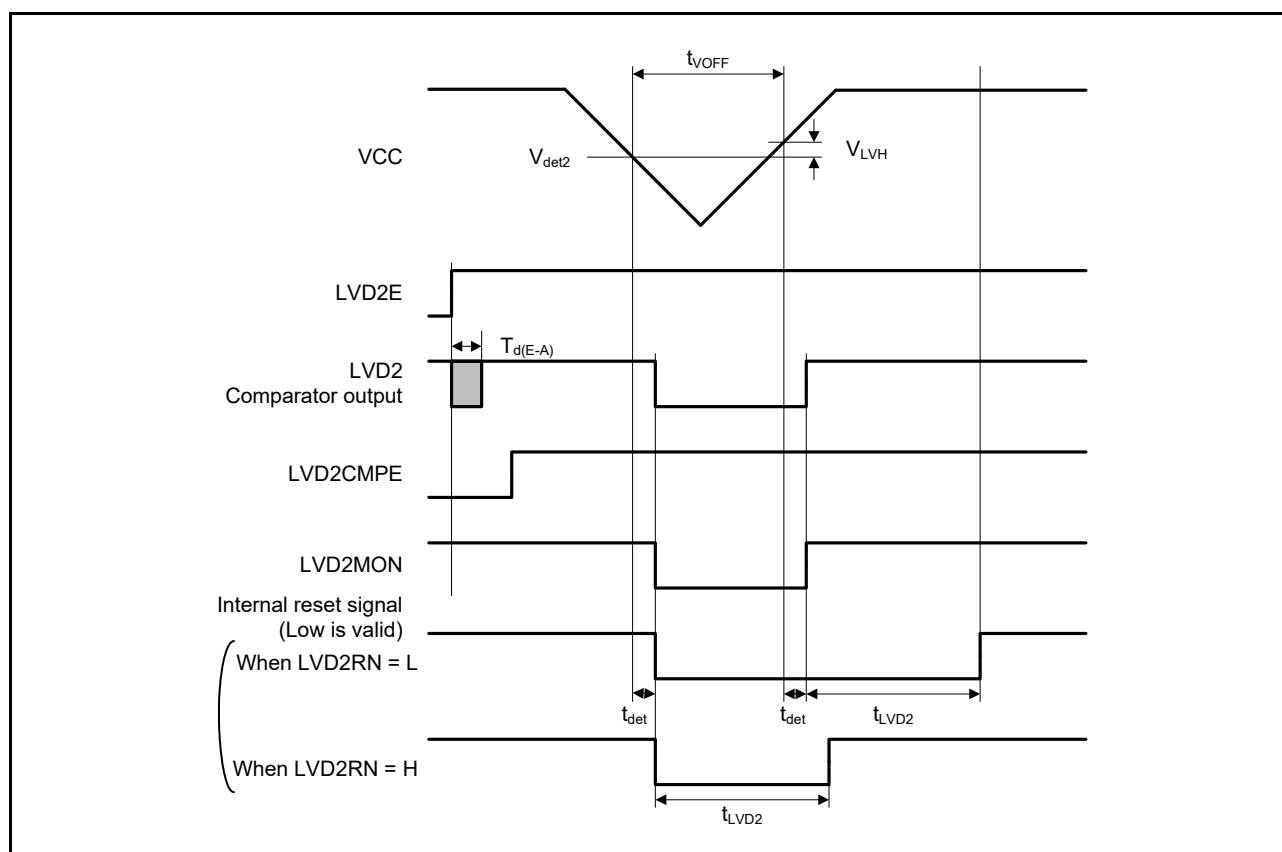


Figure 63.105 Voltage Detection Circuit Timing (V_{det2})

63.10 Oscillation Stop Detection Timing

Table 63.59 Oscillation Stop Detection Circuit Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7 \text{ to } 3.6 \text{ V}$, $2.7 \text{ V} \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0 \text{ V}$,
 $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Detection time	t_{dr}	—	—	1	ms	Figure 63.106

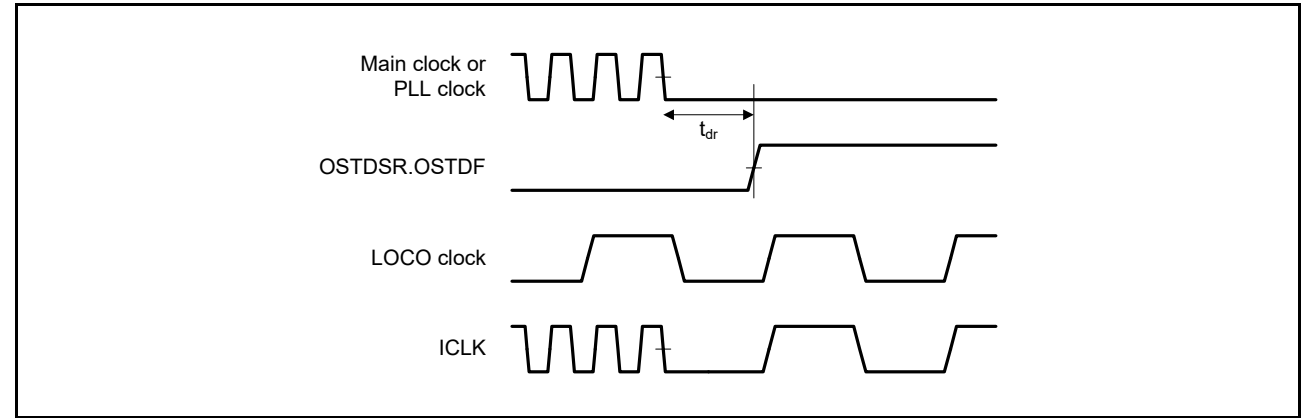


Figure 63.106 Oscillation Stop Detection Timing

63.11 Battery Backup Function Characteristics

Table 63.60 Battery Backup Function Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = 2.7 \text{ to } 3.6 \text{ V}$, $2.7 \text{ V} \leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0 \text{ V}$,
 $V_{BATT} = 1.62 \text{ to } 3.6 \text{ V}$, $T_a = T_{opr}$

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
Voltage level for switching to battery backup	V _{DETBATT}	2.50	2.60	2.70	V	Figure 63.107
Lower-limit V _{BATT} voltage for power supply switching due to VCC voltage drop	V _{BATTSW}	2.70	—	—		
VCC-off period for starting power supply switching	t _{VOFFBATT}	200	—	—	μs	

Note: The VCC-off period for starting power supply switching indicates the period in which VCC is below the minimum value of the voltage level for switching to battery backup ($V_{DET BATT}$).

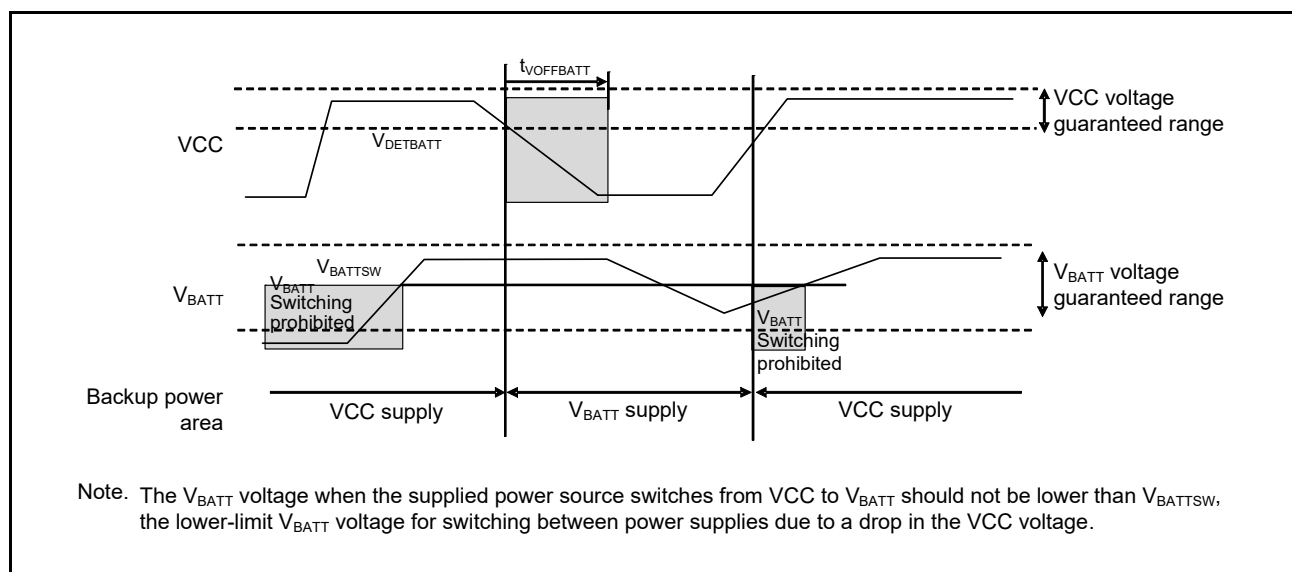


Figure 63.107 Battery Backup Function Characteristics

63.12 Flash Memory Characteristics

Table 63.61 Code Flash Memory Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 Temperature range for programming/erasure: $T_a = T_{opr}$

Item		Symbol	FCLK = 4 MHz			FCLK = 15 MHz			20 MHz $\leq$ FCLK $\leq$ 60 MHz			Unit	Conditions
			Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.		
Programming time $N_{PEC} \leq 100$ times	128 bytes	t_{P128}	—	0.75	13.2	—	0.38	6.6	—	0.34	6	ms	
	8 Kbytes	t_{P8K}	—	49	176	—	25	88	—	22	80	ms	
	32 Kbytes	t_{P32K}	—	194	704	—	97	352	—	88	320	ms	
Programming time $N_{PEC} > 100$ times	128 bytes	t_{P128}	—	0.91	15.8	—	0.46	8	—	0.41	7.2	ms	
	8 Kbytes	t_{P8K}	—	60	212	—	30	106	—	27	96	ms	
	32 Kbytes	t_{P32K}	—	234	848	—	117	424	—	106	384	ms	
Erasure time $N_{PEC} \leq 100$ times	8 Kbytes	t_{E8K}	—	78	216	—	48	132	—	43	120	ms	
	32 Kbytes	t_{E32K}	—	283	864	—	173	528	—	157	480	ms	
Erasure time $N_{PEC} > 100$ times	8 Kbytes	t_{E8K}	—	94	260	—	58	158	—	52	144	ms	
	32 Kbytes	t_{E32K}	—	341	1040	—	208	632	—	189	576	ms	
Reprogramming/erasure cycle*1		N_{PEC}	10000 *2	—	—	10000 *2	—	—	10000 *2	—	—	Times	
Suspend delay time during programming		t_{SPD}	—	—	264	—	—	132	—	—	120	μ s	
First suspend delay time during erasing (in suspend priority mode)		t_{SESD1}	—	—	216	—	—	132	—	—	120	μ s	
Second suspend delay time during erasure (in suspend priority mode)		t_{SESD2}	—	—	1.7	—	—	1.7	—	—	1.7	ms	
Suspend delay time during erasure (in erasure priority mode)		t_{SEED}	—	—	1.7	—	—	1.7	—	—	1.7	ms	
Forced stop command		t_{FD}	—	—	32	—	—	22	—	—	20	μ s	
Data retention*3, *4		t_{DRP}	20	—	—	20	—	—	20	—	—	Year	$T_a \leq 85^\circ\text{C}$
			10	—	—	10	—	—	10	—	—		$T_a \leq 105^\circ\text{C}$

Note 1. Definition of reprogram/erase cycle:

The program/erase cycle is the number of erasing for each block. When the number of program/erase cycles is n, each block can be erased n times. For instance, when 128-byte program is performed 64 times for different addresses in 8-Kbyte block and then the block is erased, the program/erase cycle is counted as one. However, the same address cannot be programmed more than once before the next erase cycle (overwriting is prohibited).

Note 2. Characteristics are degraded as the number of program/erase increases. This is the minimum value of program/erase cycles to guarantee all characteristics listed in this table.

Note 3. This shows the characteristics when a self-programming library provided by Renesas Electronics or a flash programmer is used and the program/erase cycle does not exceed the specified value.

Note 4. This result is obtained from reliability testing.

Table 63.62 Data Flash Memory Characteristics

Conditions: $V_{CC} = AVCC0 = AVCC1 = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AVCC0$,
 $V_{SS} = AVSS0 = AVSS1 = V_{REFL0} = V_{SS_USB} = 0$ V,
 Temperature range for programming/erasure: $T_a = T_{opr}$

Item		Symbol	FCLK = 4 MHz			FCLK = 15 MHz			20 MHz $\leq$ FCLK $\leq$ 60 MHz			Unit	Conditions
			Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.		
Programming time	4 bytes	t_{DP4}	—	0.36	3.8	—	0.18	1.9	—	0.16	1.7	ms	
Erasure time	64 bytes	t_{DP64}	—	3.1	18	—	1.9	11	—	1.7	10	ms	
	128 bytes	t_{DP128}	—	4.7	27	—	2.9	16	—	2.6	15	ms	
	256 bytes	t_{DP256}	—	8.9	50	—	5.4	31	—	4.9	28	ms	
Blank check time	4 bytes	t_{DBC4}	—	—	84	—	—	33	—	—	30	μ s	
	64 bytes	t_{DBC64}	—	—	280	—	—	110	—	—	100	μ s	
	2 Kbytes	t_{DBC2K}	—	—	6160	—	—	2420	—	—	2200	μ s	
Reprogramming/erasure cycle*1		N_{DPEC}	100000 *2	—	—	100000 *2	—	—	100000 *2	—	—	Times	
Suspend delay time during programming		t_{DSPD}	—	—	264	—	—	132	—	—	120	μ s	
First suspend delay time during erasure (in suspend priority mode)	64 bytes	—	—	—	216	—	—	132	—	—	120	μ s	
	128 bytes	—	—	—	216	—	—	132	—	—	120	μ s	
	256 bytes	—	—	—	216	—	—	132	—	—	120	μ s	
Second suspend delay time during erasure (in suspend priority mode)	64 bytes	—	—	—	300	—	—	300	—	—	300	μ s	
	128 bytes	—	—	—	390	—	—	390	—	—	390	μ s	
	256 bytes	—	—	—	570	—	—	570	—	—	570	μ s	
Suspend delay time during erasing (in suspend priority mode)	64 bytes	—	—	—	300	—	—	300	—	—	300	μ s	
	128 bytes	—	—	—	390	—	—	390	—	—	390	μ s	
	256 bytes	—	—	—	570	—	—	570	—	—	570	μ s	
Forced stop command		t_{FD}	—	—	32	—	—	22	—	—	20	μ s	
Data retention*3, *4		t_{DDRP}	20	—	—	20	—	—	20	—	—	Year	$T_a \leq 85^\circ\text{C}$
			10	—	—	10	—	—	10	—	—		$T_a \leq 105^\circ\text{C}$

Note 1. Definition of reprogram/erase cycle:

The program/erase cycle is the number of erasing for each block. When the number of program/erase cycles is n , each block can be erased n times. For instance, when 4-byte program is performed 512 times for different addresses in 2-Kbyte block and then the block is erased, the program/erase cycle is counted as one. However, the same address cannot be programmed more than once before the next erase cycle (overwriting is prohibited).

Note 2. Characteristics are degraded as the number of program/erase increases. This is the minimum value of program/erase cycles to guarantee all characteristics listed in this table.

Note 3. This shows the characteristics when a self-programming library provided by Renesas Electronics or a flash programmer is used and the program/erase cycle does not exceed the specified value.

Note 4. This result is obtained from reliability testing.

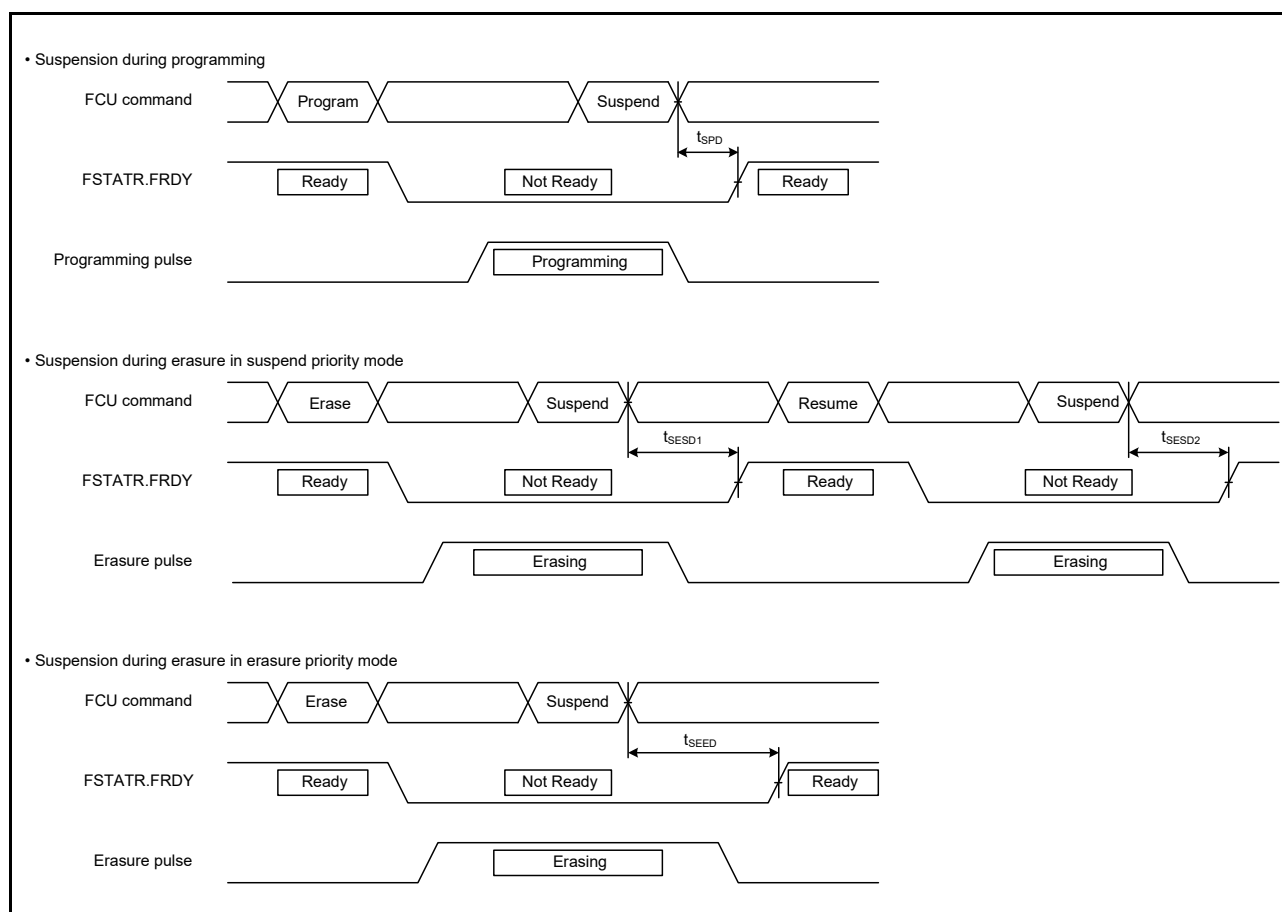


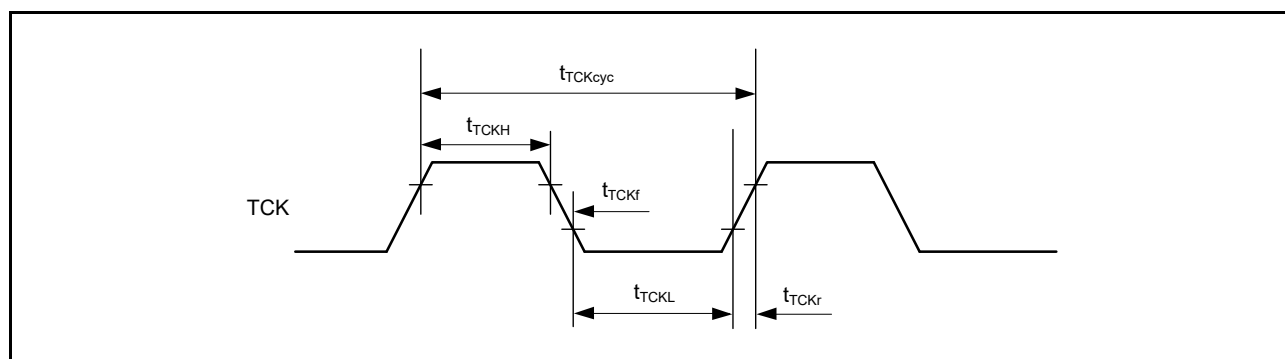
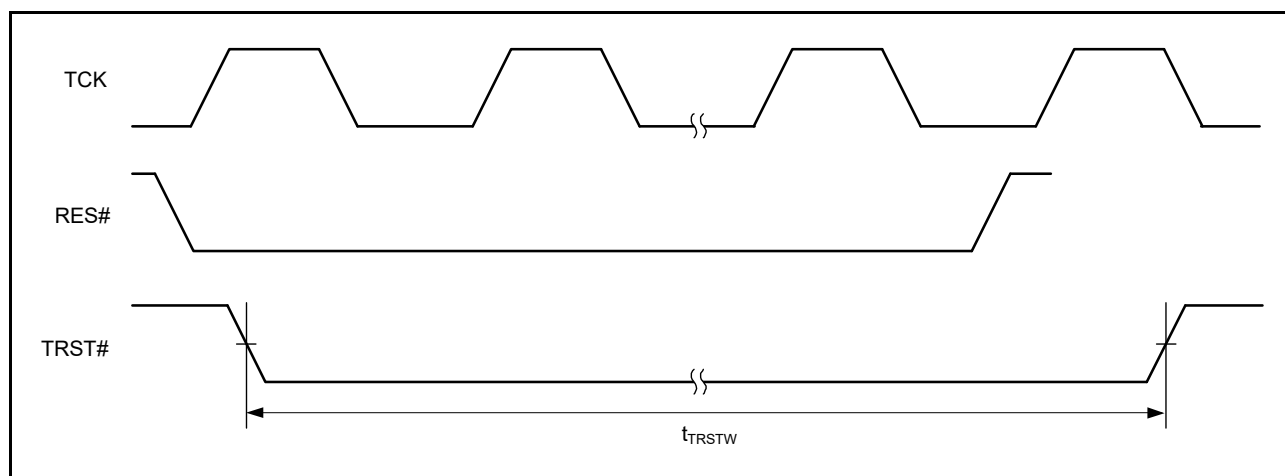
Figure 63.108 Flash Memory Programming/Erasure Suspension Timing

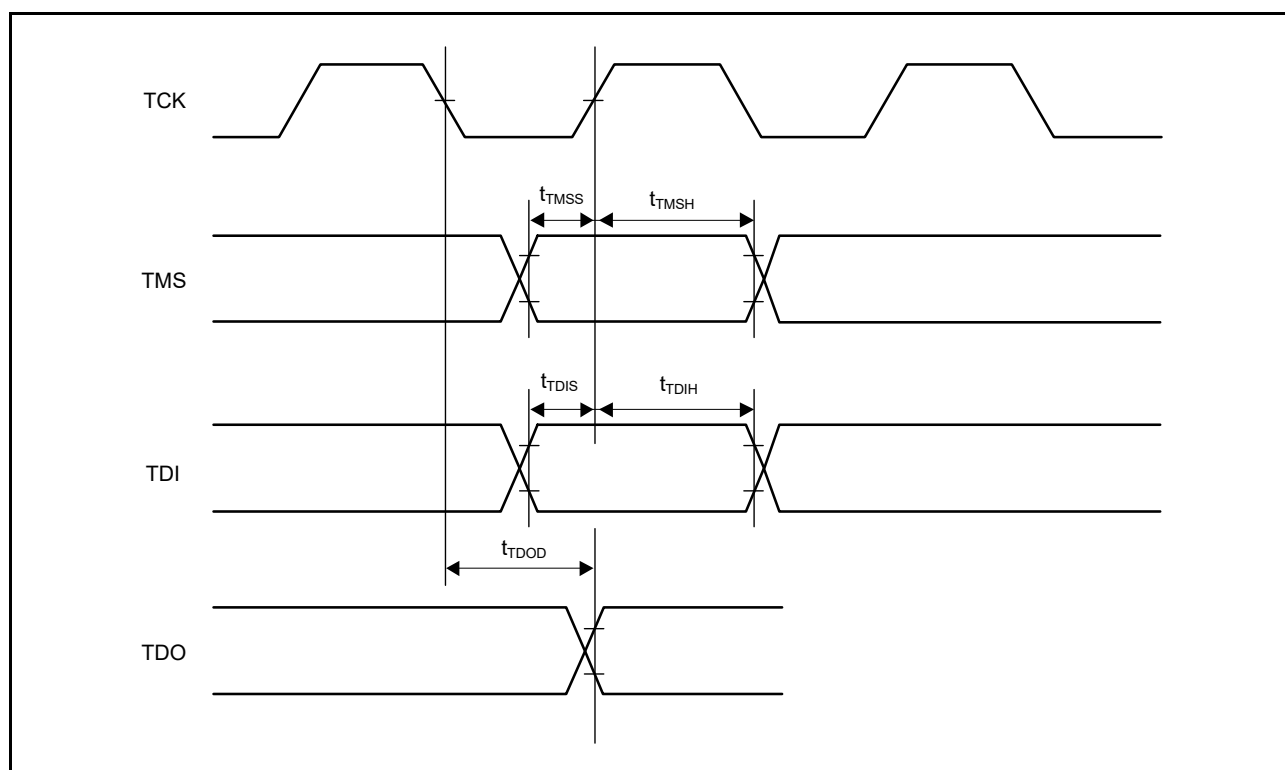
63.13 Boundary Scan

Table 63.63 Boundary Scan Characteristics

Conditions: $V_{CC} = AV_{CC0} = AV_{CC1} = V_{CC_USB} = V_{BATT} = 2.7$ to 3.6 V, 2.7 V $\leq V_{REFH0} \leq AV_{CC0}$,
 $V_{SS} = AV_{SS0} = AV_{SS1} = V_{REFL0} = V_{SS_USB} = 0$ V,
 $T_a = T_{opr}$,
 Output load conditions: $V_{OH} = 0.5 \times V_{CC}$, $V_{OL} = 0.5 \times V_{CC}$, $C = 30$ pF,
 High-drive output is selected by the drive capacity control register.

Item	Symbol	Min.	Typ.	Max.	Unit	Test Conditions
TCK clock cycle time	t_{TCKcyc}	100	—	—	ns	Figure 63.109
TCK clock high pulse width	t_{TCKH}	45	—	—	ns	
TCK clock low pulse width	t_{TCKL}	45	—	—	ns	
TCK clock rise time	t_{TCKr}	—	—	5	ns	
TCK clock fall time	t_{TCKf}	—	—	5	ns	
TRST# pulse width	t_{TRSTW}	20	—	—	t_{TCKcyc}	Figure 63.110
TMS setup time	t_{TMSS}	20	—	—	ns	Figure 63.111
TMS hold time	t_{TMSH}	20	—	—	ns	
TDI setup time	t_{TDIS}	20	—	—	ns	
TDI hold time	t_{TDIH}	20	—	—	ns	
TDO data delay time	t_{TDOD}	—	—	40	ns	

**Figure 63.109 Boundary Scan TCK Timing****Figure 63.110 Boundary Scan TRST# Timing**

**Figure 63.111 Boundary Scan Input/Output Timing**

Appendix 1. Port States in Each Processing Mode

Table 1.1 Port States in Each Processing State (1/6)

Port Name/ Pin Name	Operating Mode According to Registers Setting		Reset	Software Standby Mode		Deep Software Standby Mode IOKEEP = 1/0	After Deep Software Standby Mode is Canceled (Return to Start-up Mode)	
				OPE = 1	OPE = 0		IOKEEP = 1*1	IOKEEP = 0
P00/IRQ8, P01/IRQ9, P02/IRQ10	All		Hi-Z	Keep-O*2		Keep	Keep	Hi-Z
P03/DA0/ IRQ11	All	DA0 output (DAOE0 = 1)	Hi-Z	DA output retained		Hi-Z	Hi-Z	Hi-Z
		Other than the above (DAOE0 = 0)		Keep-O*2		Keep	Keep	
P05/DA1/ IRQ13	All	DA1 output (DAOE1 = 1)	Hi-Z	DA output retained		Hi-Z	Hi-Z	Hi-Z
		Other than the above (DAOE1 = 0)		Keep-O*2		Keep	Keep	
P07/IRQ15	All		Hi-Z	Keep-O*2		Keep	Keep	Hi-Z
P10/ IRQ0-DS/ ALE	Single-chip mode (EXBE = 0)		Hi-Z	Keep-O*2		Keep-O*3	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)			[ALE output] H [Other than the above] Keep-O*2	[ALE output] Hi-Z [Other than the above] Keep-O*2			
P11/ IRQ1-DS	All		Hi-Z	Keep-O*2		Keep-O*3	Keep	Hi-Z
P12/IRQ2/ WR3#/BC3#	Single-chip mode (EXBE = 0)		Hi-Z	Keep-O*2		Keep-O	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)			[WR3#/BC3# output] H [Other than the above] Keep-O*2	[WR3#/BC3# output] Hi-Z [Other than the above] Keep-O*2			
P13/IRQ3/ WR2#/BC2#	Single-chip mode (EXBE = 0)		Hi-Z	Keep-O*2		Keep-O	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)			[WR2#/BC2# output] H [Other than the above] Keep-O*2	[WR2#/BC2# output] Hi-Z [Other than the above] Keep-O*2			
P14/IRQ4/ USB0_OVRC URA, P15/IRQ5/ CRX1-DS, P16/IRQ6/ SCL2-DS/ USB0_VBUS/ USB0_OVRC URB, P17/IRQ7/ SDA2-DS	All		Hi-Z	Keep-O*2		Keep-O*3	Keep	Hi-Z
P20/IRQ8, P21/IRQ9, P22/USB0_ OVRCURB, P23	All		Hi-Z	Keep-O*2		Keep-O*3	Keep	Hi-Z
P24/CS4#, P25/CS5#, P26/CS6#, P27/CS7#	Single-chip mode (EXBE = 0)		Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)			[CSn# output] H [Other than the above] Keep-O	[CSn# output] Hi-Z [Other than the above] Keep-O			

Table 1.1 Port States in Each Processing State (2/6)

Port Name/ Pin Name	Operating Mode According to Registers Setting	Reset	Software Standby Mode		Deep Software Standby Mode IOKEEP = 1/0	After Deep Software Standby Mode is Canceled (Return to Start-up Mode)	
			OPE = 1	OPE = 0		IOKEEP = 1*1	IOKEEP = 0
P30/ IRQ0-DS/ RTCIC0, P31/ IRQ1-DS/ RTCIC1, P32/ IRQ2-DS/ RTCIC2, P33/ IRQ3-DS, P34/IRQ4, P35/NMI	All	Hi-Z	Keep-O*2		Keep-O*3	Keep	Hi-Z
P36, P37	All	Hi-Z	Keep-O		Keep	Keep	Hi-Z
P40/ IRQ8-DS, P41/ IRQ9-DS, P42/ IRQ10-DS, P43/ IRQ11-DS, P44/ IRQ12-DS, P45/ IRQ13-DS, P46/ IRQ14-DS, P47/ IRQ15-DS	All	Hi-Z	Keep-O*2		Keep-O*3	Keep	Hi-Z
P50/WR0#/ WR#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[WR0#/WR# output] H [Other than the above] Keep-O	[WR0#/WR# output] Hi-Z [Other than the above] Keep-O			
P51/WR1#/ BC1#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/disabled extended (EXBE = 1)		[WR1#/BC1# output] H [Other than the above] Keep-O	[WR1#/BC1# output] Hi-Z [Other than the above] Keep-O			
P52/RD#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[RD# output] H [Other than the above] Keep-O	[RD# output] Hi-Z [Other than the above] Keep-O			
P53/BCLK	All	Hi-Z	[Clock output] H [Other than the above] Keep-O		Keep	Keep	Hi-Z
P54/ALE	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[ALE output] L [Data output] Hi-Z [Other than the above] Keep-O	[ALE output] Hi-Z [Data output] Hi-Z [Other than the above] Keep-O			
P55/IRQ10	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O*2		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[Data output] Hi-Z [Other than the above] Keep-O				
P56, P57	All	Hi-Z	Keep-O		Keep	Keep	Hi-Z

Table 1.1 Port States in Each Processing State (3/6)

Port Name/ Pin Name	Operating Mode According to Registers Setting	Reset	Software Standby Mode		Deep Software Standby Mode IOKEEP = 1/0	After Deep Software Standby Mode is Canceled (Return to Start-up Mode)	
			OPE = 1	OPE = 0		IOKEEP = 1 ¹	IOKEEP = 0
P60/CS0#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS0# output] H [Other than the above] Keep-O	[CS0# output] Hi-Z [Other than the above] Keep-O			
P61/CS1#/ SDCS#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS1# output] H [Data output] Hi-Z [Other than the above] Keep-O	[CS1# output] Hi-Z [Data output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[SDCS# output] H [Other than the above] Keep-O	[SDCS# output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh enabled (SDSELF. SFEN = 1)		[SDCS# output] L [Other than the above] Keep-O				
P62/CS2#/ RAS#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS2# output] H [Data output] Hi-Z [Other than the above] Keep-O	[CS2# output] Hi-Z [Data output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[RAS# output] H [Other than the above] Keep-O	[RAS# output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh enabled (SDSELF. SFEN = 1)		[RAS# output] L [Other than the above] Keep-O				
P63/CS3#/ CAS#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS3# output] H [Data output] Hi-Z [Other than the above] Keep-O	[CS3# output] Hi-Z [Data output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[CAS# output] H [Other than the above] Keep-O	[CAS# output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh enabled (SDSELF. SFEN = 1)		[CAS# output] L [Other than the above] Keep-O				
P64/CS4#/ WE#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS4# output] H [Data output] Hi-Z [Other than the above] Keep-O	[CS4# output] Hi-Z [Data output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[WE# output] H [Other than the above] Keep-O	[WE# output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh enabled (SDSELF. SFEN = 1)						

Table 1.1 Port States in Each Processing State (4/6)

Port Name/ Pin Name	Operating Mode According to Registers Setting	Reset	Software Standby Mode		Deep Software Standby Mode IOKEEP = 1/0	After Deep Software Standby Mode is Canceled (Return to Start-up Mode)	
			OPE = 1	OPE = 0		IOKEEP = 1 ¹	IOKEEP = 0
P65/CS5#/ CKE	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS5# output] H [Other than the above] Keep-O	[CS5# output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[CKE output] H [Other than the above] Keep-O	[CKE output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh enabled (SDSELF. SFEN = 1)		[CKE output] L [Other than the above] Keep-O				
P66/CS6#/ DQM0	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS6# output] H [Other than the above] Keep-O	[CS6# output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[DQM0 output] DQM0 output retained [Other than the above] Keep-O	[DQM0 output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh enabled (SDSELF. SFEN = 1)						
P67/CS7#/ IRQ15/DQM1	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O ²		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS7# output] H [Other than the above] Keep-O ²	[CS7# output] Hi-Z [Other than the above] Keep-O ²			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[DQM1 output] DQM1 output retained [Other than the above] Keep-O ²	[DQM1 output] Hi-Z [Other than the above] Keep-O ²			
	Self-Refresh enabled (SDSELF. SFEN = 1)						
P70/SDCLK	All	Hi-Z	[Clock output] H [Other than the above] Keep-O		Keep	Keep	Hi-Z
P71/CS1#, P72/CS2#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CSn# output] H [Address output] Address output retained [Other than the above] Keep-O	[CSn# output] Hi-Z [Address output] Hi-Z [Other than the above] Keep-O			
P73/CS3#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS3# output] H [Other than the above] Keep-O	[CS3# output] Hi-Z [Other than the above] Keep-O			
P74/CS4#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CS4# output] H [Address output] Address output retained [Other than the above] Keep-O	[CS4# output] Hi-Z [Address output] Hi-Z [Other than the above] Keep-O			

Table 1.1 Port States in Each Processing State (5/6)

Port Name/ Pin Name	Operating Mode According to Registers Setting	Reset	Software Standby Mode		Deep Software Standby Mode IOKEEP = 1/0	After Deep Software Standby Mode is Canceled (Return to Start-up Mode)	
			OPE = 1	OPE = 0		IOKEEP = 1 ¹	IOKEEP = 0
P75/CS5#, P76/CS6#, P77/CS7#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[CSn# output] H [Other than the above] Keep-O	[CSn# output] Hi-Z [Other than the above] Keep-O			
Port 8	All	Hi-Z	Keep-O		Keep	Keep	Hi-Z
Port 9	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[Address output] Address output retained [Data output] Hi-Z [Other than the above] Keep-O	[Address output] Hi-Z [Data output] Hi-Z [Other than the above] Keep-O			
PA0/DQM2	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[Address output] Address output retained [Other than the above] Keep-O	[Address output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[DQM2 output] DQM2 output retained [Other than the above] Keep-O	[DQM2 output] Hi-Z [Other than the above] Keep-O			
	Self-Refresh enabled (SDSELF. SFEN = 1)						
PA1/IRQ11/ DQM3	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O ²		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[Address output] Address output retained [Other than the above] Keep-O ²	[Address output] Hi-Z [Other than the above] Keep-O ²			
	Self-Refresh disabled (SDSELF. SFEN = 0)		[DQM3 output] DQM3 output retained [Other than the above] Keep-O ²	[DQM3 output] Hi-Z [Other than the above] Keep-O ²			
	Self-Refresh enabled (SDSELF. SFEN = 1)						
PA2, PA3/ IRQ6-DS, PA4/ IRQ5-DS, PA5 to PA7	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O ²		Keep-O ³	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[Address output] Address output retained [Other than the above] Keep-O ²	[Address output] Hi-Z [Other than the above] Keep-O ²			
PB0/IRQ12, PB1/ IRQ4-DS, PB2 to PB7	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O ²		Keep-O ³	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[Address output] Address output retained [Other than the above] Keep-O ²	[Address output] Hi-Z [Other than the above] Keep-O ²			
PC0/IRQ14, PC1/IRQ12, PC2, PC3, PC4/CS3#, PC5/CS2#, PC6/IRQ13/ CS1#, PC7/IRQ14/ CS0#	Single-chip mode (EXBE = 0)	Hi-Z	Keep-O ²		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)		[Address output] Address output retained [CSn# output] H [Other than the above] Keep-O ²	[Address output] Hi-Z [CSn# output] Hi-Z [Other than the above] Keep-O ²			

Table 1.1 Port States in Each Processing State (6/6)

Port Name/ Pin Name	Operating Mode According to Registers Setting		Reset	Software Standby Mode		Deep Software Standby Mode IOKEEP = 1/0	After Deep Software Standby Mode is Canceled (Return to Start-up Mode)	
				OPE = 1	OPE = 0		IOKEEP = 1 ^{*1}	IOKEEP = 0
PD0//IRQ0, PD1//IRQ1, PD2//IRQ2, PD3//IRQ3, PD4//IRQ4, PD5//IRQ5, PD6//IRQ6, PD7//IRQ7	Single-chip mode (EXBE = 0)		Hi-Z	Keep-O ^{*2}		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)			[Data output] Hi-Z [Other than the above] Keep-O ^{*2}				
PE0, PE1, PE2/ IRQ7-DS, PE3, PE4, PE5//IRQ5, PE6//IRQ6, PE7//IRQ7	Single-chip mode (EXBE = 0)		Hi-Z	Keep-O ^{*2}		Keep-O ^{*3}	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)	8 bits in width of bus		[Data output] Hi-Z [Other than the above] Keep-O ^{*2}				
		16 bits in width of bus		[Data output] Hi-Z				
PF0, PF1, PF2, PF3, PF4, PF5//IRQ4	All		Hi-Z	Keep-O ^{*2}		Keep	Keep	Hi-Z
Port G	Single-chip mode (EXBE = 0)		Hi-Z	Keep-O		Keep	Keep	Hi-Z
	On-chip ROM enabled/ disabled extended (EXBE = 1)			[Data output] Hi-Z [Other than the above] Keep-O				
Port H	All		Hi-Z	Keep-O		Keep	Keep	Hi-Z
Port J	All		Hi-Z	Keep-O		Keep	Keep	Hi-Z
Port K	All		Hi-Z	Keep-O		Keep	Keep	Hi-Z
Port L	All		Hi-Z	Keep-O		Keep	Keep	Hi-Z
Port M	All		Hi-Z	Keep-O		Keep	Keep	Hi-Z
Port N	All		Hi-Z	Keep-O		Keep	Keep	Hi-Z
Port Q	All		Hi-Z	Keep-O		Keep	Keep	Hi-Z
USB0_DM	All		Hi-Z	Keep-O ^{*4}		Hi-Z ^{*3}	Hi-Z	
USB0_DP	All		Hi-Z	Keep-O ^{*4}		Hi-Z ^{*3}	Hi-Z	

H: High-level

L: Low-level

Keep-O: Output pins retain their previous values, and input pins become high-impedance.

Keep: Pin states are retained during periods on software standby.

Hi-Z: High-impedance

Note 1. The I/O port state is retained until the DPSBYCR.IOKEEP bit is cleared to 0.

Note 2. Input is enabled if the pin is specified as the software standby cancelling source while it is used as an external interrupt pin.

Note 3. Input is enabled if the pin is specified as the deep software standby cancelling source.

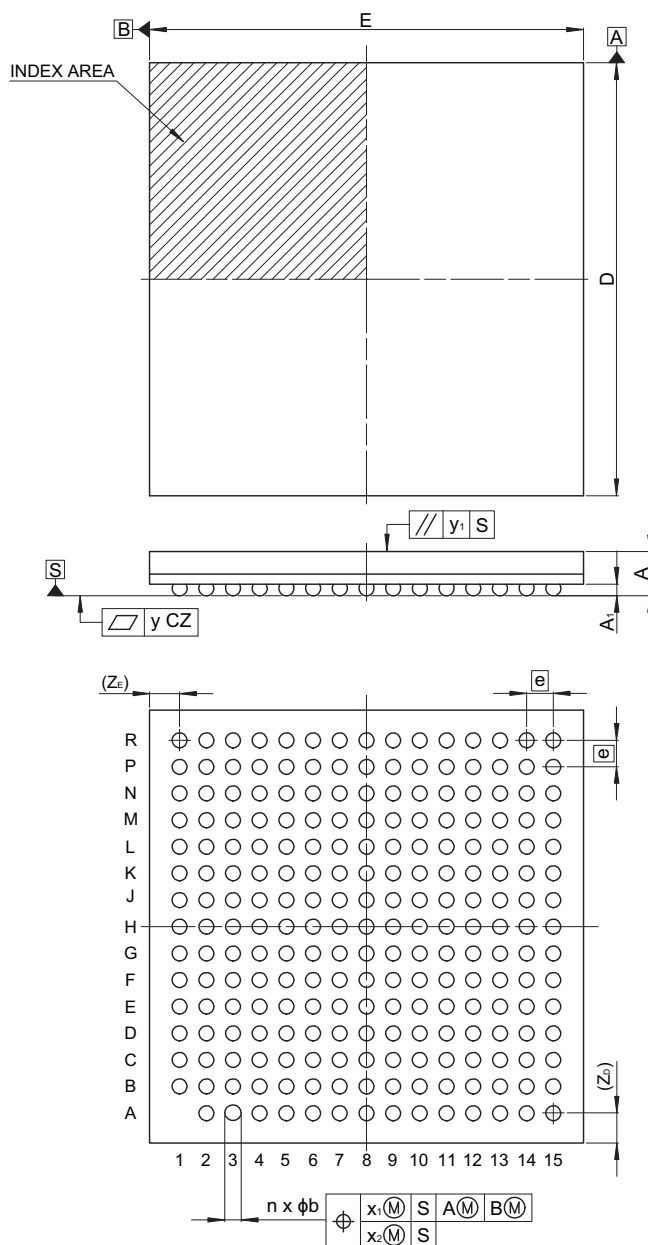
Note 4. Input is enabled while the pin is used as an input pin.

Appendix 2. Package Dimensions

Information on the latest version of the package dimensions or mountings has been displayed in “Packages” on Renesas Electronics Corporation website.

JEITA Package Code	RENESAS Code	Previous Code	MASS (Typ) [g]
P-LFBGA224-13x13-0.80	PLBG0224GA-A	224FHE	0.4

Unit: mm



Reference Symbol	Dimensions in millimeters		
	Min	Nom	Max
D	12.9	13.0	13.1
E	12.9	13.0	13.1
A	—	—	1.40
A ₁	0.30	0.35	0.40
e	—	0.80	—
b	0.40	0.45	0.50
x ₁	—	—	0.15
x ₂	—	—	0.08
y	—	—	0.10
y ₁	—	—	0.20
n	—	224	—
Z _D	—	0.90	—
Z _E	—	0.90	—

Figure A 224-Pin LFBGA (PLBG0224GA-A)

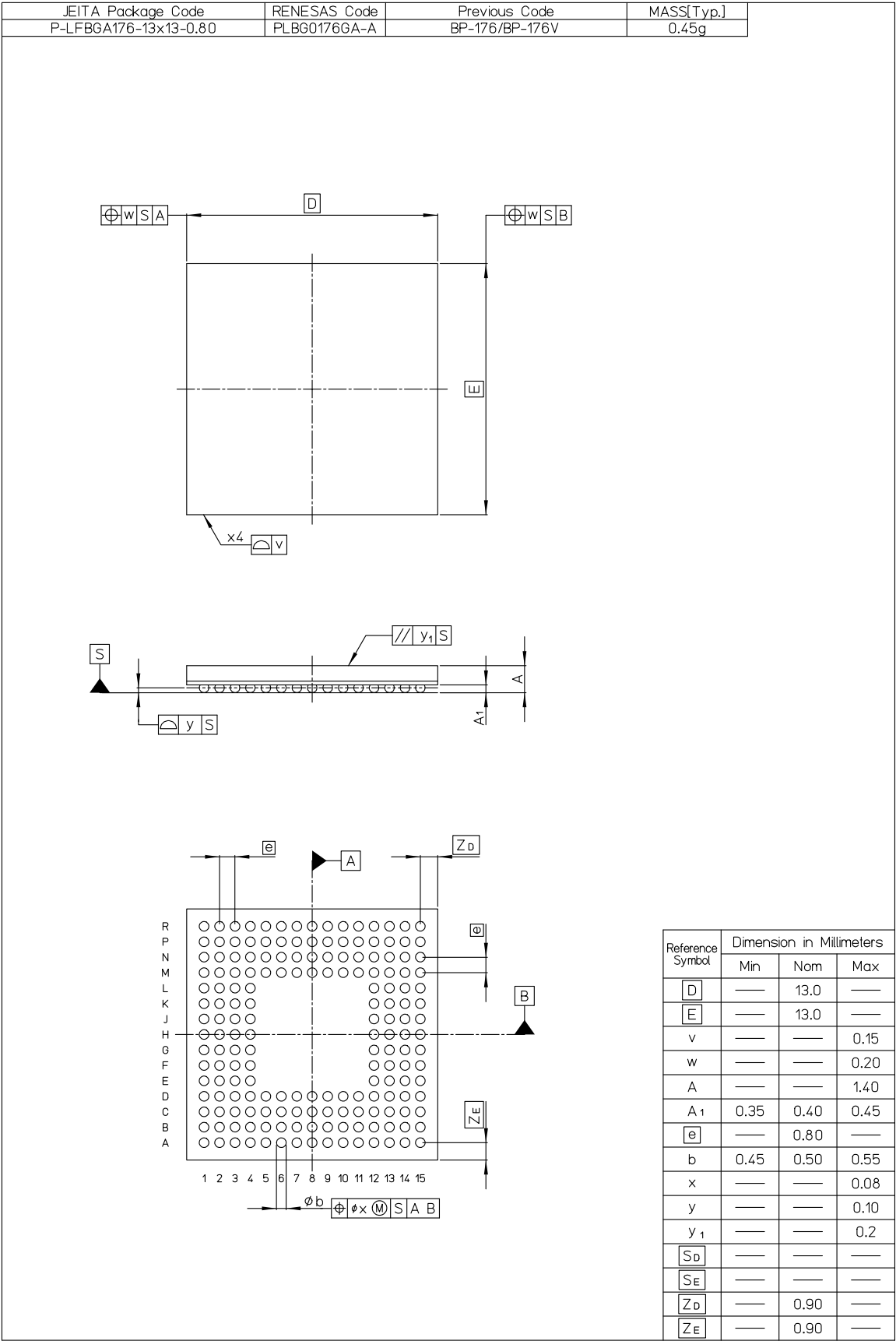


Figure B 176-Pin LFBGA (PLBG0176GA-A)

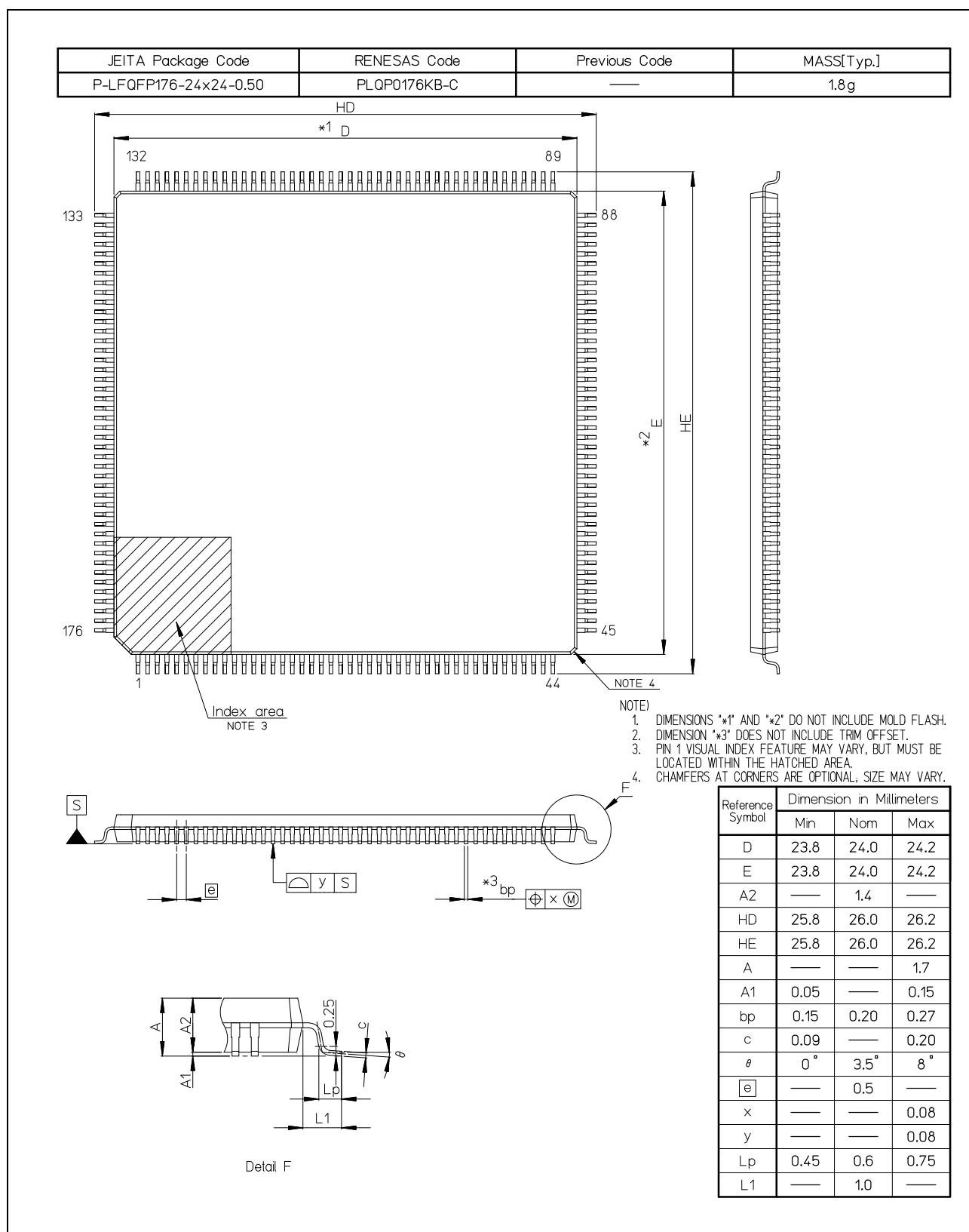


Figure C 176-Pin LFQFP (PLQP0176KB-C)

JEITA Package Code	RENESAS Code	Previous Code	MASS[Typ.]
P-TFLGA145-7x7-0.50	PTLG0145KA-A	145F0G	0.1g

Index mark (Laser mark)

Reference Symbol

Dimension in Millimeters			
Reference Symbol	Min	Nom	Max
D	—	7.0	—
E	—	7.0	—
v	—	—	0.15
w	—	—	0.20
A	—	—	1.05
e	—	0.5	—
b	0.21	0.25	0.29
b ₁	0.29	0.34	0.39
x	—	—	0.08
y	—	—	0.08
Z _D	—	0.5	—
Z _F	—	0.5	—

Figure D 145-Pin TFLGA (PTLG0145KA-A)

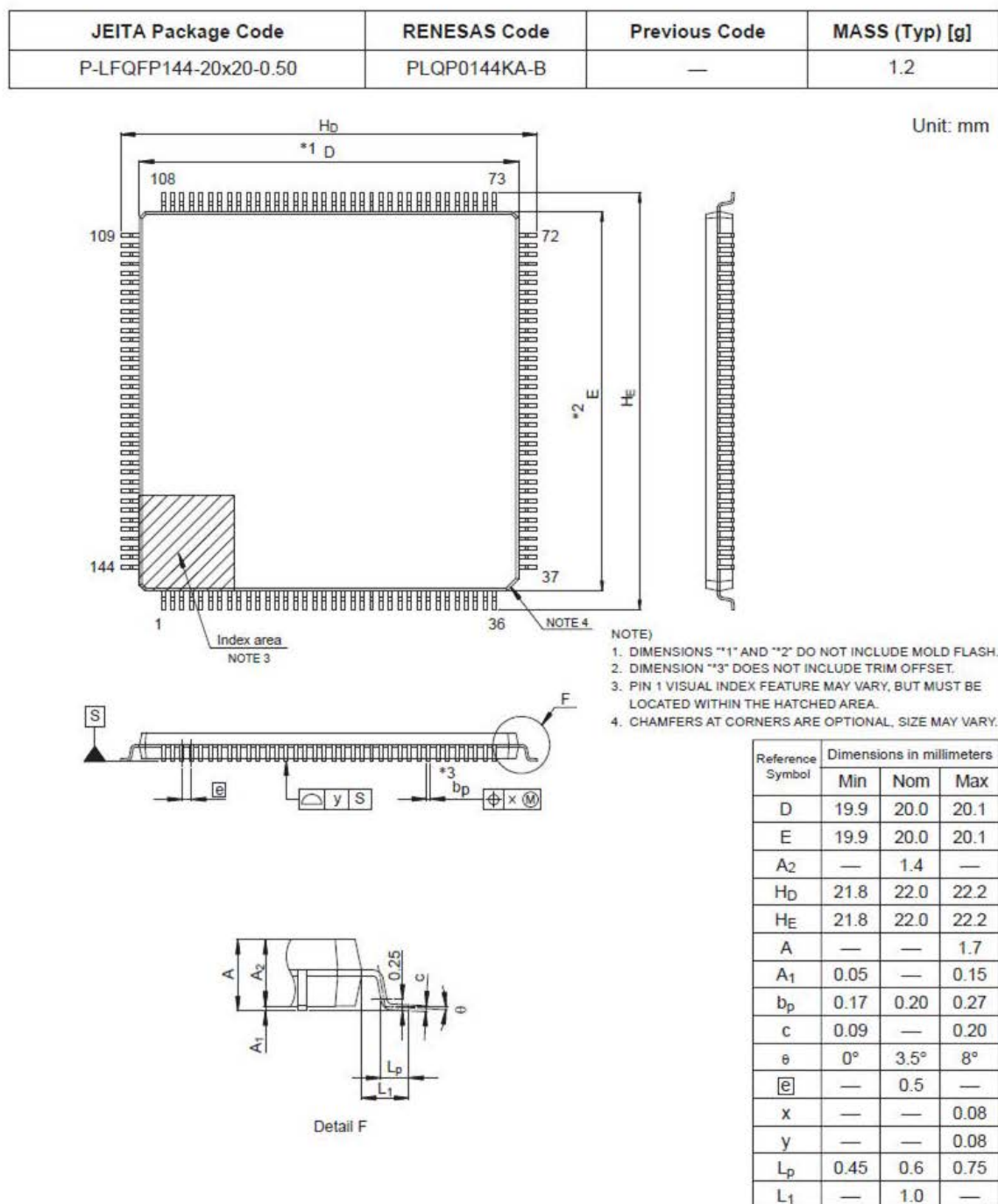


Figure E 144-Pin LFQFP (PLQP0144KA-B)

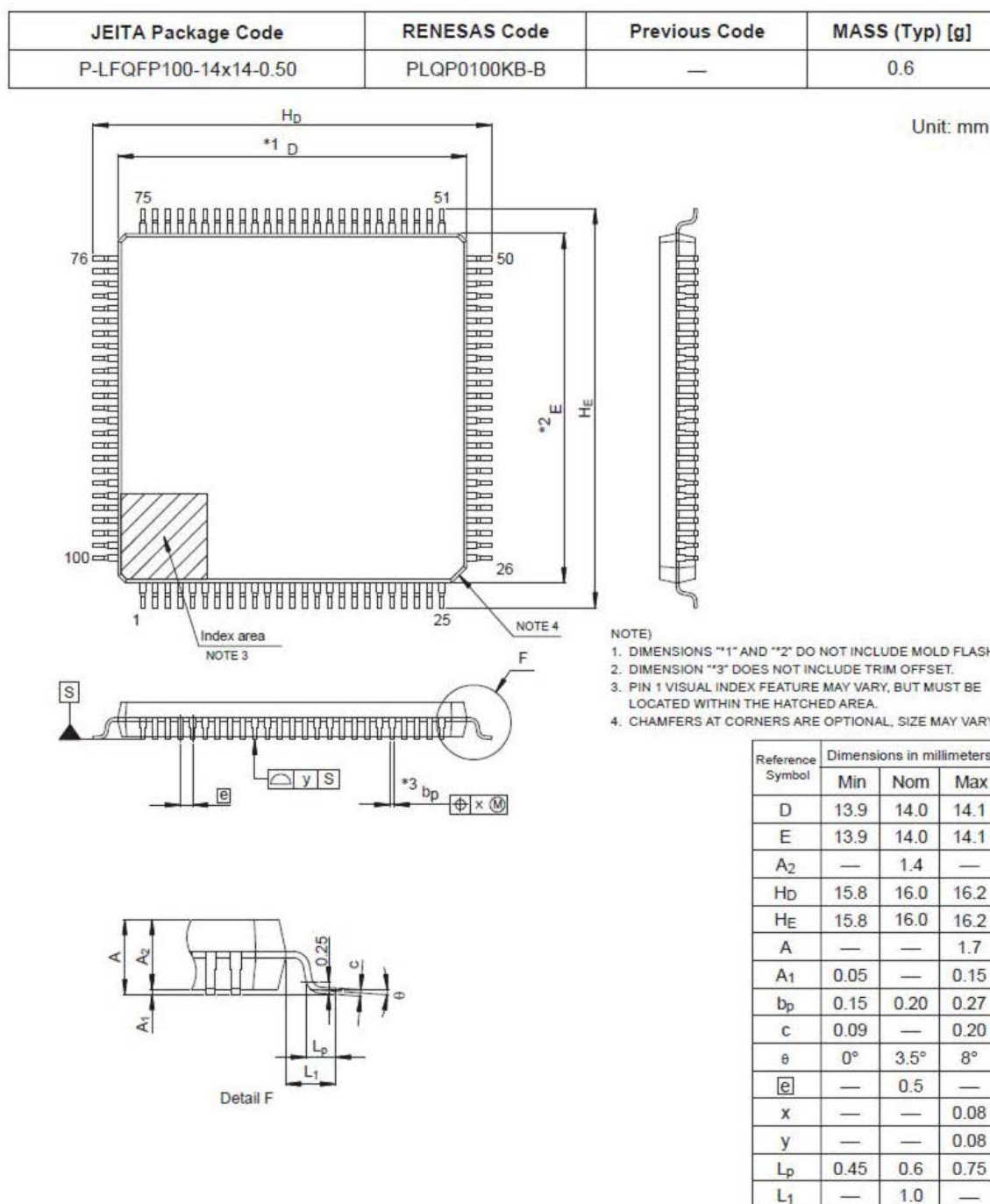


Figure F 100-Pin LFQFP (PLQP0100KB-B)

REVISION HISTORY	RX72N Group User's Manual: Hardware
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Classifications

- Items with Technical Update document number: Changes according to the corresponding issued Technical Update
- Items without Technical Update document number: Minor changes that do not require Technical Update to be issued

Rev.	Date	Description		Classification
		Page	Summary	
1.00	May 31, 2019	—	First edition, issued	
1.11	Feb 26, 2021	1. Overview		
		88, 89	Table 1.3 List of Products, changed	
		135, 136, 138	Table 1.8 List of Pin and Pin Functions (145-Pin TFLGA), changed	TN-RX*-A0222A/E
		143, 148	Table 1.9 List of Pin and Pin Functions (144-Pin LFQFP), changed	
		9. Clock Generation Circuit		
		335	Table 9.1 Specifications of Clock Generation Circuit (1/2), changed	
		376	Table 9.7 Internal Clocks and Supply Destination Modules (1/2), changed	
		383	9.10.2 Note on Rewriting the SCKCR3 Register, added	TN-RX*-A0224B/E
		10. Clock Frequency Accuracy Measurement Circuit (CAC)		
		398	10.3.1 Measuring Clock Frequency, changed	
		13. Register Write Protection Function		
		453	Table 13.1 Association between PRCR Bits and Registers to be Protected, changed	
		15. Interrupt Controller (ICUD)		
		467	Table 15.1 ICU Specifications (2/2), changed	
		496	15.2.26 Software Configurable Interrupt B Request Register k (PIBRk) (k = 0h to Bh), Note1, changed	
		497	15.2.27 Software Configurable Interrupt A Request Register k (PIARk) (k = 0h to Ah, Ch), Note 1, changed	
		502	Table 15.3 Interrupt Sources for Software Configurable Interrupt B (3/3), changed	
		519 to 521	Table 15.7 Group Interrupt Requests, changed	
		539	15.7.7.1 Polling for Software Configurable Interrupts, Note 1, changed	
		16. Buses		
		548	Table 16.6 Pin Configuration of the External Bus (1/2), changed	
		583, 584, 576	16.4.1 Data Alignment Control for CS Area, changed	
		20. Data Transfer Controller (DTCb)		
		787	Figure 20.1 DTC Block Diagram, changed	
		796	20.2.9 DTC Vector Base Register (DTCVBR), changed	
		799	20.2.13 DTC Index Table Base Register (DTCIBR), changed	
		21. Event Link Controller (ELC)		
		839, 840	Table 21.3 Correspondence between Values Set in ELSRn.ELS[7:0] Bits and Event Signals, changed	
		23. Multi-Function Pin Controller (MPC)		
		897, 898	Table 23.1 Functions Assigned to Each Multiplexed Pin (15/27) to (16/27), changed	TN-RX*-A0222A/E
		902 to 905	Table 23.1 Functions Assigned to Each Multiplexed Pin (20/27) to (22/27), changed	
		909	Table 23.1 Functions Assigned to Each Multiplexed Pin, Note2, deleted	
		914	23.2.3 P1n Pin Function Control Register (P1nPFS) (n = 0 to 7) Table 23.4 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LFQFP, changed	
		915	Table 23.5 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-/100-Pin LFQFP, changed	
		916	23.2.4 P2n Pin Function Control Register (P2nPFS) (n = 0 to 7) Table 23.6 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-Pin LFQFP, 145-Pin TFLGA, changed	TN-RX*-A0222A/E
		917	23.2.5 P3n Pin Function Control Register (P3nPFS) (n = 0 to 4) Table 23.7 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-/100-Pin LFQFP, 145-Pin TFLGA, changed	

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		Page	Summary	
1.11	Feb 26, 2021	923	23.2.10 P8n Pin Function Control Register (P8nPFS) (n = 0 to 7) Table 23.13 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-Pin LFQFP, changed	
		924	Table 23.14 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-Pin LFQFP, changed	
		925	23.2.11 P9n Pin Function Control Register (P9nPFS) (n = 0 to 7) Table 23.16 Register Settings for Input/Output Pin Function in 145-Pin TFLGA, 144-Pin LFQFP, changed	
		928	23.2.14 PCn Pin Function Control Register (PCnPFS) (n = 0 to 7) Table 23.19 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-Pin LFQFP, 145-Pin TFLGA, changed	
		930	23.2.15 PDn Pin Function Control Register (PDnPFS) (n = 0 to 7) Table 23.20 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-Pin LFQFP, 145-Pin TFLGA, changed	TN-RX*-A0222A/E
		932	23.2.16 PEn Pin Function Control Register (PEnPFS) (n = 0 to 7) Table 23.21 Register Settings for Input/Output Pin Function in 224-/176-Pin LFBGA, 176-/144-Pin LFQFP, 145-Pin TFLGA, changed	
		940	23.2.23 PMn Pin Function Control Register (PMnPFS) (n = 0 to 7) Table 23.31 Register Settings for Input/Output Pin Function in 224-Pin LFBGA, changed	
		24. Multi-Function Timer Pulse Unit 3 (MTU3a)		
		962	Table 24.2 MTU Functions, changed	
		967	Table 24.5 CCLR[2:0] (MTU1 and MTU2), changed	
		979	Table 24.15 TIOR (MTU1), changed	
		986	Table 24.29 TIOR (MTU1), changed	
		1004	24.2.16 Timer Longword General Registers (TGRALW, TGRBLW), changed	
		1025	24.2.32 Timer Waveform Control Registers (TWCRA, TWCRB), changed	
		25. Port Output Enable 3 (POE3a)		
		1219	25.2.15 MTU0 Pin Select Register 1 (M0SELR1), changed	
		1220	25.2.16 MTU0 Pin Select Register 2 (M0SELR2), changed	
		1221	25.2.17 MTU3 Pin Select Register (M3SELR), changed	
		1222	25.2.18 MTU4 Pin Select Register 1 (M4SELR1), changed	
		1223	25.2.19 MTU4 Pin Select Register 2 (M4SELR2), changed	
		1224	25.2.20 MTU6 Pin Select Register (M6SELR), changed	
		1230	Table 25.4 Correspondence between MTU Pins, changed	
		1231	25.3.2 Input-Level Detection Operation, changed (1) Falling Edge Detection, changed	
		1232	(2) Low-Level Detection, changed	
		1233	25.3.6 Additional Functions for High-Impedance Control, changed	
		30. 8-Bit Timer (TMRb)		
		All	Terms unified: baud rate clock → base clock	
		1553	Chapter title, changed	
		1558	30.2.1 Timer Counter (TCNT), changed	
		1559	30.2.2 Time Constant Register A (TCORA), changed 30.2.3 Time Constant Register B (TCORB), changed	
		1560	30.2.4 Timer Control Register (TCR), changed	
		33. Realtime Clock (RTCd)		
		1634	33.2.17 RTC Control Register 1 (RCR1), changed	
		1636, 1637	33.2.18 RTC Control Register 2 (RCR2), changed	TN-RX*-A0236B/E
		1644	33.2.23 Time Capture Control Register n (RTCCRN) (n = 0 to 2), changed	
		1651	Figure 33.3 Clock and Count Mode Setting Procedure, changed	TN-RX*-A0236B/E
		1652	Figure 33.4 Setting the Time, changed	
		1653	Figure 33.5 30-Second Adjustment, changed	
		1659	33.3.9 Time Capture Function, changed	
		1664	33.6.5 Notes on Writing to and Reading from Registers, changed	
		1665	Figure 33.14 Initialization Procedure, changed	TN-RX*-A0236B/E

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		Page	Summary	
1.11	Feb 26, 2021	40. USB 2.0 FS Host/Function Module (USBb)		
		All	Terms unified: PIPE _n → Pipe n	
		1928	40.1 Overview, changed	
			Table 40.1 USB Specifications, changed	
		1929	Table 40.2 USB Pin Configuration, changed	
		1932, 1933	40.2.2 System Configuration Status Register 0 (SYSSTS0), changed	
		1934	40.2.3 Device State Control Register 0 (DVSTCTR0), changed	
		1937, 1938	40.2.4 CFIFO Port Register (CFIFO), D0FIFO Port Register (D0FIFO), D1FIFO Port Register (D1FIFO) (2) When the MBW bit is 0, changed	
		1941, 1942	40.2.5 CFIFO Port Select Register (CFIFOSEL), D0FIFO Port Select Register (D0FIFOSEL), D1FIFO Port Select Register (D1FIFOSEL) • D0FIFOSEL, D1FIFOSEL, changed	
		1944	40.2.6 CFIFO Port Control Register (CFIFOCTR), D0FIFO Port Control Register (D0FIFOCTR), D1FIFO Port Control Register (D1FIFOCTR), changed	
		1945	40.2.7 Interrupt Enable Register 0 (INTENB0), Note 1, changed	
		1946	40.2.8 Interrupt Enable Register 1 (INTENB1), Note 1, changed	
		1988	40.2.37 Deep Standby USB Transceiver Control/Pin Monitoring Register (DPUSR0R), changed	
		1989, 1990	40.2.38 Deep Standby USB Suspend/Resume Interrupt Register (DPUSR1R), changed	
		1991	40.3.1.2 Controller Function Selection, changed	
			40.3.1.3 Controlling USB Data Bus Resistors, changed	
		1996	40.3.1.5 Release from Deep Software Standby Mode Due to USB Suspend/ Resume Interrupts, changed	
			Figure 40.6 Overview of Connection between USB Resume Detecting Unit and USB0 I/O Pins, title, changed	
		1997	Table 40.12 USB0 Suspend/Resume Interrupt Sources and Corresponding I/O Pins, title changed	
			Figure 40.7 USB Setting Flowchart for Transition to Deep Software Standby Mode as Host/Function Controller, changed	
		1998	Figure 40.8 USB Setting Flowchart for Canceling Deep Software Standby Mode as Host Controller, changed	
		1999	Figure 40.9 USB Setting Flowchart for Canceling Deep Software Standby Mode as Function Controller, changed	
		2000	Table 40.13 Interrupt Sources, changed	
		2001	Figure 40.10 Circuits Related to Interrupts in USB, changed	
		2036	40.4.1 Setting the Module-Stop Function, changed	
		41. Serial Communications Interface (SCIj, SCli, SCli)		
		2041	Table 41.3 SCli Specifications (2/2), changed	
		2055	41.2.7 Transmit FIFO Data Register (FTDR), changed	
		2056	41.2.9 Serial Mode Register (SMR) (1) Non-Smart Card Interface Mode (SCMR.SMIF = 0), Note 4, changed	
		2058	(2) Smart Card Interface Mode (SCMR.SMIF = 1), Note 2, changed	
		2060, 2061	41.2.10 Serial Control Register (SCR) (1) Non-Smart Card Interface Mode (SCMR.SMIF = 0), changed	
		2071	41.2.11 Serial Status Register (SSR/SSRFIFO) (3) Non-Smart Card Interface Mode and FIFO Mode (SCMR.SMIF = 0 and FCR.FM = 1), changed	
		2072	41.2.12 Smart Card Mode Register (SCMR), changed	
		2088	41.2.15 Serial Extended Mode Register (SEMR), Note 1, changed	
		2102	41.2.24 Line Status Register (LSR), changed	
		2105	41.2.27 Serial Port Register (SPTR), changed	
			Table 41.30 Controlling the TXD _n pin, changed	
		2119	41.3.2 Receive Data Sampling Timing and Reception Margin in Asynchronous Mode, Note 1, changed	
		2121	41.3.5 CTS and RTS Functions, changed	

Rev.	Date	Description		Classification
		Page	Summary	
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